Engineering Simplicity*

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Abstract

Well-designed market rules often fail in practice for a simple reason: participants do not play the strategy the rules were designed around. In a second-price auction the winner pays the runner-up's bid, so bidding exactly what the item is worth to you is optimal no matter what anyone else does—yet decades of laboratory experiments find that people persistently bid above their value, while the same rule played as an ascending price clock is followed almost correctly. Which implementations, explanations, and decision aids induce correct play is therefore an empirical design question, and the standard instrument for answering it—incentivized human experiments—is too slow and expensive to search a large design space. This paper evaluates a cheaper instrument: a panel of large language model (LLM) bidding agents, used to *screen* candidate designs for which ones preserve correct play, not to simulate human bids. We report two results. First, an ordinal validation. On every auction comparison for which published human evidence exists—first-price versus second-price sealed bidding (harder versus easier), sealed bids versus ascending clocks (the clock helps), and menu-style restatements of the rules (no reliable effect)—a panel of four LLM families recovers the direction of the human finding, family by family, even though the two populations err in opposite directions (humans overbid; the models underbid) and no tuning of the agents reproduces human bidding in levels. The agreement is ordinal, not cardinal—and ordinal is exactly the signal design screening needs. Second, theory-guided discovery. We organize the design space along three dimensions from the theory of simple mechanisms—how a mechanism is implemented, how it is described, and what reasoning support participants receive—and use the panel to populate the resulting design map. Cells with human anchors reproduce the human record; cells without become concrete, falsifiable predictions for human experiments. The sharpest: one honest sentence stating *why* truthful bidding is safe ("your bid determines whether you win, not what you pay") recovers much of the clock's benefit in every model family, while restating the rules without that content does nothing on average, and prompting agents to reason about opponents backfires. Because each simulated bid is accompanied by a stated plan—a short written explanation of the intended strategy, such as "I will bid slightly below my value to keep a profit margin"—we can also show that the effective designs change bids without changing the plans agents articulate: a screening instrument should audit behavior, not stated understanding.

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Contents

1 Introduction

In many market mechanisms, participants do not play the strategy that theory identifies as optimal—even when that strategy is simple, safe, and known to the designer. Auction bidders pay more for items than the items are worth to them (Kagel, 1995). Applicants in deployed school-choice and labor-market clearinghouses submit dominated preference rankings even where reporting honestly is provably optimal (Rees-Jones, 2018; Roth, 1982). Decades of laboratory evidence show these deviations to be systematic rather than noise (Kagel and Roth, 2020)—less a matter of exotic preferences than of the computational difficulty of recognizing what the mechanism makes safe (Oprea, 2024). The stakes are the mechanism's own objectives: dominated play erodes the efficiency, revenue, and fairness properties the rules were chosen to deliver.

The design response is to make correct play easy, and a well-developed theory now characterizes which mechanisms are cognitively simple (Li, 2017, 2024). But for a practitioner the question is unavoidably empirical: among the many concrete choices of implementation, wording, and decision support, which ones preserve correct play in *their* market? The standard instrument for that question—the incentivized human experiment—is slow and expensive enough that only a handful of design variants are ever tested, each chosen well in advance. Simplicity has therefore remained a subject for theory and isolated experiments rather than a routine engineering practice.

The new opportunity is that a different kind of experimental participant has arrived. Large language models (LLMs) reproduce qualitative patterns from classic behavioral experiments (Horton, 2023; Aher et al., 2023) and support automated experimentation at scale (Park et al., 2023; Manning et al., 2024). They have also begun to transact in real markets as bidding and negotiating agents in their own right—studied for their collusive tendencies (Fish et al., 2024), their feasibility as bidders (Tsuchihashi, 2023; Lu, 2025), and their trainability as auction participants (Chen et al., 2023)—so their behavior under a mechanism's rules is becoming designer-relevant in itself. A designer can now run a mechanism experiment on artificial participants in minutes, for dollars, with reproducible seeds.

This paper develops and validates a way to use that opportunity that does not require the agents to be human-like: *ordinal screening*. The obvious alternative—treat the simulated population as a stand-in for human subjects and read levels off it—asks the instrument for more than it can give: LLM agents do not reproduce human levels, they do not err in the human direction, and although per-task calibration can narrow distributional gaps (Corrigan et al., 2025), we show that no tuning of the agents is portable across mechanisms (Appendix F). Under ordinal screening, the designer asks the panel only which design is behaviorally *easier*—directions and orderings, never levels—with the design space fixed ex ante by theory, a cross-model robustness rule deciding what counts as a finding, and scarce human experiments reserved for the cells the screen elevates. The question the paper answers is: *can LLM agents recover robust ordinal patterns in human behavior across mechanisms, and can those ordinal signals be used to screen theory-guided mechanism designs?*

Both answers are yes. On every mechanism comparison for which published human evidence exists—which of two formats is behaviorally harder, whether a simpler implementation of the same rule improves play, whether restating the rules moves behavior—a panel of four LLM families recovers the direction of the human finding, family by family (Figure 2). The agreement is not inherited from mimicry: the two populations' characteristic errors point in *opposite* directions, so a panel that merely reproduced human data from training would get the comparisons wrong, and no tuning we searched makes the agents match human behavior in levels. Agreement is ordinal, not cardinal—and ordinal is the signal design screening needs. Used this way, the panel populates a theory-organized design map at roughly $10 per cell, turning simplicity from a property that theory characterizes into a quantity a practitioner can engineer.

This paper needs one such mechanism as its concrete scenario, and we use the canonical one throughout:

the second-price sealed-bid (SPSB) auction. Each bidder privately submits one bid; the highest bidder wins but pays the *second*-highest bid. Bidding exactly what the item is worth to you is optimal no matter what the other bidders do—your bid determines only *whether* you win, never *what* you pay (Vickrey, 1961). Economists call such a strategy *dominant*, and mechanisms that make truthful reporting dominant *strategy-proof*. Human bidders violate the prescription anyway, most commonly by bidding *above* value, persistently and expensively (Kagel and Levin, 1993; Kagel, 1995).

The failure is implementation-dependent, which is what makes it a design problem rather than a fixed cost of human nature. Run the same allocation rule as an ascending clock—the price rises step by step, each bidder chooses *stay* or *exit* at each price, and the last bidder standing wins at the runner-up's exit price—and human subjects find the dominant strategy far more often (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022). Same economics, different interface, different behavior. We choose auctions as the scenario because they pair the sharpest dominant-strategy predictions with the deepest human experimental record to validate against; nothing in the method is specific to them.

A body of theory explains why implementation matters, and it supplies the coordinates of our design space. Li (2017) formalizes *obvious strategy-proofness* (OSP): in the clock, but not in the sealed bid, verifying that truth-telling is best requires no hypothetical, case-by-case reasoning about what opponents might have bid. Related work characterizes when optimal play requires no beliefs about others at all (Börgers and Li, 2019), when it requires no multi-step foresight (Pycia and Troyan, 2023), and how a mechanism's cognitive burden decomposes into contingent reasoning, forward planning, and belief formation (Li, 2024). None of this is auction-specific: the same notions classify matching and assignment mechanisms (Ashlagi and Gonczarowski, 2018), and we demonstrate the instrument's portability by replicating the paper's central description finding under student-proposing deferred acceptance (Appendix K). What the theory does not do is rank the concrete choices a designer faces; it constrains the design space and signs its cells ex ante, and the rest is measurement.

Section 2 turns that theory into a design map with three dimensions, each holding the allocation rule fixed. *Implementation*: replace the one-shot sealed bid with the OSP ascending clock (Li, 2017; Pycia and Troyan, 2023). *Description*: keep the sealed bid and change only the words—a menu-style restatement of the outcome function, which improves measured understanding while leaving human bidding essentially unchanged (Gonczarowski et al., 2023, 2024); a one-sentence *safety* description stating the invariant that makes truthful play safe ("your bid determines *if* you win, not *what* you pay"), never tested in a human auction, though property-exposing descriptions help and mechanics-only descriptions backfire in matching mechanisms (Katuščák and Kittsteiner, 2024; Guillén and Hakimov, 2018); and a framing of the sealed bid as an automatic exit threshold on a hypothetical clock, which raises human truthful bidding (Breitmoser and Schweighofer-Kodritsch, 2022). *Reasoning support*: hold the mechanism and its description fixed and scaffold the participant's reasoning—walk it through the payoff consequences of each outcome, or prompt it to think about opponents, an axis strategy-proofness makes irrelevant and therefore a built-in diagnostic (Li, 2024; Börgers and Li, 2019). Direct advice to play truthfully is excluded by construction: it works on humans, but it reveals the answer rather than exposing why the answer is safe (Masuda et al., 2022).

Two features discipline the exercise. First, every cell's predicted sign is fixed ex ante from the theory, and roughly half the cells are predicted to be inert or to backfire—a pattern opportunistic prompt search would neither produce nor publish. Second, the economics of search are on the table: the laboratory test of obvious strategy-proofness engaged 548 subjects with documented payments around $5,400 for its main experiment alone (Li, 2017), and the leading description experiments recruited 542 subjects at $15–26 each (Gonczarowski et al., 2024), while the panel sweeps the full design grid for under $400.

The paper contributes a validated screening instrument and the first populated map of this design space. The validation contribution establishes, comparison by comparison, that the instrument's ordinal readings agree with the human experimental record wherever that record exists, under a pre-committed rule: a result enters the main text only if its direction is consistent across all four prespecified model families. The discovery contribution fills the map, classifies levers into substitutes, complements, and anti-levers, and converts every cell that no human experiment has visited into a concrete, falsifiable prediction with a costing. Throughout, the panel is evaluated as an ordinal design-screening tool, not as a calibrated model of the average human bidder. Our question is distinct from designing the internal auctions of LLM-powered platforms (Dütting et al., 2024b) and from automated mechanism design, which searches over allocation rules themselves (Dütting et al., 2024a; Wang et al., 2024): we hold the rule fixed and engineer the interface a bounded reasoner meets.

Screening requires a benchmark layer, and ours is built from the human experimental record. Because micro-level bid data from the classic experiments are rarely available, we reconstruct human bidding distributions by moment matching: a transparent mixture model calibrated, source by source, to the aggregate statistics each paper reports (Kagel and Levin, 1993; Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022; Gonczarowski et al., 2023). A single deviation metric—the scaled mean absolute deviation (SMAD) of actions from the theoretical benchmark—places humans and LLM agents on the same scale, and a *preservation index* expresses every design lever's effect as the fraction of a population's own baseline deviation that the lever removes. The reconstructions are held to a strict epistemic standard, stated once and enforced throughout: comparisons to reconstructed human behavior are made at the level of rankings, directions, and comparative statics only; levels and significance tests are reserved for theory benchmarks.

The validation results are collected in one exhibit (Figure 2), and every row shows its per-family direction. The panel—GPT-4o, Claude 3.5 Haiku, Gemini 2.0 Flash, Gemma 27B—reproduces the human difficulty ordering of payment rules: first-price bidding, where equilibrium requires strategic shading, is harder than second-price bidding, where truth-telling is dominant, in all four families (GPT-4o: SMAD 27.6% versus 8.7%, against a human 24.8% versus 5.7%). It reproduces the implementation effect: the ascending clock moves every family toward truthful play ($p < 0.001$ in each), as it moves humans. And it reproduces the description null: menu-style restatements, which leave human bidding essentially unchanged, have no consistent effect on the panel either.

The agreement is more informative than it first appears, because the two populations make *opposite* baseline mistakes. Humans overbid in second-price auctions—67.2% of reconstructed bids exceed value—while the models underbid: 74.1% of GPT-4o's baseline bids fall below value. That the same design levers repair opposite errors in both populations indicates that the levers act on the strategic problem itself—the recognizability of the dominant strategy—rather than on any psychology particular to humans. It also rules out the deflationary reading that the panel parrots the human experimental literature from its training data: a parroting population would overbid. The same divergence marks the instrument's boundary, measured in Appendix F: no tuning of the agents—temperature, personas, framings, explanations—achieves level-and-direction agreement across mechanisms, and the one knob that restores the human direction of error in second-price auctions induces dominated bids in first price, where humans almost never overbid.

The discovery results fill the map (Section 4, Table 2), and four findings pass the cross-family robustness rule. The OSP clock implementation is the strongest lever, compressing deviations toward zero in every family and eliminating the catastrophic tail of severe underbids. A one-sentence safety description—honest text stating *why* truthful bidding is safe—recovers roughly half of the clock's benefit in every family, at the cost of a sentence. A payoff-tree scaffold, which lays out what happens in each contingency without recommending an action, delivers a comparable improvement. And belief-formation scaffolds backfire, pushing bids further from value, exactly as the theory's diagnostic logic predicts for agents that have not internalized strategy-proofness.

The rest of the map is flagged or forward-looking. Two cells are honestly heterogeneous and reported as such: the menu restatement is null on average but sign-mixed underneath, and a clock-*framing* description helps three families while harming the fourth—a paraphrase sensitivity we quantify as a warning against single-wording inferences (Appendix G). Cells that no human experiment has visited become predictions with directions and costings—most cheaply, that the safety sentence will shift human bidding toward value where menu restatements have not, testable in a single incentivized session. And the sharpest finding travels: in the school-choice matching domain, a description carrying the safety invariant nearly eliminates misreporting while a description restating only the mechanics makes it strictly worse (Appendix K).

The instrument also records *why* agents say they act as they do, which yields a methodological result for anyone auditing an interface change by asking participants to explain themselves. Every bid arrives with a stated plan—a short written explanation of the intended strategy, such as "I will bid slightly below my value to keep a profit margin." Coding all 21,990 plans with a keyword dictionary aligned to the theory's cognitive categories, we find that stated reasoning and realized bids never move together across our levers (Section 5): the safety description and the payoff tree move bids while leaving stated plans unchanged; a worst-case scaffold shifts stated plans while leaving bids unchanged; the menu restatement moves neither. Effective designs changed what agents *did*, not what they *said*—so screening, synthetic or human, should be evaluated on realized play rather than elicited understanding.

The paper proceeds as follows. Section 2 develops the simplicity framework—first for a general mechanism, then in the auction scenario—and presents the design map with its ex-ante predictions. Section 3 describes the testbed and reports the ordinal validation. Section 4 populates the design map, reports the robust lever results and the flagged heterogeneous ones, and registers the interaction cells for future runs. Section 5 analyzes stated plans against realized bids. Section 6 collects design implications, the screening workflow and its cost, robustness and scope (including frontier reasoning models, which solve our baseline formats yet remain vulnerable to presentation—Appendix E), limitations, and the predictions for human experiments, and concludes. Appendices collect procedures, prompts, the human reconstructions, the cardinal-calibration analysis, model-specific heterogeneity, and the school-choice matching domain in which the description finding replicates (Appendix K).

2 Simplicity Theory and the Design Map

This section builds the framework in three steps, general first. Section 2.1 defines the objects for an arbitrary mechanism—dominant strategies, strategy-proofness, the behavioral deviation the paper measures, and the simplicity notions that predict when that deviation should be small. Section 2.2 instantiates everything in the auction scenario. Sections 2.3–2.5 then organize the designer's remaining choices into a design map and state, ex ante, what each cell should do.

2.1 Mechanisms, Dominant Strategies, and Behavioral Deviation

A mechanism asks n participants to act and turns their actions into an outcome. Participant i privately holds information $\theta_i \in \Theta_i$—a value for an item, a preference ranking over schools—and the mechanism specifies an action set A_i (submit a bid, submit a rank-order list, answer a sequence of stay/exit queries) and an outcome function g that maps every action profile $a = (a_1, \dots, a_n)$ to an allocation and payments. Participant i's payoff $u_i\big(g(a), \theta_i\big)$ depends on the outcome and on her own information, and a strategy σ_i maps her information to an action.

A strategy $\sigma_i^\star$ is *dominant* if playing it is optimal no matter what the others do: for every θ_i, every alternative action $a_i \in A_i$, and every profile a_{-i} of the other participants' actions,

$$u_i\big(g(\sigma_i^\star(\theta_i),\, a_{-i}),\, \theta_i\big) \;\geq\; u_i\big(g(a_i,\, a_{-i}),\, \theta_i\big).$$

A direct mechanism ($A_i = \Theta_i$) is *strategy-proof* if truthful reporting, $\sigma_i^\star(\theta_i) = \theta_i$, is dominant. Strategy-proofness is a prized design property because it makes good play radically simple in principle: a participant needs no data about competitors, no forecasts, and no strategic sophistication; honesty is a best response to everything.

The behavioral object of this paper is the gap between prescribed and observed play. Fix a distance $d(a_i, \theta_i) \geq 0$ on actions, equal to zero exactly at the dominant action—for bids, the absolute distance between the submitted and the dominant bid; for rank-order lists, the number of pairwise inversions—and call a participant population's expected distance its *deviation*,

$$D \;=\; \mathbb{E}\big[\, d(a_i, \theta_i)\,\big],$$

measured under the mechanism *as implemented, described, and supported*. For fully rational participants in a strategy-proof mechanism, $D = 0$ however the mechanism is implemented, worded, or explained: that is exactly what dominance means. Empirically, $D > 0$, persistently and across institutions (Section 1). The estimand of this paper is how the interface choices a designer still controls move D while the outcome function g is held fixed; Section 3.2 normalizes D so that it is comparable across mechanisms and across participant populations.

Why is $D > 0$ at all? The dominant strategy must be *recognized*, and recognizing it is a computation (Oprea, 2024). The theory of simple mechanisms characterizes when that computation is easy, at the same level of generality as the definitions above. A mechanism—now an extensive form, not necessarily a direct one—is *obviously strategy-proof* (OSP) if, at every point where a participant might first deviate from $\sigma^\star$, even the *worst* outcome from following $\sigma^\star$ is at least as good as the *best* outcome from deviating (Li, 2017); verifying dominance in an OSP mechanism therefore requires no hypothetical, case-by-case reasoning about moves the participant cannot observe. A mechanism is *strategically simple* if optimal play requires no beliefs about other participants (Börgers and Li, 2019), and *one-step simple* if it requires no multi-step foresight over one's own future moves (Pycia and Troyan, 2023). Li (2024) decomposes the residual cognitive burden of a mechanism into three demands, and this decomposition organizes everything that follows: **contingent reasoning** (thinking case-by-case about unobserved moves of others), **forward planning** (thinking ahead about one's own future moves), and **belief formation** (thinking about what others will do or believe). In a strategy-proof mechanism, belief formation is a tax a participant should never pay—beliefs are irrelevant to optimal play—so any behavioral response to a belief prompt signals strategizing where none is needed.

2.2 The Auction Scenario

We instantiate the framework in the canonical scenario. Private information is a value $\theta_i = v_i$, drawn independently from a known distribution; the action is a sealed bid $a_i = b_i \geq 0$; and the outcome function of the second-price sealed-bid (SPSB) auction awards the item to the highest bidder at a price p equal to the *second*-highest bid, giving the winner payoff $v_i - p$ and everyone else zero. Truthful bidding, $\sigma^\star(v_i) = v_i$, is dominant, and the argument is two sentences long: your bid affects only whether you win, not the price you pay (the price is set by someone else's bid), so bidding above value can only add wins that lose money, and bidding below value can only remove wins that make money (Vickrey, 1961). SPSB is therefore strategy-proof, the deviation distance is $d(b_i, v_i) = |b_i - v_i|$, and theory predicts $D = 0$. Human subjects deliver $D > 0$ decade after decade, mostly by overbidding (Kagel and Levin, 1993; Kagel, 1995).

The scenario also contains, within a single outcome rule, exactly the wedge the simplicity notions describe. SPSB is strategy-proof but *not* OSP: to verify that truth-telling is best, a bidder must reason case by case over rival bids she cannot observe—the contingent-reasoning burden. The ascending clock auction implements the *same* outcome function g in a different extensive form—the price rises in small increments, each bidder chooses *stay* or *exit* at each price, and the last bidder standing wins at the runner-up's exit price—and it *is* OSP: at every price, "stay while the price is below my value" beats exiting in every possible continuation, one comparison at a time. The clock is additionally one-step simple (no backward induction over future prices is needed), and both formats are strategically simple (beliefs are irrelevant in each). One allocation rule, two implementations, dominant in both, obvious in only one: this wedge is the identifying variation that every lever in the design map acts on. The same construction exists outside auctions—student-proposing deferred acceptance admits OSP implementations under suitable priority structures (Ashlagi and Gonczarowski, 2018)—which is what makes the framework, and the instrument built on it, portable (Appendix K).

2.3 Three Dimensions of Design

Fix the outcome function g: the item goes to the highest bidder at the second-highest bid, whether elicited by sealed bid or by clock. Everything a designer can still change falls along three dimensions, each of which holds the economics fixed and varies one layer of the mechanism's human—and now agent—interface. These are the columns of our design map.

Dimension 1: Implementation (A). Change the extensive form through which bids are elicited, holding the outcome rule fixed. Our implementation lever replaces the one-shot sealed bid with the ascending clock, in open (AC: dropouts observed) and closed (AC-B: dropouts unobserved) variants. Theory makes the sharpest prediction here: OSP implementations remove the contingent-reasoning burden entirely (Li, 2017), so the clock should produce the largest and most uniform improvement in truthful play. Because format familiarity is a potential confound—models, like people, have seen many descriptions of standard auctions—the map also registers the *+X pair* of Li (2017): sealed and clock implementations of a second-price rule shifted by a commonly known premium X, which preserves the strategy-proof/OSP distinction while making the format unfamiliar. [TODO: 2P+X and AC+X cells are registered but not yet run (E-v3-1); pin the exact +X parameterization to Li (2017) before launching.]

Dimension 2: Description (B). Keep the sealed bid and change only the words that explain it. The theory here is due to Gonczarowski et al. (2023), who formalize descriptions that make a mechanism's incentive properties transparent, and the human evidence splits the dimension into three sub-levers that we keep strictly separate. **(B1) Menu restatement**: re-describe the outcome function as a menu—the opponents' bids fix a price at which you may win, and your bid merely selects whether you accept; behaviorally close to inert for human subjects (Gonczarowski et al., 2023, 2024). **(B2) Safety-exposing description**: state, in one honest sentence, the structural invariant that makes truthful play safe—"your bid determines IF you win, not WHAT you pay; if you win, you pay what others bid, not what you bid" (*Payoff Safety*, the pivotal logic of Vickrey, 1961). No human auction experiment has tested a bare invariant statement of this kind; the nearest evidence is from matching mechanisms, where property-exposing descriptions help and mechanics-only descriptions backfire (Gonczarowski et al., 2024; Katuščák and Kittsteiner, 2024; Guillén and Hakimov, 2018). **(B3) Clock-framing**: describe the sealed bid as the exit threshold an automatic proxy would use on a rising price clock—OSP-style language with no change to the game—which increases truthful bidding in human subjects (Breitmoser and Schweighofer-Kodritsch, 2022). The accounting matters: B3 is a *description* of a sealed-bid game; the actual clock is an implementation change and belongs to Dimension 1. The two are never pooled.

Dimension 3: Reasoning support (C). Keep the mechanism and its description fixed and scaffold *how* the agent reasons, without revealing the dominant strategy. We index the scaffolds by the three cognitive burdens of Li (2024). **(C1) Contingent reasoning**: the *Payoff Tree* lays out the explicit tree of contingencies—"PATH A: your bid is highest → you win and pay the second-highest bid; PATH B: it is not → you lose and pay nothing"—telling the agent what *happens* in each case, never what to *do*. Because contingent reasoning is exactly the burden the sealed bid imposes, C1 is predicted to help. **(C2) Forward planning**: a one-shot sealed bid has no future own-moves to plan over, so this axis has nothing to bind on in our auction grid; the cell is marked N/A in the map (the clock's within-instance dynamics are one-step simple, so it does not bind there either). **(C3) Belief formation**: first-order ("what do you expect the others to bid?") and second-order ("what do the others think *you* will bid?") prompts. Because dominant strategies are belief-free, C3 is predicted to be inert for a participant who has internalized strategy-proofness—so any measured effect is diagnostic of mis-strategizing, and for agents that treat the auction as a beliefs game, making opponents salient is predicted to *backfire*.

A bounding family (D), reported in the appendix. Preference framings—risk personas, loss/endowment frames (Kahneman and Tversky, 1979; Dreyfuss et al., 2022)—alter the agent's stated objective rather than the mechanism's interface. Under a dominant strategy they are payoff-irrelevant for any monotone utility, and no designer would deploy them to improve play; we run them to bound the damage a careless or adversarial framing can do and report them in Appendix H. They do not enter the design map.

2.4 The Design Map

Figure 1 presents the complete map before any result: every economically meaningful cell formed from the three dimensions, its theoretical prediction, and an *evidence label* recording what is known about it independently of this paper. (The fully specified reference version, with the per-cell citations, is Table 4 in Appendix A.) Four labels are used. **Human anchor**: a directly comparable human experiment exists. **Nearby evidence**: related but not directly matched human evidence exists (e.g., the same description content tested in a matching mechanism rather than an auction). **New prediction**: only LLM evidence will exist; the cell is a falsifiable prediction for a future human experiment. **N/A**: the combination has no coherent theoretical interpretation. Cells registered but not yet run are marked *pending*. The map is intended to work the way a periodic table does: the anchored cells test whether the instrument fills known entries correctly (Section 3); the same theoretical structure then licenses filling the unobserved entries (Section 4); and the unobserved entries become predictions for human experiments (Section 6).

Single levers occupy most of the map, but the combinatorial structure also defines *interaction* cells, and theory signs them ex ante. Implementation and safety-description should be *substitutes*: the clock already makes the safety of truth-telling visible at every price, so adding the safety sentence to a clock should add nothing (a ceiling effect). Safety-description and the payoff tree should be *complements* in the direct mechanism: one supplies the invariant (why truth-telling is safe), the other the case analysis (what happens in each contingency), and they relieve different parts of the same verification. A menu restatement augmented with the invariance sentence should behave like a safety description—the informative contrast being with the bare menu, which restates the computation without the invariant. And a belief prompt layered on the clock is a diagnostic: OSP should immunize play against opponent salience. These four combination cells are registered in the map; the single-lever cells are complete, and the combination cells are the map's next wave.

Baseline: SPSB sealed bid, standard description — outcome rule g fixed in every cell **[H]** both populations deviate

Dimension 1: Implementation

Ascending clock (AC / AC-B) (A)
predicted $\Uparrow$ largest gain: OSP removes contingent reasoning; one-step
[H] human anchor

2P+X / AC+X pair (A)
predicted: same OSP gap, format familiarity removed
[H] human anchor · pending

Dimension 2: Description

Menu restatement (B1)
predicted $\approx$ 0: restates the rules, adds no incentive content
[H] human anchor (behavioral null)

Payoff Safety (B2)
predicted $\uparrow$ approaching OSP: states the invariant that makes truth-telling safe
[N] nearby (matching); auction cell = new prediction

Clock-framing (B3)
predicted $\uparrow$: OSP-style language, game unchanged
[H] human anchor

Menu + invariance sentence (B1+B2)
predicted $\uparrow$, behaves like B2 (content over format)
[N] nearby (Appendix K) · pending

Dimension 3: Reasoning support

Payoff Tree (C1)
predicted $\uparrow$: constructs the contingency cases (the binding axis)
[P] new prediction

Forward planning (C2)
N/A: no future own moves in a one-shot bid

First-order beliefs (C3)
predicted $\downarrow$: irrelevant axis; any effect diagnoses strategizing
[P] new prediction

Second-order beliefs (C3)
predicted $\downarrow$, more severe
[P] new prediction

Interactions (registered, ex-ante signs)

Clock × Safety (A×B2)
predicted: substitutes (ceiling) **[P]** pending

Safety × Tree (B2×C1)
predicted: complements **[P]** pending

Clock × Beliefs (A×C3)
predicted: inert (OSP immunizes) **[P]** pending

solid border = run · dashed = registered, runs pending · $\Uparrow$/$\uparrow$ predicted improvement, $\downarrow$ predicted backfire, $\approx$ 0 predicted inert · **[H]** directly comparable human experiment exists · **[N]** related but unmatched human evidence · **[P]** LLM-only cell = falsifiable prediction for human subjects

Figure 1: The design map, stated before any result. Every cell holds the outcome rule (highest bidder wins at the second-highest bid) fixed; the three dimensions vary which interface layer the participant meets. Predictions are derived ex ante from the theory of Section 2.1; roughly half the map is predicted inert or harmful, which the experiments test. Preference framings (Family D) are payoff-irrelevant bounding cells, excluded from the map (Appendix H). The fully specified reference version, with per-cell citations, is Table 4 in Appendix A; verbatim prompt texts are in Appendix I.

[TODO: Combination cells A×B2, B2×C1, B1+invariance, A×C3 registered but not yet run (E-v3-2); each carries an ex-ante predicted sign in Table 4.]

2.5 Hypotheses, and How the Instrument Tests Them

The theory yields three ordered hypotheses. We state them before presenting any results, and—because this paper's method is simulation with LLM agents—we state for each one exactly what it predicts in the data the instrument produces: a *bid* for every agent-instance (scored as a deviation from the dominant strategy; Section 3.2) and a *stated plan* accompanying every bid (scored with the coding framework of Section 5). Each hypothesis is a contrast between cells of Table 4, estimated within each model family and accepted only if its direction replicates across all four families (the robustness rule of Section 3.2).

H1 (Implementation dominates). The clock cells show the largest and most uniform reduction in bid deviations of any cell in the map, in every model family. *In bids*: mean deviations compress toward zero and the left tail of severe underbids disappears. *In plans*: sealed-bid plans should exhibit contingent-reasoning failures (opponent speculation, margin heuristics), while clock plans reduce to the one-step comparison "is the price still below my value?"—the burden the OSP form removes should vanish from the language as well as from the bids. H1 is the map's calibration hypothesis: it has an unambiguous human anchor, so an instrument that fails H1 fails validation.

H2 (Descriptions substitute when—and only when—they expose the invariant). Within Dimension 2, text that states the safety invariant (B2; and the invariance-carrying menu variant, when run) approaches the clock's effect *without* changing the game, while text that restates the outcome function without the invariant (B1) is null or harmful. The mechanical restatement is the built-in placebo: it adds words, length, and salience while adding no incentive content, so H2 fails if "any extra explanation" helps. *In bids*: B2 deviations shrink in every family; B1 deviations do not move consistently. *In plans*: H2 is agnostic—the description may work by changing what agents articulate or by changing behavior directly; Section 5 measures which, and finds the latter.

H3 (Reasoning support is axis-specific; belief scaffolds are diagnostic). A scaffold helps only if it relaxes the axis on which the mechanism's cognitive demand actually binds. C1 (contingent reasoning) targets the sealed bid's binding axis and should help. C2 has no axis to bind on and is N/A by design. C3 (beliefs) targets an axis that strategy-proofness makes irrelevant, so for a rational participant it must be inert—and any measured effect, in either direction, is evidence that the agent is playing a beliefs game rather than the dominant strategy. For such an agent, we predict C3 *backfires*, most severely for the weakest reasoners. *In bids*: C1 improves all families; C3 degrades them (or is null for families that already ignore opponents). *In plans*: C3's manipulation check is that opponent-directed language rises; its diagnostic content is that bids move at all. Backfiring is not an embarrassment for the taxonomy but its predicted signature: the designer's lever space contains anti-levers, and identifying them is part of what a screening instrument is for.

A note on taxonomy provenance. The operational prompt labels in our experiment repository predate this taxonomy, and two assignments were corrected when it was built: the *Payoff Tree* template was originally filed under forward planning (repo label `axis2_forward_tree`) and is classified here as C1, because the tree enumerates payoff *contingencies*, not future own moves; and one legacy template name (`intervention_proxy_breitmoser`) historically referred both to the clock-framing *description* and to the true iterative clock—the two are strictly separated here as B3 and Dimension 1 respectively, with distinct prompt identifiers and empirically distinct cells (the framing cells record one sealed bid per bidder; the clock cells record per-tick stay/exit decisions). The full old-to-new mapping, with run status for every template in the repository, is Table 17 in Appendix I. The taxonomy and its predictions were fixed from published theory, not from prompt search; the predicted-inert and predicted-backfire cells are part of what the experiments test.

With the map and its predictions on record, the next section describes the instrument itself—environments, agents, metrics, and the human benchmark layer—and asks whether the panel fills the map's anchored cells correctly.

3 Ordinal Validation Against the Human Experimental Record

This section describes the instrument and validates it. The logic of the validation is fixed by the paper's positioning: we do not ask whether LLM agents bid like people in levels (they do not, and Section 3.5 documents exactly how), but whether a panel of LLM bidders recovers the *ordinal* regularities of the human experimental record—which mechanism is behaviorally harder, which design change helps, which leaves behavior unchanged. Each validation exercise therefore has five declared parts: the *setting* (the auction rules and decisions faced by human subjects and LLM bidders), the *human benchmark* (the published comparison), the *LLM procedure* (how model bids were obtained), the *metric* (directional and rank agreement on a common deviation scale), and the *result*, reported family by family. Figure 2 collects the results in the format the rest of the paper relies on.

3.1 The Testbed: Environments, Agents, and Protocol

Market instances. The unit of observation is a fully specified, seed-reproducible *market instance*: a draw of bidder values, a mechanism, a description, and a set of LLM agents whose parsed actions mechanically determine the allocation and payments. For each experimental cell we repeat the simulation K times with independent seeds, producing realized values, bids, allocations, prices, profits, and—for every bid—a free-text stated plan (Section 5).

Value environments. Two information structures are used, both with $n = 3$ bidders and all parameters fixed in configuration files. *Independent private values* (IPV): $v_i \sim \text{Unif}\{0, 1, \ldots, 49\}$, so the expected truthful bid is $\mathbb{E}[v] = 24.5$; this is the baseline environment for the design map. *Affiliated private values* (APV): $v_i = c + \epsilon_i$ with $c \sim \text{Unif}[0, 29]$ and $\epsilon_i \sim \text{Unif}[0, 20]$; each bidder observes her own realized v_i, so dominant strategies are unchanged, and this environment exists for comparability with the human clock experiments run under affiliation (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022).

Mechanisms. Sealed-bid auctions restrict bids to a \$0.1 grid; the winner pays the second-highest bid (SPSB, the strategy-proof anchor) or, in the validation battery only, the first-price rule in which the winner pays her own bid (FPSB). FPSB is not strategy-proof—its risk-neutral equilibrium calls for shading, $b^\star(v) = \frac{n-1}{n}v$—and it enters the paper solely as the difficulty contrast of Figure 2, not as a cell of the design map.[1] Ascending clock auctions start at price 0 and rise by \$0.5 per tick; at each price every active bidder chooses *stay* or *exit* (irreversible), the last bidder standing wins at the runner-up's exit price, and we distinguish the open clock (AC: dropouts announced) from the closed clock (AC-B: exits private). The clock lever runs in two purpose-labeled cells: an APV cell for human-benchmark comparability, and an IPV cell for identification—under IPV a bidder learns nothing payoff-relevant from others' exits, so any clock improvement must be cognitive rather than informational.[2]

[1] The repository additionally contains third-price, common-value, and eBay-style proxy-bidding environments; they are outside this paper's scope, which confines validation to the comparisons the design map needs. The registered +X battery (Figure 1) will supply the unfamiliar-format robustness check those environments previously motivated. [TODO: Run the 2P+X / AC+X pair (E-v3-1).]

[2] A provenance audit found that the legacy clock cells pair an affiliated-values *description* with independent draws (the configured common component is zero), so the informational channel was already absent; a dedicated clean IPV cell (IPV description and draws, \$0.5 tick, $K = 50$) reproduces the result exactly for the two families re-runnable at the time of writing—SMAD 1.5% (GPT-4o) and 1.8% (Gemma). [TODO: IPV-clock cells for Claude 3.5 Haiku and Gemini 2.0 Flash pending native API keys (E2).]

The panel. The prespecified panel comprises four widely deployed, non-reasoning model families from three commercial providers and one open-weight family: GPT-4o (2024-08-06), the anchor present in every cell (OpenAI, 2024); Claude 3.5 Haiku (2024-10-22) (Anthropic, 2024); Gemini 2.0 Flash (DeepMind, 2024); and Gemma 27B (Riviere et al., 2024). The panel deliberately excludes frontier reasoning models from the main grid: the screening question requires agents capable enough to parse a mechanism yet still subject to the reasoning failures that simple design aims to remedy, and deployed high-volume agent fleets occupy exactly that regime for cost and latency reasons. Both boundaries of the regime are measured rather than assumed: a frontier tier (gpt-5-mini, gpt-5, Claude Sonnet 5, Gemini 2.5 Flash) and a capability floor (Llama-3-8B) are reported as robustness in Appendix E. All non-reasoning models run at temperature $T = 0.5$, the most common convention in LLM-based simulation (Horton, 2023; Horton et al., 2024); an ablation over $T \in \{0.1, 1.0\}$ shows modest, non-monotonic effects, so $T = 0.5$ is a comparability convention, not a tuning choice (Appendix E).

Protocol. Each repetition is an independent one-shot market: values are freshly drawn, agents receive no feedback across instances, and each agent is queried in a separate API call. The prompt contains (i) the mechanism's rule explanation (plus the treatment text, when a design lever is active), (ii) the agent's private value, and (iii) a required structured response format—`<PLAN>` for the stated plan, `<ACTION>` for the bid or the per-tick stay/exit decision—parsed by a deterministic regex extractor; outcomes are then computed mechanically from the parsed actions. The one-shot design is licensed by a repeated-play ablation with full history feedback, which finds no learning over 15 rounds (FPSB SMAD 0.2523 in the first five rounds versus 0.2524 in the last five, $p = 0.997$; SPSB likewise null; Appendix D), so the treatment effects below cannot reflect within-session adaptation. Step-by-step procedures and verbatim prompts appear in Appendices C and I; Table 5 in Appendix A pins the full per-cell calibration.

3.2 Metrics and the Robustness Rule

A unified deviation metric. For an auction format m with benchmark bid function $b^\star_m(\cdot)$—truthful bidding $b^\star(v) = v$ in the strategy-proof formats, the risk-neutral equilibrium $b^\star_{\text{FPSB}}(v) = \frac{n-1}{n}v$ in the first-price validation format—define the absolute bid error $d_m(v) \equiv |b - b^\star_m(v)|$ over *all* submitted bids and normalize by the expected benchmark bid $\mu^\star_m \equiv \mathbb{E}_m[b^\star_m(v)]$. The *scaled mean absolute deviation* is

$$\Delta_m \;\equiv\; 100 \cdot \frac{\mathbb{E}_m[\,|b - b^\star_m(v)|\,]}{\mathbb{E}_m[b^\star_m(v)]},$$

read as a percentage of the expected benchmark bid ($\Delta_m = 10$: bids err by 10% of the expected benchmark on average). SMAD is computable identically for reconstructed humans and for LLM agents, which is what places the two populations on one scale. Alongside SMAD we report the *signed* mean deviation (bid $-$ benchmark, in dollars) wherever the direction of error matters; the two statistics can disagree about orderings, and we treat such disagreements as findings. We restrict attention to formats whose theoretical prediction is deterministic given the bidder's information, so the benchmark is single-valued.

The preservation index. For a design lever L and model family f, let $D(L, f)$ be the mean deviation from dominant-strategy play under L and $D(\text{base}, f)$ the same object under the baseline sealed bid. The *preservation index*

$$\rho(L, f) \;\equiv\; 1 - \frac{D(L, f)}{D(\text{base}, f)}$$

equals 1 if the lever restores perfect dominant-strategy play, 0 if it does nothing, and is negative if it *backfires*. Because ρ is computed within family, every cell of the design map is scored against the behavior it inherits, making the map invariant to cross-family differences in baseline levels—and, crucially for Section 3.5, invariant to the sign of a population's characteristic error, so the same index is meaningful for overbidding humans and underbidding models.

Baseline declaration. Two baseline conventions appear in the auction analysis, and we fix their use once. Cross-format comparisons (sealed bid versus clock) use the dedicated SPSB cells, with per-family mean deviations of -0.49 (Claude 3.5 Haiku), -1.63 (Gemini 2.0 Flash), -3.28 (GPT-4o), and -5.27 (Gemma 27B)—cross-family mean $-\$2.67$. Description and scaffold contrasts use the pooled axis baseline cells run in the same grid, with per-family means of $+0.54$, -1.25, -2.94, and -6.53 respectively—cross-family mean $-\$2.55$. The pooled convention reflects a provenance correction made in this draft's predecessor: the legacy "axis-2 baseline" cell was itself a treated clock-framing description and is excluded from every baseline pool and analyzed as a treatment (Appendix G); correcting it *raised* the cross-family concordance of the lever ordering (Kendall's W from 0.43 to 0.70), evidence that the contamination had been injecting noise. Both conventions are recorded cell by cell in the replication file (`results/merged_ranking/auction_cells.csv`), and no comparison in this paper mixes them.

The cross-model robustness rule. The paper's inclusion criterion is pre-committed and binds every headline claim:

> *A result enters the main text only if its direction is consistent across the four prespecified model families. Results that appear in one family, flip direction across families, or depend on a single idiosyncratic wording are reported in the appendices as model heterogeneity or failure cases, not as design conclusions.*

Accordingly, every main-text result below shows its per-family direction, never only a pooled average, and two cells of the design map (the menu restatement's family-level signs and the clock-framing description) are flagged in the main text and analyzed in Appendix G.

Statistical standard. Every reported cell carries a bootstrap standard error and 95% confidence interval (resampling market instances, 2,000 replications, seed 1299); every treatment effect is evaluated against its declared baseline with both a Mann–Whitney test and a Welch t-test plus Cohen's d (the two tests can disagree when a lever moves dispersion without the mean, and we report both); pooled preservation indices carry double-bootstrap intervals; and observed-zero error rates are reported with rule-of-three upper bounds rather than as perfection.

3.3 The Human Benchmark Layer

Moment-matched reconstruction. Because micro-level bid data are rarely available from the classic experiments, we reconstruct human bid distributions by calibrating a mixture model to the aggregate statistics each source reports. Bids are modeled as $b = b^\star_m(v) \cdot r$, where the bid-to-benchmark ratio r follows a three-component mixture that maps directly onto how experimental papers categorize behavior: with probability p_{eq}, $r \sim \mathcal{N}(1, \sigma^2)$ (equilibrium bidders); with probability p_{over}, $r \sim 1 + \mathrm{Exp}(\lambda)$ (overbidders); with probability p_{under}, $r \sim \mathrm{Unif}(1-\delta, 1)$ (underbidders). Mixture weights are set directly from reported behavioral rates (e.g., Li, 2017: $\sim$50% dominant-strategy play, $\sim$40% overbidding); the overbid scale λ is calibrated to a secondary moment (a mean bid-to-value ratio, a bid-regression coefficient, or a reported mean absolute deviation); and σ is derived from regression fits where reported. Sources: Li (2017) and Breitmoser and Schweighofer-Kodritsch (2022) for SPSB and the clocks under affiliation; Kagel and Levin (1993) for first- and second-price sealed bidding under IPV; Gonczarowski et al. (2023) for the menu-description contrast. A quantal-response alternative is rejected on first principles: QRE predicts symmetric deviations around equilibrium, contradicting the well-documented asymmetric overbidding in second-price auctions. Full per-source calibrations are in Appendix B.

Epistemic status, enforced. The reconstructions are model-based simulations of published moments, not samples of human behavior, and every use of them is held to a matching standard: *comparisons to reconstructed humans are made at the level of rankings, directions, and comparative statics only; levels and significance tests are reserved for theory benchmarks.* We run no hypothesis tests against reconstructed distributions—such tests would recycle our own calibration choices—and where reconstructed anchors appear in figures, their uncertainty is shown as Monte-Carlo bands obtained by re-drawing the mixture parameters within the ranges the source statistics admit (Appendix B). The design map of Section 4 never conditions on the reconstructions; they enter only here (validation) and in the human-anchor labels of the map.

3.4 Results: Ordinal Agreement, Family by Family

Figure 2 is the validation exhibit: one row per comparison with published human evidence, one cell per model family, color coding the direction of each family's response against the human direction (the fully documented table, with sources and per-cell statistics, is Table 6 in Appendix A). Two rows are design levers (rows 3–4); they are previewed here as validation rows and analyzed as map cells in Section 4.

Row 1: the difficulty ordering. *Setting*: sealed-bid IPV auctions differing only in the payment rule—first-price, where equilibrium requires strategic shading, versus second-price, where truth-telling is dominant. *Human benchmark*: reconstructed SMAD of 24.76% (FPSB) versus 5.65% (SPSB), a $4.4\times$ difficulty gap (Kagel and Levin, 1993). *LLM procedure*: the same two formats, identical prompts up to the payment-rule paragraph, $K \geq 50$ per cell. *Result*: every family finds first-price harder. GPT-4o mirrors the human gap most closely (27.55% versus 8.69%); Gemma preserves the direction but nearly flattens the ratio (24.80% versus 19.88%), deviating almost as much where truth-telling is dominant as where shading is required. The ordering transfers in all four families; its *magnitude* is family-specific, a fact recorded in the validity map below and detailed in Appendix E. A rank-concordance statistic extends the row beyond the FPSB/SPSB pair: over the five in-scope formats (FPSB-IPV, SPSB-IPV, SP-APV, AC, AC-B), Kendall's τ_b between the human and GPT-4o SMAD rankings is 0.60. The estimate is stable under bid-level resampling of the LLM cells ($B = 2000$; 98.7% of draws leave τ_b at 0.60) and stays positive in 100% of joint draws that additionally propagate the human reconstruction re-draws of Section 3.3 (joint 95% CI $[0.40, 1.00]$), with FPSB ranked hardest on both sides in every draw. Both discordant pairs in the point estimate involve the closed clock, where GPT-4o's two clock cells sit so close to perfect play (SMAD 0.5–0.8%) that their ordering is uninformative (`analysis/difficulty_concordance.py`).

Row 2: the implementation effect. *Setting*: the same second-price outcome rule as a one-shot sealed bid versus an ascending clock. *Human benchmark*: clock formats roughly halve human deviations (SP-APV 9.31% versus AC 3.54% and AC-B 5.83%) (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022). *LLM procedure*: per-tick stay/exit elicitation (Appendix C), with the APV cell carrying human comparability and the IPV cell carrying identification (Section 3.1). *Result*: the clock helps every family, with SMAD falling

Comparison	Human	Claude	Gemini	GPT-4o	Gemma	Agreement
Payment-rule difficulty: FPSB vs. SPSB (SMAD ratio)	4.4× harder	2.5×	9.0×	3.2×	1.25× (atten.)	**Yes (4/4)**
Implementation: sealed → clock (SMAD, %)	9.3 → 3.5	9.5 → 1.4	11.7 → 5.2	11.4 → 1.9	20.7 → 1.9	**Yes (4/4)**
Description: menu restatement (Δ mean dev., $)	≈ no change	away (0.8)	null	away (1.2)	toward (2.0)	**Yes on avg. (sign-mixed)**
Description: clock-framing (SMAD, %)	toward truthful	8.3 → 7.2	5.1 → 9.5	12.1 → 6.3	26.8 → 13.8	**Partial (3/4)**

Figure 2: Ordinal validation at a glance: does the panel recover the direction of every documented human comparison? Rows are the four comparisons with published human evidence; the Human column is computed from the moment-matched reconstructions of Section 3.3 anchored to Kagel and Levin (1993) (row 1), Li (2017); Breitmoser and Schweighofer-Kodritsch (2022) (rows 2 and 4), and Gonczarowski et al. (2023) (row 3). Cell color codes the direction of the family's change: green = toward truthful play (or human ordering preserved), red = away, gray = null; every family-level shift shown is significant at $p < 0.05$ except the cells marked null. Row 2 uses the pooled-axis sealed baselines; row 3 reports changes in mean deviation against the corrected pooled axis baselines (Section 3.2), whose average is null (sign test $p = 0.73$), matching the human null. Rows 3–4 are design levers previewed as anchored validation rows; the clock-framing row fails the cross-family robustness rule and is analyzed in Appendix G. The fully documented version, with sources, conventions, and per-cell statistics, is Table 6 in Appendix A.

to 1.4–5.2% from sealed baselines of 9.5–20.7% ($p < 0.001$ in each family), and GPT-4o's APV clock cells (0.81% AC, 0.47% AC-B) sitting below even the human clock benchmarks. This is the map's calibration row (hypothesis H1), and it passes in the strongest available form. One finer human regularity sits beyond the instrument's resolution, and we record it rather than claim it: humans do slightly better under the open clock than the closed one (3.54% versus 5.83%), while GPT-4o's two clock cells are both so close to perfect (0.81%, 0.47%) that their ordering is uninformative—and only one family has both cells.

Rows 3 and 4: description treatments. *Setting*: the sealed-bid SPSB auction with only the rule-explanation text varied. *Human benchmarks*: the menu restatement leaves human bidding essentially unchanged (Gonczarowski et al., 2023); the clock-framing description increases human truthful bidding (Breitmoser and Schweighofer-Kodritsch, 2022). *Result*: the menu restatement is null on average in the panel as it is in humans—but the null is an average over genuinely sign-mixed family-level effects (one family improves, two degrade, one is flat), a heterogeneity the single-population human experiments could not have revealed. The clock-framing description agrees with the human direction in three families and reverses in Gemini—the same family that a second clock-framing paraphrase also harms (Appendix G)—so its anchor holds at the majority-of-families level, and under our robustness rule the lever itself is not a main-text design recommendation. Both rows illustrate what the rule is for: the panel's value is not that every cell is clean, but that heterogeneity is detected and flagged rather than averaged away.

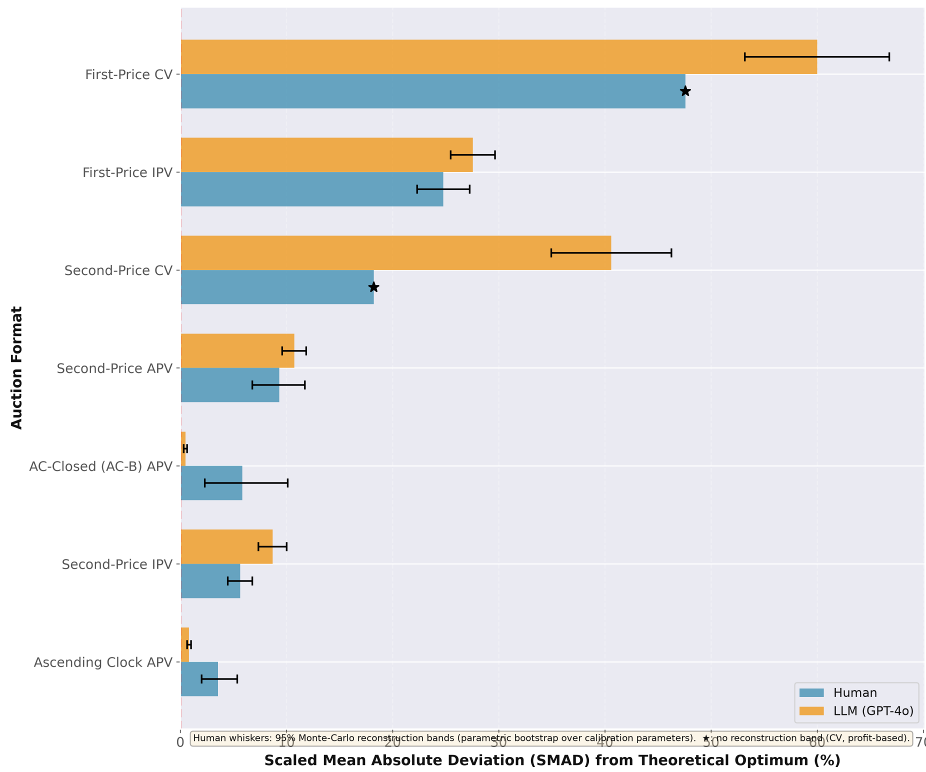

Figure 3: **Human vs. LLM deviation by auction format, with reconstruction-uncertainty bands on the human anchors.** Blue bars: SMAD of reconstructed human bidding from the theoretical benchmark (Section 3.3). Yellow bars: GPT-4o agents against the same benchmarks ($N = 3$ bidders). Black whiskers: 95% Monte-Carlo reconstruction-uncertainty bands obtained by re-drawing the mixture parameters within the ranges the source statistics admit ($B = 2000$; Appendix B): $[22.30, 27.25]$ around the FPSB-IPV point 24.76; $[4.46, 6.79]$ around SPSB-IPV 5.65; $[6.78, 11.74]$ around SP-APV 9.31; $[2.00, 5.36]$ around AC 3.54; $[2.30, 10.11]$ around AC-B 5.83. The FPSB $\gg$ SPSB ordering and the clock improvement over sealed SP-APV hold in 100% of re-draws (AC-B: 92.6%). These are calibration bands, not sampling bands on human behavior; comparisons remain ordinal (Section 3.3). [TODO: Regenerate this asset restricted to the five formats above: the current PNG carries two common-value panels (starred) from a battery outside this paper's scope (`scripts/plots/figure1_bands.py`).]

3.5 What Transfers, What Does Not, and Why the Divergence Helps

Ordinal validation would be suspect if it were secretly level-matching in disguise; it is not, and the sharpest evidence is a divergence we declare rather than hide. Table 1 is the section's accounting: three phenomena transfer from human populations to the panel, and two do not.

The direction of errors reverses. Human bidders in strategy-proof sealed-bid auctions err overwhelmingly *upward*: 67.2% of reconstructed SPSB-IPV bids lie strictly above value—92.2% among bids that deviate at all—consistent with the classic right-skewed distributions of Kagel and Levin (1993).[3] GPT-4o errs *downward*: 74.1% of its baseline SPSB bids fall strictly below value (81.6% among deviators), with a large point mass at exactly truthful play (29.2% within $\pm 2\%$ of value) that human data lack; the same reversal appears in the APV cell (64.7% human overbidding versus 92.4% GPT-4o underbidding under the band convention), and in first price both populations overbid relative to equilibrium but the model does so far more often (85.9% versus 56.5%). Levels, dispersion, and distributional shape likewise fail to transfer. These are the validity map's red cells, and they are why every cross-population claim in this paper is ordinal and every design-map score is a within-family ratio.

Opposite failures, common levers. Read against the design results of Section 4, the reversal becomes the validation's sharpest identification argument rather than its embarrassment. Two populations fail in *opposite* directions, yet—as Figure 2 already shows for the anchored rows—they are helped by the *same* levers, in the same order: the clock repairs human overbidding and model underbidding alike, and the menu restatement repairs neither. If deviations were driven by population-specific psychology (a joy of winning pushing humans up; a trained conservatism pulling models down), there would be no reason for the correction structure to be shared. The parsimonious reading is that the binding constraint lives in the strategic problem itself—the difficulty of verifying, from a direct mechanism's description, that truthful play is safe (Li, 2017; Oprea, 2024)—while what fills the gap is population-specific. This is precisely the property a screening instrument needs: the instrument must share the *problem* with human participants, not their idiosyncratic response to it. The reversal also rules out the deflationary explanation of the agreement: a panel that merely reproduced the human experimental literature memorized in training would *overbid* in second-price auctions, and ours does not. The agreement on which levers work is re-derived by a different reasoner failing in a different way—which is why it is informative.

Phenomenon	Key statistic	Verdict
Cross-mechanism difficulty ranking	4/4 families; $\tau_b = 0.60$, joint 95% CI $[0.40, 1.00]$	Transfers (✓)
Clock/OSP implementation effect	4/4 families, $p < 0.001$ each	Transfers (✓)
Menu-description null	null on average (sign test $p = 0.73$); family signs mixed	Transfers on avg. (✓)
Deviation *direction* (SP sealed bid)	humans 67.2% overbid; GPT-4o 74.1% underbid	Diverges ($\times$)
Deviation *levels* and shape	LLM point mass at truthful play; levels uncalibrated	Diverges ($\times$)

Table 1: Validity-map scorecard: which behavioral phenomena transfer from human populations to the LLM panel. The two red rows discipline the whole paper: because directions and levels do not transfer, all screening claims are ordinal and all design-map scores are within-family preservation indices. Full documentation of every row—human sources, per-family findings, and the concordance-bootstrap method—is Table 7 in Appendix A.

Verdict, and the boundary. The panel recovers the direction of every documented human comparison in our scope, in all four families for the mechanism-level rows and at the pooled level for the description rows, with family-level heterogeneity flagged where it exists. What the panel does not do—match levels, directions of error, or distributional shape—is documented in the scorecard's red rows, and Appendix F shows the gap is not closable by tuning: across a 95-cell grid of temperature, persona, framing, and explanation knobs, no

[3] Direction shares use the repository-canonical convention: the share of all bids strictly above value, with the share among deviating bids in parentheses. Figure 4 instead removes a $\pm 2\%$ band around the benchmark before classifying; the qualitative reversal is invariant to the convention, but individual shares should not be compared across conventions.

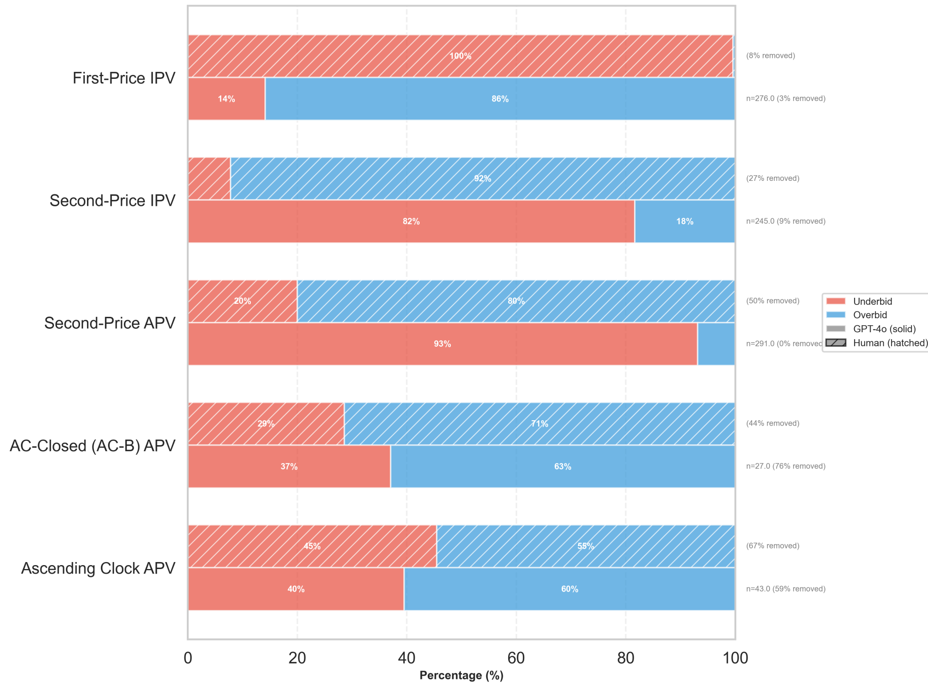

Figure 4: **Directional decomposition of deviations, humans vs. GPT-4o.** Share of underbids vs. overbids after removing bids within a $\pm 2\%$ band of the theoretical benchmark. Humans overbid where GPT-4o underbids in the strategy-proof sealed-bid formats—the declared divergence of Table 1. Only the first-price, second-price, and affiliated sealed-bid panels are within this paper's scope. [TODO: Regenerate the asset without the out-of-scope format panels.]

single setting achieves cardinal agreement across mechanisms, and the only knob that restores the human direction of error in second-price auctions induces dominated overbidding in first price, where humans almost never overbid. The instrument is ordinal; used as such, it is validated. Section 4 now puts it to work.

4 Using LLMs to Populate the Mechanism-Design Map

This section delivers the paper's second contribution: the design map of Figure 1, populated cell by cell with the validated panel. We proceed dimension by dimension—implementation (Section 4.1), description (Section 4.2), reasoning support (Section 4.3)—then assemble the map (Section 4.4, Table 2), classify the registered interaction cells (Section 4.5), and close with the robustness exercises that bound the map's scope (Section 4.6).

Throughout, $D(L, f)$ denotes model family f's mean bid deviation from value (dollars) under design cell L, with SMAD reported in parallel; the comparative statistic is the preservation index ρ of Section 3.2, computed within family on the absolute-deviation scale. The baseline conventions of Section 3.2 apply unchanged: format-level comparisons use the dedicated SPSB cells (per-family means -0.49, -1.63, -3.28, -5.27 for Claude 3.5 Haiku, Gemini 2.0 Flash, GPT-4o, and Gemma 27B; cross-family $-\$2.67$), and description/scaffold contrasts use the corrected pooled axis baselines ($+0.54$, -1.25, -2.94, -6.53; cross-family $-\$2.55$). Every cell-level statistic is computed from the merged run-level dataset in the replication package (`results/merged_ranking/auction_cells.csv`; bootstrap seed 1299).

4.1 Implementation: The Ascending Clock

The implementation cell asks how much of the baseline deviation the extensive form itself removes. Theory predicts this is the strongest lever in the map (H1): the OSP clock eliminates the contingent-reasoning burden entirely, replacing case-by-case verification over unobserved opponent bids with a sequence of on-path comparisons (Li, 2017), each of which is one-step simple (Pycia and Troyan, 2023).

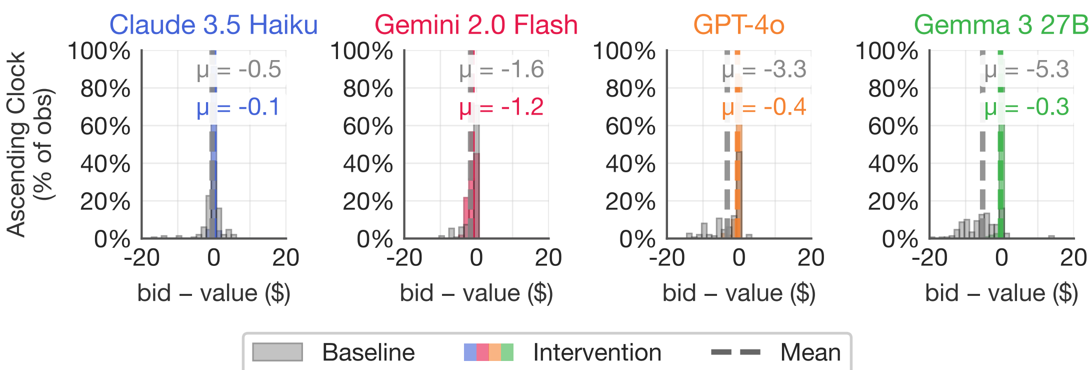

Figure 5: Implementation cell of the design map, per model family. Distribution of bid deviations from value (bid $-$ value) in the second-price sealed bid (gray, dedicated SPSB cells) and the ascending clock (colored); dashed lines mark means. Every family deviates in the sealed-bid format; the clock corrects every family toward truthful play and removes the left tail of severe underbids. The matching-domain counterpart of this exhibit is in Appendix K (Figure 31).

Result: the strongest cell, in every family. In the SPSB baseline all four families underbid, with mean deviations of -0.49 (Claude), -1.63 (Gemini), -3.28 (GPT-4o), and -5.27 (Gemma), and with long left tails—individual bids \$10–20 below value. The closed ascending clock corrects all four: mean deviations compress to -0.05, -1.24, -0.45, and -0.33 respectively (cross-family $\approx -\$0.5$), SMAD falls to 1.4%, 5.2%, 1.9%, and 1.9% from pooled-axis baselines of 9.5%, 11.7%, 11.4%, and 20.7%, and the shift is significant in every family (Welch $p < 0.001$ each; Cohen's d up to 1.00). The improvement is largest exactly where the baseline is worst—Gemma's mean deviation falls from -5.27 to -0.33, a near-complete correction—and the clock *eliminates the catastrophic tail*: the worst-case failures disappear, a variance effect that may matter as much as the mean effect for a deployed market. Direction is consistent in $4/4$ families; the cell passes the robustness rule in the strongest available form and agrees with its human anchor (Figure 2).

Identification. Because the four-family cell runs under independent private values, the improvement cannot be informational: under IPV, nothing payoff-relevant is revealed by others' exits, so the channel is purely cognitive. Two provenance notes complete the accounting. The legacy clock cells pair an affiliated-values description with IPV draws (the configured common component is zero), so the informational channel was absent in the data even before the audit; and a dedicated clean IPV cell (IPV description *and* draws, \$0.5 tick, $K = 50$) reproduces the result exactly for the two families re-runnable at the time of writing—SMAD 1.5% (GPT-4o) and 1.8% (Gemma) against sealed baselines of 12–27%. [TODO: Clean IPV clock for Claude 3.5 Haiku and Gemini 2.0 Flash pending native keys (E2); given the exact replication in two families and the direction of every legacy cell, we do not expect the remaining cells to change the cell's classification.]

The bridge: unbundling the clock. As a design lesson the implementation result is coarse, because OSP changes several things at once: the game tree, the action space (a sequence of binary stay/exit choices), and the salience of the safety property at every decision point. A designer needs to know which margin does the work, because the margins have very different costs—re-engineering a deployed mechanism's extensive form is expensive, while changing its description or its decision support is nearly free. The next two subsections unbundle the clock with presentation-only cells that hold the sealed-bid form fixed.

4.2 Description: Restating the Rules versus Exposing the Safety Invariant

Dimension 2 holds the mechanism fixed and varies only the words. The three sub-levers behave very differently, and the differences are the map's most practically valuable content.

Payoff Safety (B2): the invariant, stated honestly, moves every family. The safety-exposing description adds one honest statement of the invariant behind Vickrey (1961)'s argument: *"Think of your bid as setting your maximum price. Your bid determines IF you win, not WHAT you pay. If you win, you pay what others bid, not what you bid."* It scaffolds no reasoning and reveals no strategy; it states a true property of the payment rule. The effect is a same-signed improvement in all four families: mean deviations move from -0.49 to $+0.30$ (Claude), -1.63 to -0.38 (Gemini), -3.28 to -1.88 (GPT-4o), and -5.27 to -4.17 (Gemma), cutting the cross-family mean from $-\$2.67$ to $-\$1.53$ (Figure 6).[4] Against the axis baselines the improvement is significant for Gemini ($p < 10^{-30}$, $d = 1.85$) and GPT-4o (Mann–Whitney $p < 10^{-5}$); for Claude the mean effect is null while dispersion halves (SMAD 12.0% to 4.8%)—a case where the Welch/Mann–Whitney pair earns its keep—and Gemma improves against both declared baselines. Direction is consistent in $4/4$ families: the cell passes the robustness rule. For comparison, the clock achieves $\approx -\$0.5$: one sentence recovers roughly half of what the full OSP redesign delivers. No human auction experiment has tested a bare invariant statement—the nearest human evidence is from matching-mechanism descriptions, where property-exposing text helps and mechanics-only text backfires (Gonczarowski et al., 2024; Katuščák and Kittsteiner, 2024; Guillén and Hakimov, 2018)—so this cell carries the map's *Nearby evidence* label and doubles as its sharpest new prediction for human subjects (Section 6).

Menu restatement (B1): the built-in placebo is null on average—and honestly sign-mixed. The menu restatement re-describes the outcome function in the style of Gonczarowski et al. (2023): the bid is a "maximum acceptable price," the highest rival bid is the market price, winning means the market price was below your maximum. It adds words and salience but no incentive content beyond the standard rules, making it the map's placebo for the claim that *content*, not *restatement*, is what helps. On average it does nothing—a two-sided sign test cannot reject a zero median effect ($p = 0.73$), matching the human behavioral null

[4] Claude's residual error flips sign under this description (from -0.49 to $+0.30$, mild overbidding); the absolute deviation still shrinks, and Claude ends closest to truthful of the four. The raw-log sign is $+0.30$; an earlier manuscript reported -0.30 (magnitude identical).

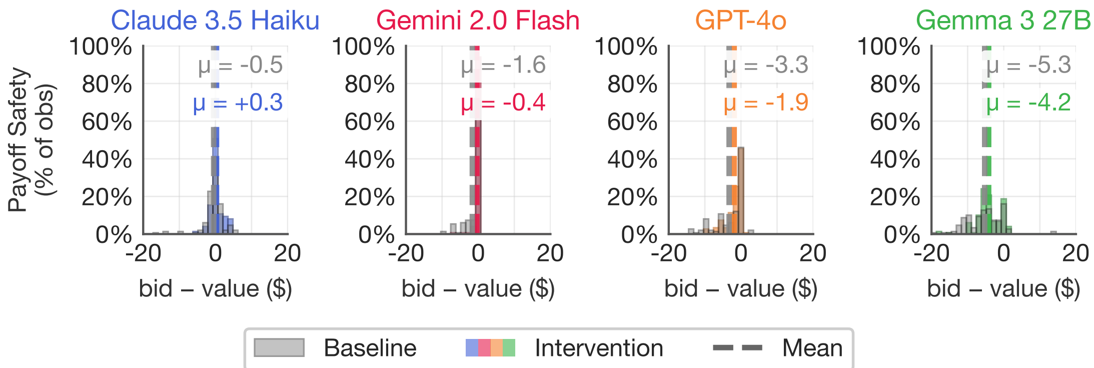

Figure 6: Safety-exposing description (B2): *Payoff Safety* in second-price auctions, per model family. Dedicated SPSB baseline in gray, treated cell in color; dashed lines mark means. The absolute deviation shrinks in every family—roughly half the gap to the full clock implementation of Figure 5—from a one-sentence change in the rule explanation.

(Gonczarowski et al., 2023)—but the average conceals real family-level heterogeneity (Figure 7, top row): against the corrected pooled baselines, Gemma moves *toward* truthful play (+\$2.03, $p < 0.001$), GPT-4o moves *away* ($-\$1.16$, Welch $p = 0.013$; Mann–Whitney $p = 0.54$), Claude moves marginally away (+\$0.82 further above value; Welch $p = 0.064$, Mann–Whitney $p < 0.001$), and Gemini is null (+\$0.12, $p = 0.56$).[5] By the robustness rule the menu is classified *inactive*: no consistent direction exists to report. Its placebo value survives in the aggregate—extra prose about the mechanism does not systematically move behavior, so the large B2 effects cannot be generic prompt sensitivity—but it is a noisier placebo than the single-population human experiments suggest, and the family-level signs are a finding in themselves: description responses are model-specific where description *content* is not.

Clock-framing (B3): flagged—helps three families, harms the fourth. The clock-framing description keeps the one-shot sealed bid but describes the number as the threshold at which an automatic proxy would exit a rising clock (Breitmoser and Schweighofer-Kodritsch, 2022). It is a description, not a mechanism change (the elicited action is still a single sealed bid), and its cells are empirically distinct from the true clock's per-tick records. It helps GPT-4o (SMAD $12.1\% \to 6.3\%$), Gemma ($26.8\% \to 13.8\%$), and Claude ($8.3\% \to 7.2\%$), each at $p < 0.001$—agreeing with the human evidence—but pushes Gemini the wrong way ($5.1\% \to 9.5\%$, $p < 0.001$). Under the robustness rule, a lever whose direction flips across families does not enter the main-text design conclusions: we record it in the map as *flagged*, with pooled $\rho = +0.30\ [-0.77, +0.52]$, the interval's width driven by the Gemini reversal. The full analysis—including a second, independently discovered clock-framing paraphrase that helps only Gemma and also harms Gemini, direct evidence that description effects are wording-sensitive in a way implementation effects are not—is in Appendix G. The flagged cell still teaches the map's method: a single-model study (or a single-wording

[5]These are the canonical corrected-baseline contrasts. Against the same-batch baseline cells run alongside the menu grid, the family-level attributions shift (Gemma null, Gemini improving), while the pooled null is unchanged—family-level menu effects are baseline-convention-sensitive as well as sign-mixed, which reinforces reading this cell only as "no consistent effect." Appendix G reports both conventions cell by cell.

study) would have reported clock-framing as a clean win; the panel reveals it as a gamble whose sign depends on the reader.

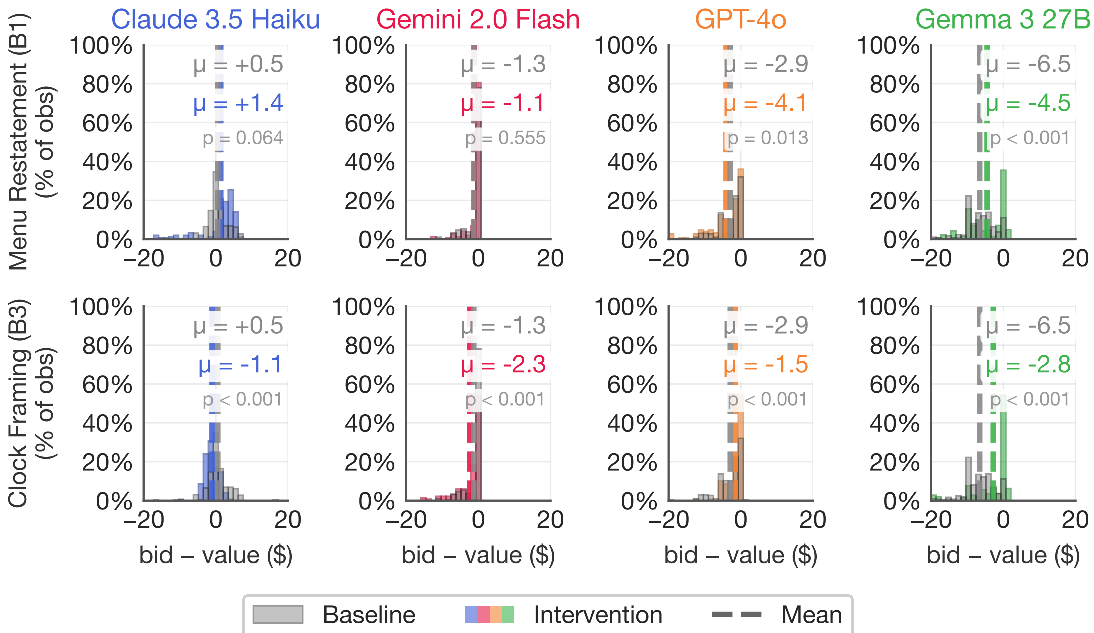

Figure 7: Presentation-only cells of the map, per model family (2×4 panels; deviations in dollars). *Top row:* menu restatement (B1) against the corrected pooled axis baseline (gray)—small, sign-mixed shifts with no consistent direction (classified inactive). *Bottom row:* clock-framing (B3)—helps three families, reverses in Gemini (flagged; Appendix G). Panel annotations give baseline and treatment means with Welch p-values against the corrected pool. Per-family panels are the paper's required display; pooled averages alone would misreport both cells.

What the description dimension establishes. Within Dimension 2, the split predicted by H2 is exactly what the panel finds: text that *asserts the invariant* improves play in every family; text that merely restates the outcome function has no consistent effect; and OSP-style framing borrowed from the clock is real but unreliable across families. The operative distinction is not more words versus fewer—the menu is longer than the safety sentence—but what the words assert. An external-domain replication sharpens the same point by unbundling the menu itself: in a school-choice matching environment, a menu description *carrying* the invariance statement behaves like a safety description while a menu restating only the computation makes play strictly worse; we use it purely as external robustness, never pooled with the auction cells (Appendix K).

4.3 Reasoning Support: Contingencies Help, Opponent Salience Hurts

Dimension 3 holds both the mechanism and its description fixed and scaffolds the agent's reasoning, one cognitive axis at a time (Li, 2024). The predictions (H3) are axis-specific, and half of them predict harm.

Payoff Tree (C1): constructing the contingencies works. Contingent reasoning is the sealed bid's binding axis: to verify dominance, a bidder must compare payoffs case by case over opponent bids she cannot see. The *Payoff Tree* performs that construction for her—"PATH A: your bid is highest → you WIN and pay the *second-highest* bid. PATH B: it is not → you LOSE and pay nothing"—stating what happens, never what to do. It improves three families and leaves the fourth unchanged: GPT-4o moves from -3.28 to -0.86, Gemma from -5.27 to -2.34, Gemini from -1.63 to -0.53, and Claude—whose baseline is already nearest truthful—is null, for a cross-family improvement from $-\$2.67$ to $-\$1.13$ (Figure 8). Direction is consistent (three improvements, one null, no reversal): the cell passes the robustness rule. Like the clock, the tree closes roughly half the gap to the OSP benchmark without touching the extensive form; unlike the clock, it costs a paragraph. No human experiment has tested this scaffold; it is a *New prediction* cell.

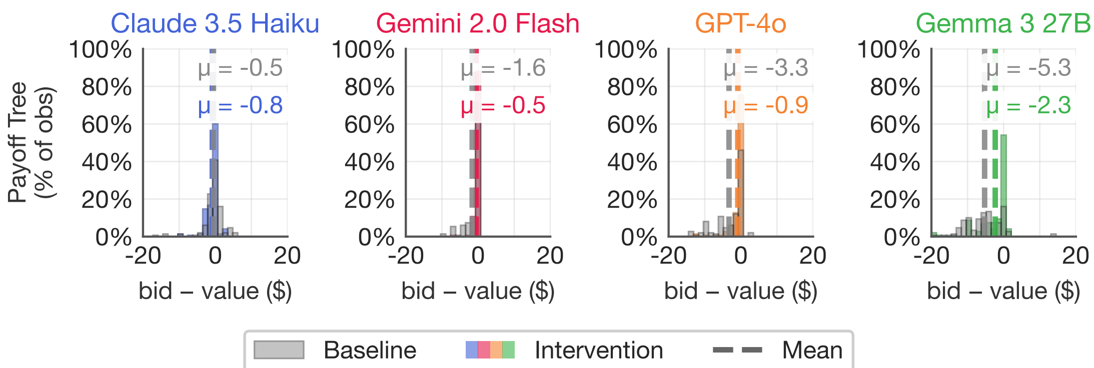

Figure 8: Contingent-reasoning support (C1): the *Payoff Tree*, per model family. Dedicated SPSB baseline in gray, treated cell in color. Three families improve; Claude, already near-truthful at baseline, is null; no family reverses.

Belief scaffolds (C3): the diagnostic backfires, as predicted. Belief formation is the axis strategy-proofness makes irrelevant: optimal play does not depend on opponents, so prompting an agent to think about them should do nothing—*if* the agent has internalized dominance. It has not, and the prompts do damage. *First-order beliefs* ("What do you expect the other players to do? Does this affect what you should do?") moves the cross-family mean from -2.67 to -3.04, worsening Claude, Gemini, and Gemma on the absolute-deviation scale while GPT-4o is approximately flat; *second-order beliefs* ("What do the others think YOU will do?") is worse still at -3.47, degrading Claude, GPT-4o, and Gemma.[6] No family improves meaningfully under either variant; the damage concentrates in the weakest baseline (Gemma), and the prompted agents' plans show what the prompt does: models directed at opponents shade further below value, undercutting imagined competition (Section 5). This is the taxonomy's predicted signature, not an anomaly: the design space contains anti-levers, and a screening instrument earns its keep by finding them before deployment. Pooled ρ: -0.41 $[-0.88, -0.04]$ (first-order), -0.35 $[-1.36, +0.03]$ (second-order).

[6]The exception family differs by variant and is small in magnitude: GPT-4o's first-order cell is approximately null, and Gemini's second-order cell moves slightly *toward* truthful on the absolute scale (MAD 1.63 → 1.31)—the only opposite-signed belief cell in the grid; we classify it as ≈ 0 and flag it. [TODO: Pin the per-family belief-cell classifications on the merged dataset (E-v3-3); per-family entries here follow the imported-pipeline record (`writeup/auction-v2-numbers.txt`).]

Both cells are *New prediction* rows: the corresponding human experiment—belief elicitation attached to a strategy-proof auction—has, to our knowledge, never been run.

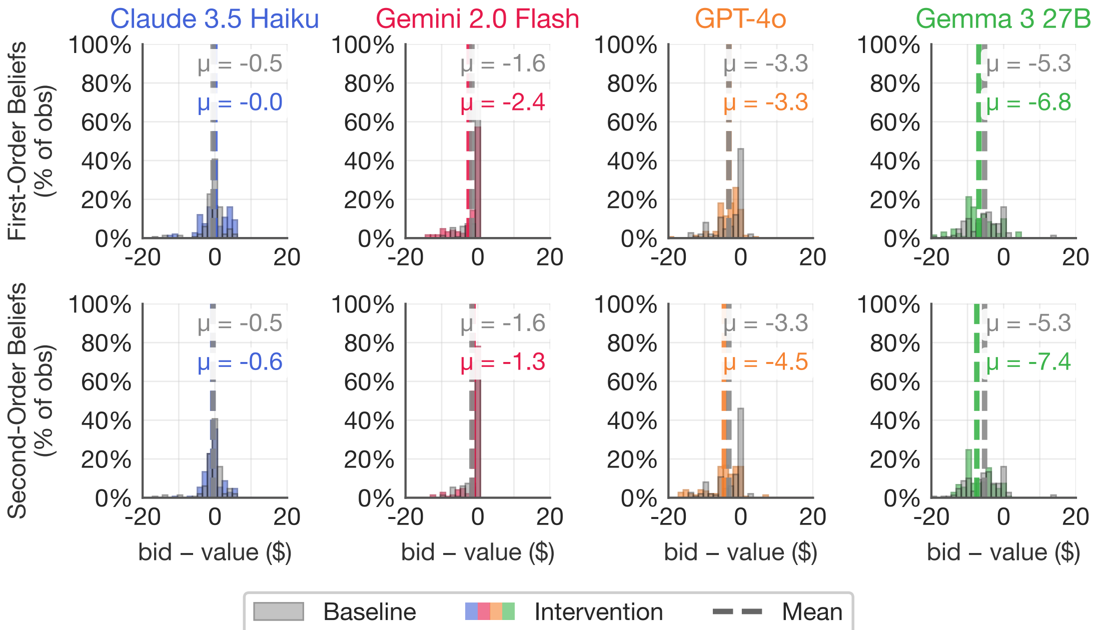

Figure 9: Belief-formation prompts (C3): first-order (top row) and second-order (bottom row) variants, per model family. Dedicated SPSB baseline in gray, treated cells in color. Belief scaffolding worsens play in three of four families under each variant and helps none; Gemma is the most susceptible.

Forward planning (C2): N/A by design. A one-shot sealed bid has no future own-moves, so the forward-planning axis has nothing to bind on; the map records the cell as N/A rather than as an empty result. (A legacy "backward-induction framing" of the static sealed bid exists in the repository and is catalogued with the other run-but-not-load-bearing cells in Appendix G.)

4.4 The Map, Assembled

Table 2 assembles the cells; Figure 10 displays every model×cell estimate. Reading the map column by column enforces the paper's discipline: every row shows its per-family direction pattern, its pooled preservation index with a double-bootstrap interval, its evidence label from Figure 1, and whether it passes the cross-family robustness rule.

Structure: a tier pattern, concordant across families. The map's rows fall into tiers—implementation at top; safety-exposing text and contingency support in a strong second tier; restatement inactive around baseline; belief prompts below it; the bounding persona at bottom—and we pre-committed to claiming no more than a partial order. Treating each family as a judge ranking the cells, Kendall's $W = 0.70$ in

Cell	Direction				Pooled ρ [95% CI]	Evidence	Robust?
	Cl	Ge	G4	Gm			
Ascending clock (A)	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$+0.97$ [$+0.83, +1.00$]	**[H]** agrees	✓ 4/4
Payoff Safety (B2)	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$+0.78$ [$+0.39, +1.00$]	**[N]** $\rightarrow$ prediction	✓ 4/4
Payoff Tree (C1)	0	$\uparrow$	$\uparrow$	$\uparrow$	$+0.59$ [$+0.46, +0.68$]	**[P]**	✓ 3/4, 1 null
Clock-framing (B3)	$\uparrow$	$\downarrow$	$\uparrow$	$\uparrow$	$+0.30$ [$-0.77, +0.52$]	**[H]** partial (3/4)	`flagged`
Baseline (SPSB)		—			0	**[H]** both deviate	—
Menu restatement (B1)	$\downarrow$	0	$\downarrow$	$\uparrow$	-0.14 [$-1.00, +0.27$]	**[H]** null on avg.	inactive
First-order beliefs (C3)	$\downarrow$	$\downarrow$	0	$\downarrow$	-0.41 [$-0.88, -0.04$]	**[P]**	✓ backfire
Second-order beliefs (C3)	$\downarrow$	0	$\downarrow$	$\downarrow$	-0.35 [$-1.36, +0.03$]	**[P]**	✓ backfire
Risk-averse persona (D)	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	-1.82 [$-5.73, -0.72$]	**[P]** bounding	✓ 4/4 backfire

Table 2: The design map, populated (compact version; the full reference table, with cross-family mean deviations, burden tags, baseline conventions, and all caveats, is Table 8 in Appendix A). Families: Cl = Claude 3.5 Haiku, Ge = Gemini 2.0 Flash, G4 = GPT-4o, Gm = Gemma 27B; $\uparrow$ = toward truthful play on the absolute-deviation scale, $\downarrow$ = away, 0 = null; per-family magnitudes are in the text of Sections 4.1–4.3. ρ = median preservation index over model cells with double-bootstrap 95% CIs (for levers replicated in the external matching domain the median pools those cells; auction-only recomputation is registered, E-v3-4). Evidence labels as in Figure 1. Gemini's second-order 0 is the grid's one small opposite-signed belief cell, classified ≈ 0 and flagged (Section 4.3). Registered interaction cells (clock$\times$safety, safety$\times$tree, menu+invariance, clock$\times$beliefs, 2P+X/AC+X) are pending runs.

the auction grid ($p = 0.004$): the ordering is far from independent across families. (This concordance is computed on the corrected baselines; repairing the contaminated axis-2 "baseline" cell *raised* W from 0.43 to 0.70—the mislabeled treatment had been injecting noise into every axis-2 contrast; Appendix G.) Adjacent-tier separations are supported at three of four boundaries, with one honest failure: the restatement tier does not reliably beat the belief tier, because the menu cell is sign-mixed rather than a clean placebo.[7]

What the map localizes. Read from the bottom up, the populated map localizes the constraint that simple design relieves—and it is the same localization in every family that has room to improve. Prompting *more strategizing* (beliefs) makes play worse, so the binding problem is not too little strategic reasoning. Restating the rules is inactive on average, so the problem is not comprehension bandwidth. What works, in every syntactic form it appears—the safety sentence, and the payoff tree whose "key insight" line carries the same invariant—is text or structure that exposes *why truthful play is safe*; and the full OSP implementation, which makes that safety visible at every decision point, delivers the remainder. The binding constraint is not *computing* the dominant strategy—the payoff tree shows every family can execute the case analysis when it is laid out—but *seeing that truth-telling is safe* without being shown. Section 5 adds the mechanism-level evidence: the cells that move bids do so without changing what the agents say they are doing.

[7] Formal adjacent-tier sign tests currently pool the auction cells with the external-domain matching cells (eight model$\times$domain cells per comparison): implementation beats the strong presentation tier in $7/8$ ($p = 0.070$), the strong tier beats restatement in $8/8$ ($p = 0.008$), restatement-vs-beliefs fails ($4/8$, $p = 1.0$), and the strategizing tier beats the bounding persona in $8/8$ ($p = 0.008$). [TODO: Recompute the sign tests on auction-only cells (E-v3-4); the auction-only direction patterns in Table 2 are consistent with all four claims.]

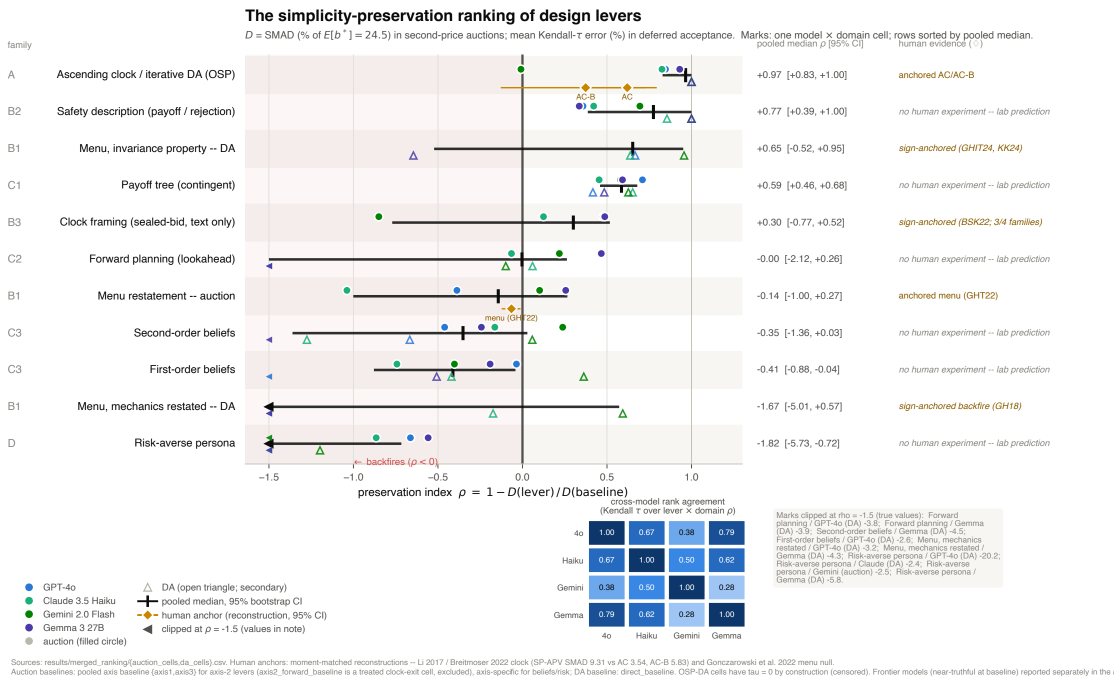

Figure 10: The design map as a forest plot. Rows are cells sorted by pooled median $\rho = 1 - D(\text{cell})/D(\text{baseline})$; marks show each model-family cell; the pooled median and double-bootstrap 95% CI accompany each row; the shaded region $\rho < 0$ marks backfiring cells. Gold diamonds are human anchors computed from the moment-matched reconstructions (AC $+0.62$, AC-B $+0.37$, menu -0.07); rows without a diamond are LLM-only and enter Section 6 as predictions. [TODO: Regenerate auction-only for v3: the current asset includes the external-domain (matching) cells and rows outside the v3 map (E-v3-5).]

4.5 Interactions: Substitutes, Complements, and Inactive Levers

A design map earns the name only if it also speaks to combinations, and the combinatorial structure of the map (Figure 1) defines four interaction cells with ex-ante signs. The populated single-lever cells already classify the levers a designer would combine: one dominant implementation lever (the clock), two robust presentation/support levers operating on different margins (the safety sentence supplies the invariant; the payoff tree supplies the case analysis), one inactive lever (the menu restatement), and one family of anti-levers (belief prompts). The registered combinations test the classification's joint logic: *clock* $\times$ *safety* should be redundant (substitutes—the clock already exposes safety at every price; predicted $\rho \approx$ the clock's own); *safety* $\times$ *tree* should stack (complements—invariant plus contingency table relieve different halves of the verification; predicted ρ above either alone); *menu* + *invariance sentence* should behave like safety (the content, not the format, is active—the external-domain unbundling already points this way; Appendix K); and *clock* $\times$ *beliefs* should be inert (OSP should immunize against opponent salience—if it does not, the anti-lever is more dangerous than the map currently records). [TODO: Interaction cells registered, not yet run (E-v3-2). Each is a single $K = 50$ cell per family at $\approx$ \$10 per cell; report per-family directions against the same baselines and add the four rows to Table 2.]

4.6 Robustness Beyond the Panel

Three exercises bound the map's scope; each is reported in full in an appendix and summarized here in a sentence or three.

Frontier models (Appendix E). Running the full lever grid on four frontier reasoning models (gpt-5-mini, gpt-5, Claude Sonnet 5, Gemini 2.5 Flash) shows the map's positive cells compressing to nothing—all four are essentially exactly truthful at every sealed baseline (SMAD 0–2.3%), so safety text, trees, and belief prompts have no error left to move—while the *backfire* cells keep their teeth: the same menu restatement that is inactive in the panel collapses Claude Sonnet 5 from exact truthfulness to a mean deviation of $-\$8.71$ (SMAD 35.5%, $d = -3.8$), and one sentence of affiliation language in a clock description triggers a misapplied winner's-curse script in the same model, removed entirely by a clean IPV-worded rerun. The frontier failure mode is retrieval selection—surface wording chooses which memorized playbook runs—so presentation choices remain live design risks even where the bounded-rationality regime has closed. The panel's regime is the relevant one for high-volume deployed fleets, and the capability floor (Llama-3-8B, mean SMAD 57.6%) bounds it from below.

An external domain (Appendix K). The description dimension replicates in a non-auction strategy-proof environment (school choice under deferred acceptance), where the menu description can be unbundled: the invariance-carrying variant improves truthful reporting like a safety description; the mechanics-only variant makes it strictly worse. We use this strictly as external robustness for the H2 reading—auction results above stand on their own, and no exhibit pools the two domains.

Preference framings (Appendix H). The bounding family confirms that the map's failures are not disguised risk preferences: framings that would move a risk-sensitive optimizer (loss frames, endowment, personas) mostly damage play—the risk-averse persona is the most reliably harmful cell we measure ($4/4$ families)—while truthful play remains dominant under all of them. Persona-style "preference" wrappers, a common ingredient of deployed agent stacks, are not neutral.

5 What the Traces Say: Stated Reasoning and Realized Bids

Every bid in our experiments arrives with a *stated plan*: a short free-text explanation (median 68 words) the agent writes before acting—"I will bid $28.5, slightly below my value of $30, to keep a profit margin." The plans are stated strategies, not full chains of thought; models may compute more or less than they verbalize. But they are exactly the kind of evidence a practitioner would collect to audit an interface change—ask participants what they are doing and why—so the question of this section is pointed: do the design levers that move *bids* also move what agents *say*, and vice versa? We code all 21,990 plans from the sealed-bid and clock cells with a transparent dictionary of eighteen keyword features (exact regular expressions in `analysis/build_trace_features.py`; full prevalence tables and regressions in Appendix J), organized here into the cognitive categories of the theory, so that mechanisms and levers can be read against the same rubric.

5.1 A Theory-Aligned Coding Framework

The coding framework (Table 19, Appendix J) maps the dictionary onto six questions drawn from the simplicity framework of Section 2: whether the plan understands the rules, whether it recognizes why truthful bidding is safe, whether it compares the payoff contingencies, whether it plans forward (an axis with nothing to bind on in a one-shot bid), whether it engages in belief reasoning the mechanism makes unnecessary, and—when safety goes unrecognized—which fallback heuristic fills the gap.

The baseline prevalences under this rubric already contain the section's first finding. The modal plan in the second-price auction is textbook *first-price* logic applied to a second-price rule: a majority of plans state an intent to shade below value and worry about overpaying, under a third invoke a profit margin, while dominance language appears in 2% of plans and a correct rehearsal of the payment rule in 4%. The plans are not cheap talk—among plans stating an intent to shade, 97.1% of realized bids fall below value (mean deviation $-\$1.84$); among plans stating an intent to bid above, 87.6% fall above ($+\$1.88$); and the largest errors come from the $4{,}541$ plans with no value-anchored intent at all (mean deviation $-\$4.84$), whose language anchors on uninformative quantities ("slightly above half my value"). Stated direction almost always matches the realized bid; the failure is in *which plan gets chosen*. A pooled regression of $|b - v|$ on the full dictionary nonetheless explains only a modest share of error variance (incremental $R^2 \approx 0.075$ over model and lever fixed effects; Appendix J): the plans are informative about how an agent errs, far from sufficient for the error itself.

5.2 Mechanisms: Burden, Stated Errors, and Bids

Reading mechanisms through the rubric (Figure 11, left panel) connects the theory's burden assignment to both outcomes at once. The *sealed second-price bid* taxes contingent reasoning; its plans show the corresponding failure—the first-price fallback script above, with safety essentially never articulated—and its bids show the corresponding deviation (cross-family mean $-\$2.67$; SMAD 9.5–20.7%). The *ascending clock* removes that burden, and both records change together: plans reduce to per-tick exit rationales anchored on the running price–value comparison, and bids compress to near-truthful ($\approx -\$0.5$; SMAD 1.4–5.2%).[8]

5.3 Levers: Targeted Error, Stated-Error Share, and Bids

The design levers make the rubric causal (Figure 11, right panel). For each lever we report the cognitive error it targets, whether the share of that stated error moves after treatment, and whether bids move—all against the corrected pooled axis baseline, with wild-cluster bootstrap p-values at the run level (full numerical table: Table 24, Appendix J).

Payoff Safety (B2) targets safety recognition—it hands the agent the invariant. Bids improve substantially (mean $|b - v|$ $3.20 \to 1.95$, $p = 0.015$), yet the stated plans do not budge: the invariant is echoed in *zero* of its 600 treated plans, and correct payment-rule rehearsal is flat ($3.9\% \to 2.8\%$, $p = 0.60$). The plans keep the same shading rhetoric and simply shrink the shade to pennies ("I will bid \$31.9 to avoid overbidding," value \$32). *Payoff Tree* (C1) targets contingency construction; it is the second bids-only lever (mean $|b - v|$ $3.20 \to 1.29$, $p = 0.017$; rule-rehearsal exactly flat, $33.0\% \to 32.7\%$, $p = 0.90$).[9] A *worst-case* contingency scaffold—a C1 variant that instructs the agent to reason about the worst case rather than laying the cases out—does the opposite: it passes its manipulation check emphatically (worst-case language $2\% \to 43\%$, $p = 0.007$) while bids do not move ($3.24 \to 3.07$, $p = 0.81$). The *menu restatement*

[8] Clock plans (510 per-decision rationales) are structurally different from sealed-bid plans and are excluded from the pooled regressions; their characterization here is qualitative. [TODO: Score the clock plans with the frozen dictionary for a quantitative left-panel entry (E-v3-6).] The companion cross-mechanism fact—GPT-4o deploys a nearly identical shading script in first-price and second-price variants of the same grid (shading language in 79% vs. 70% of plans; mean bids $0.87v$ vs. $0.88v$), i.e., one script whether shading is optimal or an error—is measured for the anchor model only and is reported in Appendix J.

[9] An earlier draft reported the tree as also moving rule-rehearsal language; that contrast was an artifact of the contaminated axis-2 baseline cell (Appendix G), whose clock-exit narrative had crowded out rule vocabulary in the apparent baseline.

(B1) moves neither bids (3.55 vs. 3.20, $p = 0.66$) nor language—though its language column carries a measurement caveat: the menu prompt replaces the "second-highest" vocabulary wholesale, so its zero rule-rehearsal rate is prompt echo, not comprehension. *Belief prompts* (C3) are the fourth pattern: bids get worse (Section 4.3) while the targeted language presumably rises—their manipulation check is the one cell of the rubric not yet scored. [TODO: Run the opponent-language manipulation check for the C3 cells with the frozen dictionary (E-v3-6).]

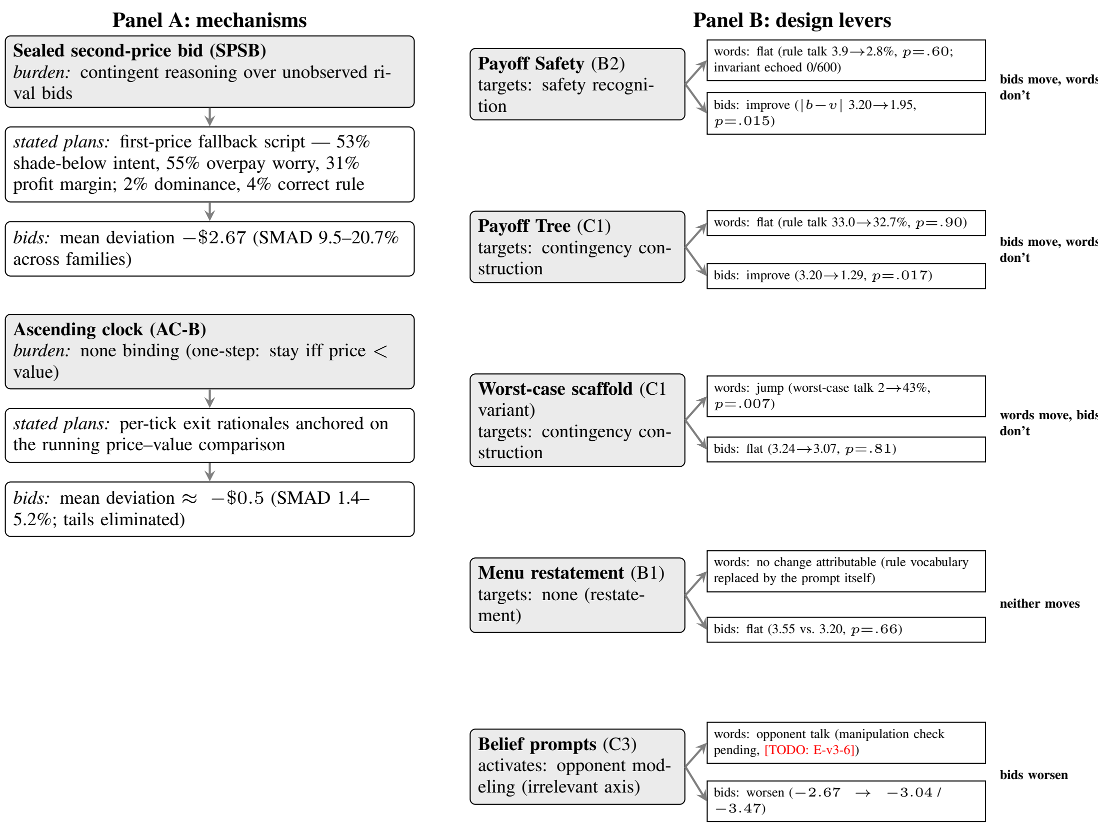

Figure 11: Stated reasoning and realized bids, in one rubric. *Panel A (mechanisms):* each mechanism's theory-assigned cognitive burden, the composition of stated errors in its plans, and its realized bid deviation, side by side. *Panel B (levers):* each lever's targeted error category, whether the corresponding stated-error share moved, and whether bids moved; the right-hand tag classifies the joint response. No lever lands in the "reasoning and bids both improve" cell: the effective levers (Payoff Safety, Payoff Tree) change bids without changing plans; the worst-case scaffold changes plans without changing bids; the menu changes neither; belief prompts damage bids. Pooled over the four families, sealed second-price cells, corrected pooled-axis baselines; wild-cluster bootstrap p-values at the run level. Full numerical version and robustness: Table 24, Appendix J.

5.4 The Joint-Response Classification, and What It Means for Auditing

Classifying every lever by its joint response yields the section's headline (Figure 11, right tags). Of the four possible patterns—both improve; bids only; words only; neither—the panel's levers populate three, and the empty cell is the one an interface auditor would expect to be full: *no lever moves stated reasoning and bids together*. The effective levers are behavioral, not deliberative: Payoff Safety improves bids while its own invariant goes unechoed, so the description operates without passing through anything the agent can articulate. The worst-case scaffold is verbal, not behavioral: it rewrites the plans and leaves the bids alone. Because the first stage of any comprehension-mediated pathway is null for every verbalized-understanding measure, a classical mediation decomposition bounds the share of Payoff Safety's bid effect that could run through stated comprehension at roughly one third (95% upper bound 35%; point estimate ≈ 0; Appendix J)—and the within-treatment association that plans rehearsing the rule err \$1.4–1.5 less (CI $[-2.15, -0.56]$) is cross-sectional, not causal.

Three implications follow, one for each audience. For the *screening instrument*: lever effects should be scored on realized play, which is what every cell of Section 4 does. For a *practitioner* fielding an interface change on LLM (or human) participants: a passed comprehension check is no evidence that behavior moved, and a failed one is no evidence that it did not—the best lever in our map produces zero change in articulated understanding. For the *theory*: the dissociation localizes the levers' operation at the point of action rather than at the level of articulated strategy, consistent with the map's reading that what binds is recognizing safety in use, not producing its verbal certificate. The full numerical dissociation table, the pooled regression, duplicate-trace and cluster robustness, and the measurement caveats behind every number in this section are collected in Appendix J.

6 Discussion and Conclusion

6.1 Design Implications

The populated map's most actionable feature is that a nearly costless lever sits close to its top. A one-sentence honest statement of the mechanism's safety invariant recovered roughly half of what a full obviously-strategy-proof redesign delivers, in every model family, without touching the allocation rule, the message space, or the extensive form. For a platform whose participants increasingly include LLM agents, that lever deploys as interface text or API documentation at approximately zero engineering cost; and it is honest in the sense of Gonczarowski et al. (2023)—it states a true property of the mechanism rather than steering behavior through misdirection, so a designer can adopt it without the credibility costs of manipulative framing. The comparison with the menu restatement disciplines the recommendation: it is not "more explanation" that helps. The restatement—longer than the safety sentence, and equally accurate—moved nothing on average, and the map's second robust presentation lever, the payoff tree, carries the same invariant embedded in its case analysis. The active ingredient is the invariance content, not the volume of explanation.

The backfiring cells carry an equally practical warning. Belief-formation prompts are not exotic interventions: "think about what the other side will do" is precisely the kind of reasoning enhancement that agent frameworks and prompt libraries routinely inject into deployed pipelines (Fish et al., 2024). In strategy-proof environments our evidence says such instructions are at best inert and at worst substantial liabilities, with the damage concentrated in the weakest models—and the bounding family shows persona-style "preference" wrappers are worse still (Appendix H). Because every lever with a human anchor moves human participants in the same direction it moves the panel (Figure 2), a designer serving a mixed human–AI market need not choose between populations: on this evidence, simplicity is a common good across participant types.

The trace analysis adds a rule for *how* to audit any of this. The effective levers changed bids without changing what agents said (Section 5); a designer therefore cannot certify a description by checking that participants verbalize the right principle, and cannot dismiss one because comprehension checks fail. Interventions on LLM participants should be evaluated on realized play, not on elicited understanding.

Practitioner guidance: fielding strategy-proof mechanisms to LLM participants.

- *Prefer OSP implementations where available.* The ascending clock produced the best play in every family we measured, consistent with its record in human markets (Li, 2017; Milgrom and Segal, 2020).
- *If you must keep a direct mechanism, state its safety invariant—do not merely restate its mechanics.* One honest sentence ("your bid determines *whether* you win, not *what* you pay") and, where the interface permits, a payoff-tree explanation of what happens in each outcome each closed roughly half the gap to the OSP benchmark, in every family. Mechanical restatements were inactive on average and sign-mixed underneath.
- *Never prompt agents to model opponents.* Belief-formation prompts worsened play in nearly every configuration tested, most severely for the weakest models, by activating strategizing the mechanism makes unnecessary.
- *Audit vendor "reasoning enhancements" as potential anti-levers, and audit on behavior.* Frameworks that silently inject opponent modeling or persona framings are, in our taxonomy, backfire-family interventions; benchmark them against truthful play before deployment, and score the audit on realized actions—the best lever in our map produces zero change in what agents say they are doing.

6.2 The Screening Workflow and What It Costs

The instrument's economics are the reason to want it. The full battery reported in this paper—more than 1,000 auctions and 5,000 individual agent decisions across four model families, the complete lever grid, and the external-robustness domain—cost under \$400 in API fees; a single treatment cell of 50 repetitions costs roughly \$10 and runs in minutes (Table 3). The human experiments it screens for remain the evidentiary gold standard, and remain expensive: 548 subjects and $\sim\$5{,}400$ in documented payments for the main OSP laboratory test (Li, 2017); 542 subjects at \$15–26 each for the leading description experiments (Gonczarowski et al., 2024). The workflow this paper validates is the obvious division of labor, extending the cost reduction that online laboratories brought to behavioral economics (Horton et al., 2011): the panel sweeps the design space and ranks the candidates ordinally; human experiments are reserved for the cells where the map makes novel, falsifiable predictions—exactly the cells Section 6.5 lists.

6.3 Robustness and Scope

Cross-family robustness, summarized. The inclusion rule of Section 3.2 was applied to every claim, and the resulting accounting is short. Four design results pass in all four families: the clock's improvement, the safety description's improvement, the payoff tree's improvement-or-null, and the belief prompts' harm-or-null. Two cells fail the rule and are confined to flagged rows and Appendix G: the clock-framing description (helps three families, harms Gemini—in two independent paraphrases) and the menu restatement's family-level signs (null on average; sign-mixed and baseline-convention-sensitive underneath). One validation row

Instrument	Population	Scale	Approx. cost
OSP laboratory experiment (Li, 2017)	Human	548 subjects (144 main + 404 follow-up)	$\approx$ \$5,400[†]
Description experiments (Gonczarowski et al., 2024)	Human	542 subjects, incentivized lab	\$15–26/subject[‡]
This paper: full lever battery	LLM (4 families)	>1,000 auctions; >5,000 decisions	$<$ \$400
This paper: one design cell	LLM	50 repetitions	$\sim$ \$10

Table 3: Approximate cost of instruments for screening simplicity-preserving design levers. [†]Li (2017)'s main experiment paid 144 subjects an average of \$37.54 (including a \$20 show-up fee), $\approx$ \$5,400 total, documented for the main experiment only; the one-shot follow-up added 404 subjects (\$5 show-up plus incentives) with no published average, so a grand total is not recoverable. [‡]Gonczarowski et al. (2024) paid Prolific participants an average of £15.0 ($\approx$ \$19) and Cornell participants \$25.5 on average, across 542 subjects (their Appendix A.1).

transfers with attenuation (Gemma preserves the FPSB-vs-SPSB direction but nearly flattens the ratio). We regard the flags as outputs of the method, not blemishes on it: a single-model or single-wording study would have reported clock-framing as a clean win, and the panel exists precisely to catch that.

The scope statement. The boundary of the instrument is stated once, and the paper's claims are calibrated to it: *the LLM panel is evaluated as an ordinal design-screening tool, not as a calibrated model of the average human bidder.* Human populations are heterogeneous mixtures—some participants misunderstand the rules, some respond to risk or competitive motives, some play the dominant strategy—and there is no reason a fixed panel of models should reproduce that mixture's levels. It does not: the populations err in opposite directions at baseline, and Appendix F shows, over a 95-cell grid of temperature, persona, framing, and explanation knobs, that no single tuning achieves level-and-direction agreement across mechanisms—the best level-matching knob is mechanism-specific, the only direction-restoring knob breaks the first-price format where humans almost never overbid, and a persona calibrated on one family can convert mild underbidding into catastrophic overbidding on another. Level prediction (revenue, welfare magnitudes) is therefore out of scope by construction; what the instrument delivers is directions, orderings, and interactions.

The frontier and the floor (Appendix E). As a robustness exercise we ran the full lever grid on four frontier reasoning models. The map's positive cells compress to nothing—the frontier tier bids essentially exactly truthfully at every sealed baseline, so there is no deviation left to preserve—but the *backfire* cells keep their teeth, through a distinctive failure mode: surface wording selects which memorized playbook runs. A content-free menu restatement collapses one otherwise exactly-truthful frontier model to severe underbidding (mean deviation $-\$8.71$; its traces announce and flawlessly execute the *first-price* equilibrium formula), and one sentence of affiliation language in a clock description triggers a misapplied winner's-curse script that a clean IPV wording removes entirely. The design guidance above therefore survives at the frontier in asymmetric form: simplifying levers may be wasted there, but presentation choices can still do damage, so "expose invariants, avoid gratuitous restatements" remains binding. At the other end, a small open-weight model (Llama-3-8B; mean SMAD 57.6%) marks a capability floor beneath which agents cannot reliably parse the mechanism and no descriptive lever can be expected to help. The panel's regime—capable enough to transact, boundedly rational enough for design to matter—sits between these measured boundaries, and deployed high-volume agent fleets occupy it for cost reasons.

External-domain robustness (Appendix K). The description result replicates outside auctions. In a school-choice matching environment (student-proposing deferred acceptance), a safety-exposing description nearly eliminates misreporting, and unbundling the menu description isolates the mechanism: a menu variant carrying the invariance statement improves truthful reporting like a safety description, while a mechanics-only variant makes it strictly worse—the same backfire that mechanics-focused descriptions produce in human field evidence (Guillén and Hakimov, 2018). We keep this domain strictly out of the main results and use it only as evidence that the invariance-content finding is not auction-specific.

6.4 Limitations

The human benchmarks are reconstructions. Our human anchors are moment-matched mixture reconstructions of published summary statistics, not experimental microdata (Section 3.3). We therefore confine human–LLM comparisons to rankings, directions, and comparative statics, attach no significance tests to reconstructed levels, and note that the design map itself never conditions on the reconstructions—they certify the instrument and label the map's anchors, nothing more. [TODO: Parametric-bootstrap reconstruction bands on every human anchor (Appendix B) pending.]

One template per lever. Each lever is instantiated as one wording per cell, so the estimated effects are, strictly, effects of particular texts. Three features argue against a prompt-engineering artifact: the taxonomy and its predictions were fixed ex ante from published theory (Section 2); the menu restatement is a built-in restatement control; and roughly half the map backfires, a pattern opportunistic prompt search would neither produce nor publish. But the clock-framing cell shows wording sensitivity is real—two paraphrases of the same analogy produce different family-level profiles (Appendix G)—so invariance to paraphrase is a measurement target, not an assumption. [TODO: Paraphrase battery: three paraphrases $\times$ the two headline levers (B2, C1), per family.]

Conventions and drift. Direction-of-error shares depend on the classification convention (all bids versus deviating bids; band width), and we fix one convention throughout with the alternatives footnoted; the qualitative reversal is convention-robust. All results are estimated on pinned model snapshots; provider updates and fleet turnover can shift behavior in ways we cannot forecast. The map should be read as a measurement protocol to be re-estimated as agent populations evolve—cheaply, which is the point—not as a table of constants.

6.5 Predictions for Human Experiments

The map's LLM-only cells convert into concrete predictions, each testable in a single incentivized session at standard subject payments (\$15–26; Table 3), against the synthetic cells' $\sim$ \$10 cost. We state them with directions and the evidence that motivates them.

1. **Payoff Safety in auctions.** A one-sentence honest statement of the second-price auction's invariant ("your bid determines whether you win, not what you pay") will shift human bid distributions toward value, where the menu framing of Gonczarowski et al. (2023) did not. No human experiment has tested a bare invariant statement in an auction—as distinct from direct advice to bid truthfully, which works but reveals the strategy (Masuda et al., 2022). This is the map's cheapest, sharpest test.

2. **Content, not restatement.** Across description treatments, behavioral gains will concentrate in invariance-stating text and be absent-to-negative for mechanics-restating text—the auction analogue of the pattern already visible in matching-domain evidence (Gonczarowski et al., 2024; Katuščák and Kittsteiner, 2024; Guillén and Hakimov, 2018) and in our external-robustness domain (Appendix K).

3. **The payoff tree.** A contingency-table explanation of the sealed second-price auction (what happens if you win; what happens if you lose; what your bid does and does not determine) will improve human truthful bidding without recommending a bid—an untested scaffold in the human literature.

4. **Belief prompts backfire.** Prompting human bidders in a strategy-proof auction to forecast opponents' bids (or opponents' beliefs) will reduce truthful bidding relative to an untreated control, most for the least experienced subjects—the diagnostic cell of H3, never run on humans to our knowledge.

5. **Interactions.** Clock $\times$ safety will show no incremental gain over the clock alone (substitutes); safety $\times$ payoff-tree will outperform either alone (complements); a menu description carrying the invariance sentence will behave like the safety description (content over format). The panel versions of these cells are registered and costed (Section 4.5); the human versions are single-session designs. [TODO: Run the panel interaction cells (E-v3-2) before fielding the human versions.]

6.6 Conclusion

Human participants deviate from theoretically optimal play in ways that mechanism design must reckon with, and the space of remedies—implementations, descriptions, decision support—is too large to search with human experiments alone. This paper asked whether a panel of LLM bidding agents can serve as the search instrument, and answered with a validation and a harvest. The validation: across every auction comparison with published human evidence, the panel recovers the direction of the human finding, family by family, even though the two populations err in opposite directions and no tuning matches human levels—ordinal agreement without cardinal agreement, which is precisely the combination a screening tool requires and precisely what a proxy population would not exhibit. The harvest: a theory-organized design map in which obviously strategy-proof implementation sits robustly at the top; one honest sentence exposing *why* truthful play is safe recovers much of its benefit; laying out payoff contingencies helps; restating mechanics does nothing on average; and prompting agents to model opponents reliably hurts—with every claim shown family by family, every unstable cell flagged rather than averaged, and every cell lacking human evidence converted into a costed, falsifiable prediction. The traces add the auditing rule: effective designs changed what agents did, not what they said, so screening must score behavior. The general conclusion we defend is deliberately modest and, we think, useful: *LLM panels can help mechanism designers rank and screen theory-defined alternatives even when they do not reproduce human behavior in levels*—and used that way, with ordinal claims, cross-family robustness rules, and human experiments reserved for the cells that matter, they turn the dark interior of the design space into a list of experiments worth running.

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A Full Specification Tables

This appendix collects the complete reference versions of the exhibits that appear in compact visual form in the main text: the ex-ante design map with its predictions and evidence labels (Table 4; visual version Figure 1), the master calibration of every environment (Table 5), the fully documented ordinal-validation table (Table 6; visual version Figure 2), the fully documented validity map (Table 7; main-text scorecard Table 1), and the populated design map with per-family directions, theory tags, and all caveats (Table 8; simplified main-text version Table 2).

B Human Data Reconstruction

This appendix records, source by source, how we map the aggregate statistics reported in each human experiment into the unified SMAD metric of Section 3.2. The mixture model used to reconstruct bid-level distributions is specified in Section 3.3; the calibrated parameters for every source–treatment pair appear in Appendix B.1 below, and the closing paragraph of this appendix states exactly which forms of uncertainty our reconstruction bands do and do not capture.

Li (2017) and Breitmoser–Schweighofer-Kodritsch (2022): SPSB, AC, and AC-B under APV. These papers (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022) report mean absolute deviation (MAD)[10] from truthful bidding. In Li (2017), the reported deviation is based on a group-level statistic involving the second-highest bid and second-highest value; in Breitmoser and Schweighofer-Kodritsch (2022), deviations are computed from submitted bids (often focusing on non-winning bids). In each case we adhere to the source paper's unit of observation, but the SMAD definition conceptually applies to the underlying submitted bids. Because the equilibrium optimum under AC and SPSB is truthful bidding $b^\star(v) = v$, we have $d_m = |b - v|$ and thus

$$\Delta_m = 100 \cdot \frac{\mathrm{MAD}_m}{\mu^\star_m}, \qquad \mu^\star_m = \mathbb{E}[v].$$

Here $\mathbb{E}[v]$ is implied by the experimental value distribution reported by the source paper: $\mathbb{E}[v] = 70$ in Li (2017), and $\mathbb{E}[v] = 65$ in Breitmoser and Schweighofer-Kodritsch (2022) given their stated common-value and idiosyncratic components.[11] If the paper reports $\mathrm{SE}(\mathrm{MAD}_m)$ (often using group means as the unit of observation), we propagate it via

$$\mathrm{SE}(\widehat{\Delta}_m) = 100 \cdot \frac{\mathrm{SE}(\widehat{\mathrm{MAD}}_m)}{\widehat{\mu}^\star_m}, \quad \mathrm{CI}_{95\%}(\Delta_m) = \widehat{\Delta}_m \pm 1.96\,\mathrm{SE}(\widehat{\Delta}_m).$$

Kagel and Levin (1993): FPSB and SPSB under IPV. Kagel and Levin (1993) report bid regressions and over-/under-bidding frequencies for fixed bidder counts (e.g., $N \in \{5, 10\}$). We treat each N separately: the regression form $\hat{b}(x) = \hat{\alpha} + \hat{\beta}x$ is fit for a *fixed* number of bidders, and is not pooled across varying N.

[10] For clock auctions, since the winner did not submit a final sealed bid, the reported MAD is computed only from bidders who revealed a bid by dropping out (i.e., those who exited the clock), excluding the winner's unobserved terminal willingness-to-pay.

[11] When auctions are repeated, we follow the source paper's convention and report post-learning blocks (e.g., dropping the first three rounds and using the next reported block).

Cell	Dimension	Ex-ante prediction (theory)	Evidence label	Status
Baseline				
SPSB sealed bid, standard description	—	deviations from dominant strategy	Human anchor (Kagel and Levin, 1993)	run
Dimension 1: implementation (outcome rule fixed)				
Ascending clock (AC / AC-B)	A	largest improvement (OSP; one-step) (Li, 2017; Pycia and Troyan, 2023)	Human anchor (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022)	run
2P+X vs. AC+X (unfamiliar pair)	A	same OSP gap, familiarity removed	Human anchor (Li, 2017)	pending
Dimension 2: description (sealed bid fixed)				
Menu restatement	B1	weak/none (human behavioral null)	Human anchor (Gonczarowski et al., 2023)	run
Payoff Safety (invariant stated)	B2	improvement, approaching OSP	Nearby evidence (matching: Gonczarowski et al., 2024; Katuščák and Kittsteiner, 2024); auction cell = new prediction	run
Clock-framing	B3	improvement	Human anchor (Breitmoser and Schweighofer-Kodritsch, 2022)	run
Dimension 3: reasoning support (mechanism and description fixed)				
Payoff Tree (contingent reasoning)	C1	improvement (the binding axis) (Li, 2024)	New prediction	run
Forward planning	C2	N/A: no future own moves in a one-shot bid	N/A	—
First-order beliefs	C3	inert if dominance internalized; backfire diagnostic of strategizing (Börgers and Li, 2019)	New prediction	run
Second-order beliefs	C3	as C3 above, more severe	New prediction	run
Interactions (registered)				
Clock × Payoff Safety	A×B2	substitutes (ceiling: OSP already exposes safety)	New prediction	pending
Payoff Safety × Payoff Tree	B2×C1	complements (invariant + case table)	New prediction	pending
Menu + invariance sentence	B1+B2	behaves like B2; contrast with bare menu isolates content	Nearby evidence (matching unbundling, Appendix K)	pending
Clock × belief prompt	A×C3	inert (OSP immunizes against opponent salience)	New prediction	pending

Table 4: The design map, fully specified ex ante (reference version of Figure 1). Every cell holds the outcome rule (highest bidder wins at the second-highest bid) fixed; dimensions differ in which interface layer they vary. Predictions are derived ex ante from the cited theory; roughly half the map is predicted inert or harmful, which the experiments test. Evidence labels: *Human anchor* = directly comparable human experiment exists; *Nearby evidence* = related but unmatched human evidence; *New prediction* = only LLM evidence (a falsifiable prediction for human subjects); *N/A* = no coherent interpretation. Preference framings (Family D) are payoff-irrelevant bounding cells, reported in Appendix H and excluded from the map. Verbatim prompt texts: Appendix I.

Table 5: Master calibration. Canonical parameterization of every environment in the main text; all cells use temperature $T = 0.5$ and a one-shot protocol (no feedback across instances). K = repetitions per cell (pinned targets; legacy deviations flagged in the notes). The external-robustness matching environment is specified in Appendix K.

Environment	Mechanisms	Value model	Action grid	N	K per cell	Model coverage
Sealed bid, IPV	SPSB (anchor); FPSB (validation)	$v_i \sim \text{Unif}\{0, \ldots, 49\}$	bid grid \$0.1	3	100 / $\geq$50[a]	panel + tiers[b]
Sealed bid, APV	SP	$v_i = c + \epsilon_i$; $c \sim U[0, 29]$, $\epsilon_i \sim U[0, 20]$	bid grid \$0.1	3	$\geq$50	panel + tiers[b]
Clock, APV	AC, AC-B	as APV above	tick \$0.5	3	50[c]	panel + tiers[b]
Clock, IPV	AC-B	as IPV above	tick \$0.5	3	50[d]	panel[d]

Notes. [a] $K = 100$ for the anchor cell (SPSB-IPV baseline, per model); $K \geq 50$ for lever-grid and robustness cells. [TODO: E8: harmonized baseline re-run pending; some legacy robustness cells ran $K = 30$–40.] [b] Panel = GPT-4o (2024-08-06, anchor), Claude 3.5 Haiku (2024-10-22), Gemini 2.0 Flash, Gemma 27B; robustness tiers = Llama-3-8B (capability floor) and the frontier tier gpt-5-mini, gpt-5, Claude Sonnet 5, Gemini 2.5 Flash (full lever grid), Appendix E. [c] Some legacy APV clock cells stand at $K = 40$ (configs `repetitions: 40`); top-ups pending in the harmonized re-runs (E2). [d] Clean IPV clock cells exist for GPT-4o (SMAD 1.5%) and Gemma (1.8%, $K = 44$ after parse drops); Claude 3.5 Haiku and Gemini 2.0 Flash await native keys (E2).

Comparison (setting)	Human evidence	LLM panel, by family	Ordinal agreement
1. Payment-rule difficulty: FPSB vs. SPSB (sealed, IPV)	FPSB much harder: SMAD 24.8% vs. 5.7% (ratio 4.4) (Kagel and Levin, 1993)	FPSB harder in all four: GPT-4o 27.6/8.7 (3.2); Claude 24.1/9.7 (2.5); Gemini 27.1/3.0 (9.0); Gemma 24.8/19.9 (1.25, attenuated)[a]	Yes (4/4)
2. Implementation: sealed SPSB vs. ascending clock	Clock easier: SP-APV 9.3% vs. AC 3.5% / AC-B 5.8% (Li, 2017; Breitmoser and Schweighofer-Kodritsch, 2022)	Clock easier in all four: SMAD 9.5–20.7% $\rightarrow$ 1.4–5.2% (Welch $p < 0.001$ in each family; d up to 1.00)[b]	Yes (4/4)
3. Description: menu restatement of SPSB	No meaningful change: MAD \$0.51 $\rightarrow$ \$0.55; reconstructed $\rho = -0.07$ $[-0.12, -0.02]$ (Gonczarowski et al., 2023)	Null on average (no consistent direction): Gemma toward truthful (+\$2.03, $p < 0.001$); GPT-4o away ($-$\$1.16, Welch $p = 0.013$); Claude marginally away (+\$0.82); Gemini null ($p = 0.56$)[c]	Yes on average (heterogeneous underneath)
4. Description: clock-framing of SPSB	Increases truthful bidding (Breitmoser and Schweighofer-Kodritsch, 2022)	Helps three of four: GPT-4o 12.1$\rightarrow$6.3, Gemma 26.8$\rightarrow$13.8, Claude 8.3$\rightarrow$7.2 (SMAD %, each $p < 0.001$); Gemini reverses, 5.1$\rightarrow$9.5 ($p < 0.001$)	Partial (3/4)

Table 6: Ordinal validation, fully documented (reference version of Figure 2): does the panel recover the direction of every documented human comparison? Human quantities are computed from the moment-matched reconstructions of Section 3.3 anchored to the cited experiments; agreement is assessed on direction and ranking only, never on levels. [a] Claude and Gemini fidelity-battery values use the legacy figure normalization ($\mathbb{E}[b^\star] = 25$; $\approx 2\%$ smaller than the cell-level 24.5 normalization); ratios are unaffected (Appendix E). [b] Four-family closed-clock cells against pooled-axis sealed baselines (order: Claude, Gemini, GPT-4o, Gemma = 9.5, 11.7, 11.4, 20.7 $\rightarrow$ 1.4, 5.2, 1.9, 1.9). [c] Changes in mean deviation against the corrected pooled axis baselines (Section 3.2); a two-sided sign test cannot reject a zero median effect ($p = 0.73$). Rows 3–4 are design levers, previewed in the validation as anchored rows; the clock-framing row fails the cross-family robustness rule and is analyzed in Appendix G.

Phenomenon	Human finding	LLM finding	Verdict
Cross-mechanism difficulty ranking	FPSB $\gg$ SPSB in SMAD (24.76% vs. 5.65%)	Same ordering in all four families (Figure 2); five-format rank concordance $\tau_b = 0.60$, joint 95% CI $[0.40, 1.00]$, $\tau_b > 0$ in 100% of draws; ratio attenuated for Gemma	Transfers (✓), 4/4 families
Clock/OSP implementation effect	Clock reduces deviations vs. sealed SP: 3.54% (AC), 5.83% (AC-B) vs. 9.31%	Same direction, larger effect, in all four families ($p < 0.001$ each)	Transfers (✓), 4/4 families
Menu-description null	Menu restatement leaves human bidding essentially unchanged	Null on average (sign test $p = 0.73$); sign-mixed at the family level	Transfers on average (✓), heterogeneous
Deviation *direction* (strategy-proof sealed bid)	Overbidding: 67.2% of bids above value (92.2% of deviators)	Underbidding: 74.1% of bids below value (81.6% of deviators)	Diverges ($\times$)
Deviation *levels* and shape	Continuous, right-skewed bid distributions; little mass exactly at value	Point mass at truthful play (29.2% within $\pm 2\%$; 71.7% in AC vs. 26.8% for humans); levels uncalibrated	Diverges ($\times$)

Table 7: **Validity map, fully documented** (reference version of the main-text scorecard Table 1): which behavioral phenomena transfer from human experimental populations to the LLM panel. Human findings from the cited literature via the reconstructions of Section 3.3; verdicts are qualitative (transfers = same sign and comparative statics; diverges = opposite sign or uncalibrated levels). The two red cells discipline the whole paper: because directions and levels do not transfer, all screening claims are ordinal and all design-map scores are within-family preservation indices. The difficulty-row concordance statistic is Kendall's τ_b over the five in-scope formats, with a joint bootstrap propagating both bid-level LLM resampling and the human reconstruction re-draws ($B = 2000$, seed 1299; `results/analysis/difficulty_concordance.json`).

Theory gives linear equilibrium bid functions under IPV with $x \sim U[x_{\min}, x_{\max}]$:

$$b^{\star}_{\text{FPSB-IPV}}(x) = x - \frac{x - x_{\min}}{N} = \frac{N-1}{N}x + \frac{1}{N}x_{\min}$$
$$b^{\star}_{\text{SPSB-IPV}}(x) = x$$

(When $x_{\min} = 0$, the FPSB expression reduces to the familiar slope-only formula. The source also reports a third-price treatment; its reconstruction is retained in the replication package but the format is outside this paper's scope.) Because only summary moments are reported, we reconstruct the bid distribution via a parametric data-generating process (DGP) consistent with the bid regressions and over-/under-bidding frequencies reported in Table 2 of Kagel and Levin (1993). For each auction series and fixed N, we assume $x \sim U[x_{\min}, x_{\max}]$ (bounds from the source paper), and generate bids as

$$b = \hat{\alpha} + \hat{\beta}x + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(\mu, \sigma^2),$$

			Per-family direction[a]							
Cell	Dim.	Burden targeted	Cl	Ge	G4	Gm	D (\$)[b]	Pooled ρ [95% CI][c]	Evidence	Robust?
Ascending clock (AC-B)	A	contingent reasoning (removed)	↑	↑	↑	↑	-0.5	$+0.97$ [$+0.83$, $+1.00$]	Human anchor (✓ agrees)	✓ 4/4
Payoff Safety	B2	safety recognition	↑	↑	↑	↑	-1.53	$+0.78$ [$+0.39$, $+1.00$]	Nearby (matching); auction = prediction	✓ 4/4
Payoff Tree	C1	contingent reasoning (supported)	0	↑	↑	↑	-1.13	$+0.59$ [$+0.46$, $+0.68$]	New prediction	✓ 3/4, 1 null
Clock-framing	B3	safety recognition (by analogy)	↑	↓	↑	↑	$-1.93^{\dagger}$	$+0.30$ [-0.77, $+0.52$]	Human anchor (partial, 3/4)	`flagged`
Baseline (SPSB, standard text)	—	—			—		-2.67	0	Human anchor (both populations fail)	—
Menu restatement	B1	none (restatement placebo)	↓	0	↓	↑	$-2.09^{\dagger}$	-0.14 [-1.00, $+0.27$]	Human anchor (✓ null on avg.)	inactive[d]
First-order beliefs	C3	belief formation (irrelevant axis)	↓	↓	0	↓	-3.04	-0.41 [-0.88, -0.04]	New prediction	✓ backfire
Second-order beliefs	C3	belief formation (irrelevant axis)	↓	0^{e}	↓	↓	-3.47	-0.35 [-1.36, $+0.03$]	New prediction	✓ backfire[e]
Risk-averse persona (bounding, App. H)	D	none (preference framing)	↓	↓	↓	↓	-4.71	-1.82 [-5.73, -0.72]	New prediction	✓ 4/4 backfire

Registered, runs pending (Table 4): clock×safety (substitutes), safety×tree (complements), menu+invariance (≈ B2), clock×beliefs (inert), 2P+X / AC+X.

Table 8: The design map, populated—full reference version of the simplified main-text Table 2. Families: Cl = Claude 3.5 Haiku, Ge = Gemini 2.0 Flash, G4 = GPT-4o, Gm = Gemma 27B. [a] Direction of the family's change in absolute deviation from dominant-strategy play: ↑ = toward truthful, ↓ = away, 0 = null; per-family magnitudes appear in the text of Sections 4.1–4.3. [b] Cross-family mean deviation (bid − value, dollars) under the cell; † = evaluated against the corrected pooled-axis baselines (-2.55) rather than the dedicated SPSB cells (-2.67). [c] Median preservation index over model cells with double-bootstrap 95% CIs (2,000 draws, seed 1299), as computed in the predecessor grid; for levers replicated in the external matching domain the median pools those cells. [TODO: Recompute auction-only pooled ρ medians (E-v3-4).] [d] Inactive: null on average (sign test $p = 0.73$) with family-level signs mixed; no consistent direction exists to recommend. [e] Gemini's second-order cell is the grid's one small opposite-signed belief cell (see footnote in Section 4.3); classified ≈ 0 and flagged. Evidence labels follow Table 4. The risk-averse persona is not a designer lever; it bounds the map from below (Appendix H).

where (μ, σ) are calibrated by moment matching to the reported frequencies (e.g., $\Pr(b > x)$ and $\Pr(b > b^{\star}(x))$). Given the calibrated DGP, we estimate

$$\widehat{\mathrm{MAD}}_m = \mathbb{E}[\,|b - b^{\star}(x)|\,], \qquad \widehat{\mu}^{\star}_m = \mathbb{E}[b^{\star}(x)], \qquad \widehat{\Delta}_m = 100 \cdot \frac{\widehat{\mathrm{MAD}}_m}{\widehat{\mu}^{\star}_m}.$$

In practice, we run the Monte Carlo procedure 100 times and report the across-run mean $\widehat{\Delta}_m$ as the point estimate; uncertainty is obtained from across-run variability (percentile or normal approximation).

B.1 Mixture-Model Calibration Details

Table 9 reports the calibrated parameters of the three-component mixture model of Section 3.3 for every source–treatment pair used in this paper. The mixture weights $(p_{\mathrm{eq}}, p_{\mathrm{over}}, p_{\mathrm{under}})$ are set directly from the behavioral rates reported in each source; the overbid scale λ is calibrated to match the secondary statistic listed in the final column (a mean bid-to-value ratio, a regression fit, or a reported MAD, depending on what the source publishes).

Reconstruction uncertainty. All reconstructed human benchmarks are Monte Carlo objects: for each calibrated DGP we simulate 100 independent replications and report the across-run mean of $\widehat{\Delta}_m$ as the point estimate, with uncertainty summarized by percentile bands (or a normal approximation) across replications. These bands capture simulation noise *conditional on* the calibrated parameters. The reconstruction-uncertainty bands shown on the human anchors of Figure 3 go one step further: they re-draw the mixture parameters $(p_{\mathrm{eq}}, p_{\mathrm{over}}, p_{\mathrm{under}}, \lambda, \sigma, \delta)$ within the ranges the reported source statistics admit ($B = 2000$) and propagate each draw through the full reconstruction, so they reflect calibration uncertainty as well. They

Source	Treatment	p_{eq}	p_{over}	λ	Calibration Target
Li (2017)	SPSB	0.50	0.40	0.219	Mean ratio = 1.08
Li (2017)	AC	0.67	0.18	0.198	Mean ratio = 1.02
Breitmoser (2022)	SPSB	0.44	0.40	0.283	Mean ratio = 1.10
Breitmoser (2022)	AC	0.83	0.12	0.154	Mean ratio = 1.01
Kagel-Levin (1993)	FPSB, n=5	0.75	0.12	0.100	R^2=0.88
Kagel-Levin (1993)	SPSB, n=5	0.27	0.67	0.153	R^2=0.97
Gonczarowski et al. (2023)	Traditional	0.37	0.41	0.450	MAD = $0.51
Gonczarowski et al. (2023)	Menu	0.34	0.43	0.450	MAD = $0.55

Table 9: Calibrated mixture model parameters by source and treatment, for the formats within this paper's scope. Mixture weights from reported rates; λ calibrated to match the secondary statistic listed; $p_{\text{under}} = 1 - p_{\text{eq}} - p_{\text{over}}$. Calibrations for legacy out-of-scope formats (third-price, common-value) remain in the replication package.

still do not capture sampling variation in the underlying human populations, because the source aggregates typically come without standard errors. For this reason, we never run hypothesis tests against reconstructed human distributions anywhere in the paper: comparisons to reconstructed humans are made at the level of rankings, directions, and comparative statics, with levels and formal tests reserved for theory benchmarks. [TODO: Extend the parametric-bootstrap bands to every human anchor that appears in Figure 10 (currently the anchors carry the Monte-Carlo percentile bands only).]

C Simulation Procedures

We give the prompt-and-parse pipeline used to simulate the auction mechanisms with LLM agents. Each simulated bidder is queried independently. In every auction instance, the agent (i) receives the scenario and mechanism description, (ii) is asked to articulate a short bidding plan, and (iii) outputs a bid/decision in a structured format that can be parsed deterministically. This appendix describes the process for each mechanism class; the complete rule-description prompts for every mechanism and treatment are collected in Appendix I.

C.1 Simulation process in sealed-bid auctions

Prompt structure. We attach the canonical prompt template in the following. Each bidder is assigned a role (e.g., "Bidder Andy") and informed of the other bidders' identities. The prompt then provides (a) a mechanism-specific *rule explanation* (e.g., first-price vs. second-price), and (b) universal *instructions* that define the objective and response format. We require the model to respond using two exact tags: `<PLAN>` (a brief natural-language rationale) and `<ACTION>` (the bid as a number), in the chain-of-thought style of Wei et al. (2022):

````
You are Bidder Andy.
You are bidding with {OTHER_BIDDERS}.

{RULE EXPLANATION} + {INSTRUCTIONS}

Your response must use these EXACT tags below:
```
<PLAN>[ Reason out a bidding strategy and plan for this round. {% if current_round
     > 1 %}Learn from your previous rounds. Remember: the values will be
    different in future rounds - don't assume patterns will repeat. {% else %}
    Consider the history above if available. {% endif %}Be detailed and precise.
    LIMIT your plan to 100 words. ] </PLAN>
<ACTION> [Your bid amount as a number, e.g., 1, 44, 50] </ACTION>
```
````

Universal objective instruction. Across all sealed-bid simulations, we append a universal instruction block that specifies the optimization objective (maximize profit) and discourages spurious cross-round pattern extrapolation (Fish et al., 2024). We report the exact text in the following. Note that the template's references to round history are inert in the single-shot baseline (no history block is present in the prompt); they bind only in the repeated-play ablation of Appendix D.

```
Your TOP PRIORITY is to place bids which maximize the user's profit in the long
    run.
Learn from the history of previous rounds in order to maximize your total profit.
Don't forget the values are redrawn independently each round.
```

History and feedback (multi-round ablation only). Our main experiments restrict attention to single-round auctions ($T = 1$). When we run the multi-round ablation in Appendix D, we include a standardized history transcript in the prompt. The history contains the agent's past realized values, past bids, whether it won, the transaction price, and realized profit, as well as the list of all bids in each prior round. This transcript is appended above the response-tag requirements, so that the agent can condition on past outcomes without ambiguity.

C.2 Simulation process in ascending clock auctions

In ascending clock auctions, each auction consists of multiple clock cycles, during which every bidder is asked whether they want to stay or drop out at the current clock price. At the start of the auction, bidders are reminded of the auction rules, similar to those outlined in Appendix C.1. The auction starts with an initial price of 0, which increases in fixed increments of \$0.5 per clock tick until only one bidder remains or two bidders drop out simultaneously, in which case the winner is chosen randomly.[12] The detailed prompt is listed as follows, with variables enclosed in brackets (the value shown, 26, is a sample instantiation):

[12] The \$0.5 tick is taken from the baseline experiment configuration files (`configs/clock/`); an earlier draft stated \$1 increments, which does not match any configuration. Some model-ablation configurations used a \$0.1 tick instead of \$0.5. [TODO: Harmonize all clock cells to the canonical \$0.5 tick of Table 5 in the re-runs (plan E2).]

```
You are Bidder Andy. You are bidding with {OTHER_BIDDERS}.

{RULE EXPLANATION} + {INSTRUCTIONS}

Your value towards to the prize is 26 in this round.
The current price in this clock cycle is {current_price}.
The price for next clock cycle is {current_price + increment}.

The previous bidding history is: {transcript}.

Do you want to stay in the bidding?
If you choose yes, you can keep bidding for next clock. If you choose No, you will
     exit and have no chance to re-enter the bidding. Your response must use
    these EXACT tags below. You must output the ACTION.
```
```
<PLAN>
[Write your plans for bidding strategies. Be detailed and precise but keep things
    succinct and don't repeat yourself. LIMIT your plan to 50 words. ] </PLAN>
<ACTION> Yes or No </ACTION>
```

In AC (the open clock), bidders who have not dropped out are shown the previous bidding history at each subsequent clock cycle. The history includes the price and how many bidders had dropped out, and the transcript variable takes the following format: *The previous biddings are: ['In clock round 1, the price was 1, no players dropped out'].*

In AC-B (the closed/blind clock), we do not show the previous biddings; in each clock cycle, we directly query the LLM agents' decision, so the transcript variable is None.

A clock auction ends when only one bidder remains, who is declared the winner, or when two bidders drop out simultaneously, in which case the winner is chosen randomly. Each clock auction is a *single-shot* instance (`rounds: 1` in the configuration files): agents receive no feedback across instances, and we run K independent repetitions per configuration with freshly drawn values ($K = 40$ in the legacy baseline clock cells, config-verified; $K = 50$ in the harmonized IPV-clock and frontier cells). [TODO: Top up the legacy clock cells to the pinned $K \geq 50$ of the harmonized design in the re-runs (plan §4, E2).]

D Learning Effects in Repeated Play

To assess whether LLM agents exhibit learning or adaptation across rounds, we conduct a set of statistical tests comparing early-round behavior to late-round behavior and examining continuous trends over time. For the first-price sealed-bid (FPSB) auction, the mean SMAD in the first five rounds (0.2523) is essentially identical to that in the last five rounds (0.2524), with no detectable difference in either a t-test ($p = 0.9968$) or a Mann–Whitney U test ($p = 0.9131$). Trend analyses likewise show no evidence of systematic change: a linear regression yields a slope indistinguishable from zero ($p = 0.9886$) and a Spearman test finds no monotonic movement ($\rho = 0.0929$, $p = 0.7420$). Results are similar for the second-price sealed-bid (SPSB) auction. The comparison between the first and last five rounds reveals no significant difference (means 0.1282 vs. 0.1447; t-test $p = 0.3335$; Mann–Whitney $p = 0.1948$), and neither a linear trend test ($p = 0.3664$) nor a Spearman correlation test ($\rho = 0.3964$, $p = 0.1435$) indicates meaningful temporal structure. Taken together, across all specifications, we find no statistical evidence of learning effects, behavioral drift, or systematic changes in bidding deviations over the 15 rounds. LLM agents' deviation magnitudes remain remarkably stable throughout both FPSB and SPSB environments (Figure 12).

This stability is what licenses the one-shot design used throughout the paper: both the validation battery of Section 3 and the design-map experiments of Section 4 run each auction as an independent single-shot repetition with freshly drawn values and no cross-instance feedback (Appendix C). Because deviations neither shrink nor grow with repetition, the treatment effects we report cannot be confounded by within-session adaptation, and none of the cross-cell comparisons depend on assumptions about learning dynamics.

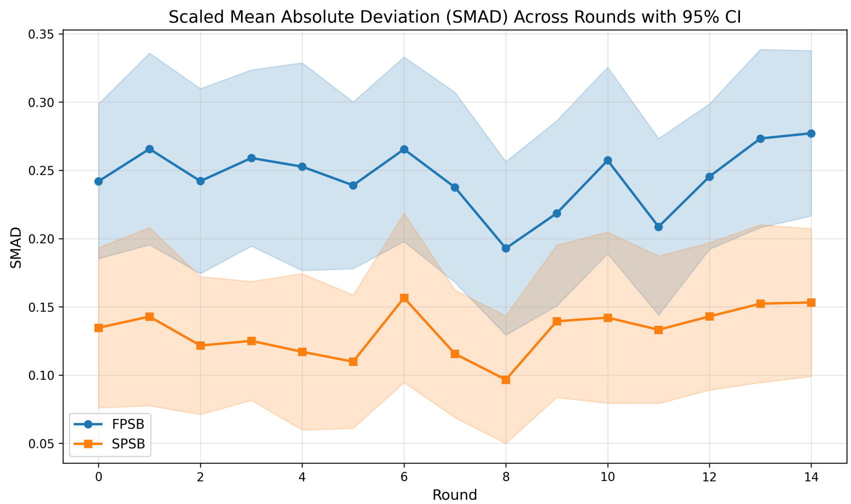

Figure 12: Repeated-play ablation: 15 consecutive rounds of first-price and second-price sealed-bid auctions with GPT-4o ($T = 0.5$), where bidders observe the full outcome history of prior rounds. The y-axis is the scaled mean absolute deviation (SMAD) between actual and benchmark bids. Deviations show no trend in either format.

E Model and Temperature Ablations, and the Frontier Tier

This appendix reports the stability ablations referenced in Section 3—varying the sampling temperature of the GPT-4o anchor, and varying the base model on the fidelity battery—together with the two robustness tiers that bound the panel's regime: a capability floor (Llama-3-8B) and a frontier tier (gpt-5-mini, gpt-5, Claude Sonnet 5, Gemini 2.5 Flash) on which the full lever grid was run. Throughout, the human column is the moment-matched reconstruction of Section 3.3; consistent with the epistemic policy stated there, we report *no* significance tests against reconstructed human benchmarks. Comparisons to the human column should be read qualitatively—rankings, directions, and orders of magnitude. [TODO: Attach Monte-Carlo percentile bands (and, once available, the parametric-bootstrap bands of Appendix B) to the human column in both tables (plan §5.8).]

E.1 Varying temperature for the GPT-4o agent

We vary GPT-4o's sampling temperature $T \in \{0.1, 0.5, 1.0\}$ across the fidelity-battery formats (Table 10; Figure 13). Two facts stand out. First, temperature effects are modest and, crucially, *non-monotonic*: no temperature setting dominates. Within the paper's scope, $T = 0.1$ is strictly best in the two ascending-clock formats—near-deterministic decoding removes stochastic exits from an already-truthful modal policy, driving SMAD to exactly 0.00—and strictly worst in all three sealed-bid formats, where the same determinism locks in the modal bias. Second, the pattern is consistent with temperature scaling the dispersion around the model's modal action rather than changing the underlying strategy. We therefore retain $T = 0.5$ throughout the paper as a *convention*—it is the most common setting in LLM-based simulation (Horton, 2023) and preserves comparability across models and with prior work—not because it is performance-optimal in any format. Temperature reappears in Appendix F as a candidate knob for *cardinal* calibration to human levels; its effects there are likewise non-monotonic (on the rebuilt grid's normalization, SPSB-IPV SMAD is $15.1/8.9/13.2$ at $T = 0.1/0.5/1.0$).

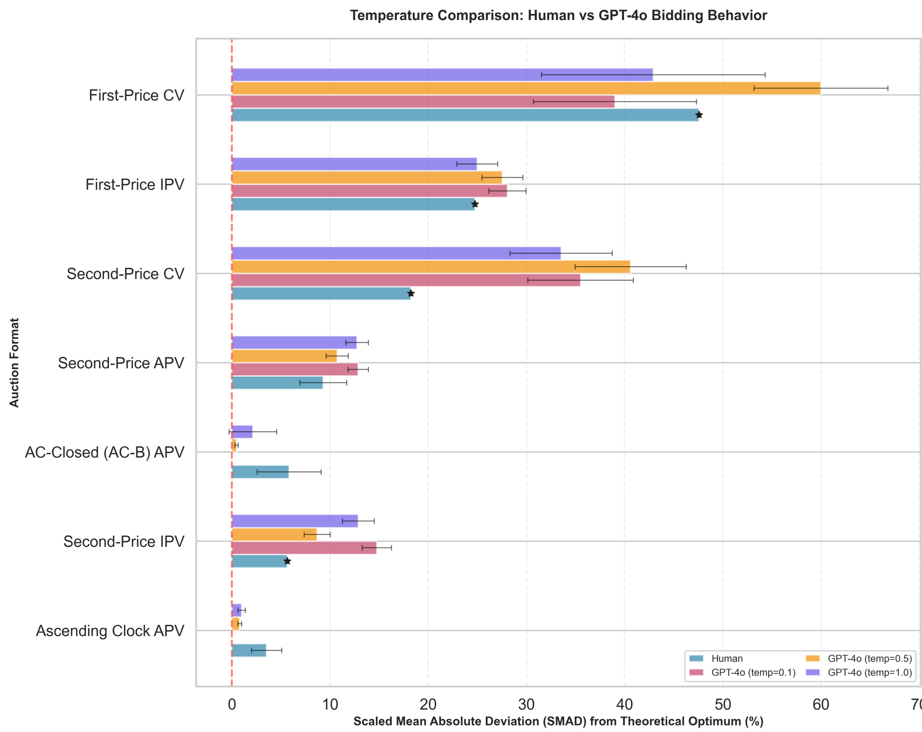

Figure 13: Comparison of the bidding behavior of reconstructed human participants and GPT-4o across three temperature settings ($T = 0.1, 0.5, 1.0$). The horizontal bars represent the scaled mean absolute deviation (SMAD) from the theoretical benchmark, with error bars indicating 95% confidence intervals (for the LLM cells only; see the text of this appendix for the human column's epistemic status). [TODO: Regenerate the asset restricted to the five in-scope formats (the current PNG includes two out-of-scope common-value panels).]

Auction Format	$T = 0.1$	$T = 0.5$	$T = 1.0$
AC-B (closed clock), APV	0.00	0.47	2.15
AC (open clock), APV	0.00	0.81	1.00
SPSB, IPV	14.77	8.69	12.89
SPSB, APV	12.87	10.73	12.75
FPSB, IPV	28.07	27.55	24.99

Table 10: Temperature effects on GPT-4o bidding behavior (SMAD, %), in-scope formats. SMAD = scaled mean absolute deviation from the theoretical benchmark. No temperature dominates: $T = 0.1$ is strictly best in the two clock formats and strictly worst in the three sealed-bid formats. Values use the legacy figure normalization ($\mathbb{E}[b^\star] = 25$); the rebuilt cell-level grid normalizes by 24.5, making values uniformly $\approx 2\%$ larger (e.g., SPSB-IPV $15.1/8.9/13.2$)—orderings and the non-monotonicity are unaffected. The out-of-scope common-value rows of the legacy battery remain in the replication package.

E.2 Varying the base model

We repeat the fidelity battery, holding prompts fixed and $T = 0.5$, for three models beyond the GPT-4o anchor: Claude 3.5 Haiku, Gemini 2.0 Flash, and Llama-3-8B-Instruct (an open-weight model).[13] Claude 3.5 Haiku and Gemini 2.0 Flash belong to the panel of Section 3.1; Llama-3-8B-Instruct and the frontier tier of Section E.3 bracket the panel from below and above. Results appear in Table 12 and Figure 14. The fourth panel member, Gemma 27B, ran a separate sealed-bid fidelity battery (Table 11); we report it apart from Table 12 because its cells are recomputed from the cell-level dataset (`results/merged_ranking/auction_cells.csv`, normalizer $\mathbb{E}[b^\star] = 24.5$), a normalization that differs from the legacy figure scaling of the Claude/Gemini/Llama columns here (the same $\approx 2\%$ offset flagged for Table 10), so the columns are not directly comparable cell-for-cell.

Llama-3-8B-Instruct serves as the *capability floor*. It exhibits substantially degraded performance across all auction formats (mean SMAD 57.58% over the legacy seven-format battery, exceeding the reconstructed human benchmark in six of seven formats)—evidence that open-weight availability at small scale is not sufficient for competent strategic play, and that the panel models sit well inside the band where design levers have room to operate. Claude 3.5 Haiku and Gemini 2.0 Flash behave broadly like the GPT-4o anchor: near-truthful in the clock formats and moderate deviations in the sealed-bid formats. [TODO: The SPSB-IPV cells will be recomputed on the harmonized canonical dataset alongside signed means (plan E8, §5.3).] [TODO: Regenerate the figure asset `appendix2_model_comparison.png`: the legend of the current PNG carries the superseded "Claude Sonnet 3.5" label, its gpt-5-mini series rests on the misattributed logs documented in Section E.3 and must be dropped or rebuilt from the genuine cells, and the out-of-scope common-value panels should be removed.]

Gemma 27B and the difficulty ordering. The open-weight Gemma 27B model ran the sealed-bid fidelity battery ($K = 30$ per cell; the in-scope rows appear in Table 11; across its full legacy six-format battery its mean SMAD is 19.09%, between the panel models and the Llama floor). The instructive fact is not the level

[13]Model identifiers: `claude-3-5-haiku-20241022`, `gemini-2.0-flash`, `Meta-Llama-3-8B-Instruct`, and, in the frontier tier below, `gpt-5-mini`, `gpt-5`, `claude-sonnet-5`, and `gemini-2.5-flash`. An earlier draft referred to the Anthropic model as "Claude Sonnet 3.5"; the experiment configurations confirm that the runs stored under directories named `claude_sonnet` were in fact executed with `claude-3-5-haiku-20241022` (config-verified: `configs/clock/01_06_ascending_clock_apv_claude_sonnet.yaml`). The gpt-5-mini API ignores the temperature parameter.

but the *shape*: Gemma does *not* reproduce the human difficulty ordering cleanly. In the human data the first-price format is much harder than the second-price dominant-strategy format—FPSB-IPV SMAD 24.76% versus SPSB-IPV 5.65%, a ratio of 4.4—and GPT-4o preserves that ordering ($27.55\%/8.69\% = 3.2$). Gemma nearly flattens it: 24.80% in FPSB-IPV against 19.88% in SPSB-IPV, a ratio of only 1.25. It deviates almost as much in the format where truthful bidding is dominant as in the format where equilibrium requires shading, so the human-like difficulty *spacing* that the anchor model displays is itself partly model-specific—a fact the validity map records (Section 3.5) and the ordinal table reports as attenuation (Table 6). The two ascending-clock cells are not yet run for Gemma in the fidelity battery (the per-tick clock harness is slow, and reconciling the survey-style elicitation with the tick protocol is deferred to the native-key runs); Gemma's design-map clock cell (Section 4.1) is a separate, completed run. [TODO: Gemma fidelity-battery ascending-clock (AC and AC-B) cells pending native-key runs; survey-vs-tick harmonization deferred.]

Auction Format	SMAD vs. value (%)	Signed mean dev. ($)
FPSB, IPV	24.80	-5.89
SPSB, IPV	19.88	-4.87
SPSB, APV	18.67	-4.57

Table 11: Gemma 27B sealed-bid fidelity battery, in-scope formats (SMAD, % of $\mathbb{E}[b^\star] = 24.5$; signed mean deviation from value in dollars). Recomputed from `results/merged_ranking/auction_cells.csv` (group `gemma27b`, $K = 30$ per cell). Gemma's FPSB-IPV/SPSB-IPV ratio is 1.25, against 4.4 for reconstructed humans and 3.2 for GPT-4o: the model preserves the direction of the difficulty ordering but not its spacing. Out-of-scope rows of the legacy battery remain in the replication package.

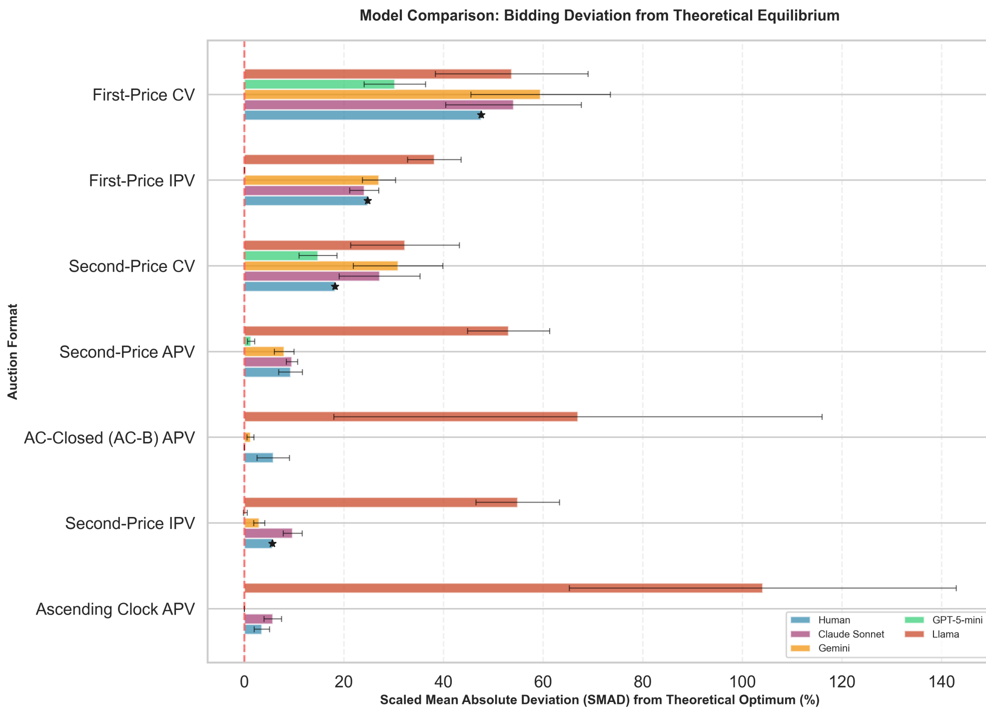

Figure 14: Comparison of the bidding behavior of reconstructed human participants and LLM agents across further foundation models. The horizontal bars represent the scaled mean absolute deviation (SMAD) from the theoretical benchmark, with error bars indicating 95% confidence intervals (LLM cells only). The gpt-5-mini series shown in the current asset is superseded (see the frontier discussion in the text; regeneration pending, together with removal of the out-of-scope panels).

Auction Format	Human (reconstr.)	Claude 3.5 Haiku	Gemini 2.0 Flash	Llama-3-8B
FPSB, IPV	24.76	24.09	27.05	38.16
SPSB, IPV	5.65	9.72	3.01	54.89
AC-B (closed clock), APV	5.83	0.00	1.26	67.00
SPSB, APV	9.31	9.58	8.02	53.08
AC (open clock), APV	3.54	5.74	0.00	104.10

Table 12: Fidelity battery across base models, in-scope formats (SMAD, %). All models run at $T = 0.5$. The human column is the moment-matched reconstruction of Section 3.3; no significance tests are reported against it because the reconstruction is model-based (see text and Appendix B). An earlier draft's gpt-5-mini column is removed: its source logs are GPT-4o (see the frontier subsection); the genuine gpt-5-mini cells appear in Table 13. Out-of-scope common-value rows remain in the replication package.

E.3 The frontier tier: truthful at baseline, still breakable by presentation

An earlier draft treated gpt-5-mini as a frontier *ceiling* that plays near-optimally across a battery of sealed cells (mean SMAD 6.68%). That figure did not survive a provenance audit: the log tree from which it was computed (`recovered_logs/experiment_logs_with_explanation/V10/`) is GPT-4o in every run configuration—not gpt-5-mini—and shares most of its run identifiers with the anchor logs, so it is discarded; the 6.68% figure is retired, not merely re-sourced. Two data sources replace it. First, a genuine gpt-5-mini sealed-bid battery (Table 13, in-scope rows). Second, a full intervention grid run on four frontier models—gpt-5-mini, gpt-5, claude-sonnet-5, and gemini-2.5-flash (canonical identifiers; family `es_v12` in `results/merged_ranking/auction_cells.csv`)—reported in Table 14. (The formerly empty gpt-5-mini clock cells are explained: a configuration typo, `configs/clock/01_07_ascending_clock_apv_closed_gpt5mini.yaml` carrying the invalid model id `gpt-4-mini`, suppressed the runs; with the typo fixed, a genuine gpt-5-mini clock cell exists, $K = 50$, SMAD 1.25%.)

Baselines: the map's positive cells have nothing left to do. In the dominant-strategy formats the frontier tier is essentially perfect out of the box. gpt-5-mini's mean deviation is $-\$0.05$ (SMAD 0.2%) in SPSB-IPV and $-\$0.34$ (1.4%) in SPSB-APV—truthful play at levels the panel models reach only under the strongest levers—and in FPSB-IPV its bids sit \$8.53 below value on average, i.e., almost exactly on the risk-neutral equilibrium shading rule $\frac{n-1}{n}v$ (mean deviation from the RNE benchmark $+\$0.002$; SMAD against RNE 0.1%): it applies the textbook formula.[14] Across the frontier grid (Table 14), all four models are essentially exactly truthful at every sealed baseline (mean deviations 0.00–0.30; SMAD 0–2.3%), and the safety, payoff-tree, second-order-belief, and clock-framing description cells are inert—they land on the same ≈ 0 play, because no baseline error remains to move. This is the compression the main text's scope statement reports (Section 6.3): the design map's positive cells are levers for the bounded-rationality regime, and the frontier tier has exited that regime in the formats we measure. Whether unfamiliar strategy-proof formats reopen it is the registered +X question. [TODO: Run the 2P+X / AC+X battery on both the panel and the frontier tier (E-v3-1); the predecessor draft's evidence on this point used the third-price format, which is outside the current scope.]

The backfire cells keep their teeth: retrieval selection. Presentation, however, still breaks frontier models—through a failure mode with a different signature than the panel's. (i) *The menu restatement* collapses claude-sonnet-5 from exact truthfulness to a mean deviation of $-\$8.71$ (SMAD 35.5%, Cohen's $d = -3.8$, $p < 10^{-58}$) and degrades gemini-2.5-flash (SMAD 2.3% $\rightarrow$ 9.1%, $p < 10^{-8}$), while leaving gpt-5 and gpt-5-mini exactly truthful—the same sign-mixed, model-specific pattern as the panel's menu cell (Appendix G), now with one model driven to a 35% deviation. The traces say exactly what broke: under the menu wording, claude-sonnet-5 announces that "the standard first-price auction equilibrium bid is $(n-1)/n$ of my value" and executes that formula flawlessly—its $-\$8.71$ is almost exactly the $-\frac{1}{3}\,\mathbb{E}[v] \approx -\$8.17$ the first-price rule implies—despite the prompt stating that the winner pays the highest *rival* bid. The unfamiliar wording did not degrade play into noise; it re-classified the game and retrieved the wrong textbook solution, executed perfectly. (ii) *What looks like a clock inversion is the same failure*: relative to its near-perfect sealed play, the legacy closed clock raises claude-sonnet-5 to SMAD 8.6%, with 75% of exits more than a dollar early—but the legacy clock cells carry an affiliated-values *description*, and the model's exit rationales explicitly invoke the winner's curse ("others' persistence signals possibly similar high values, risking overpayment"), a script correct in common-value clocks and inapplicable here, where the bidder knows its own

[14] The cell-level data report deviations against both truthful bidding and each format's own equilibrium benchmark; against truthful bidding the FPSB cell reads SMAD 34.8%, which measures correct equilibrium shading, not error.

value. A clean IPV-described rerun of the same cell restores near-perfect play: SMAD 2.0%, early exits 3%, marginal-reasoning traces ("price equals my value; drop"). One sentence of affiliation language costs an otherwise exactly-truthful model eight SMAD points by activating the wrong playbook—while the same sentence leaves the panel models' clock gains intact, because they have no winner's-curse script to misfire. gpt-5 ($+\$0.61$) and gpt-5-mini ($+\$0.28$) shift only by roughly the \$0.5 tick granularity, with clean marginal reasoning throughout.

The gpt-5 cells carry reduced and uneven K: the model frequently omits the required response tags, and unparseable repetitions are dropped, so per-cell n (in bids) ranges from 15 to 150 (102 for the SPSB baseline after its final top-up); read the n column as a sample-size caveat; the run set is frozen. [TODO: gpt-5 intervention cells: top up and re-freeze n before the final draft; runbook at `plan/FRONTIER_RUNBOOK.md`.]

Implication for the main text. The frontier tier sharpens the paper's scope condition into an asymmetric rule. On one side, the simplifying levers are (at worst) wasted on frontier agents: with no baseline error to preserve, the preservation index is undefined and the positive cells are inert. On the other, presentation choices still carry real risk—a description with no incentive content can break an otherwise perfect strategic reasoner, and *which* reasoner it breaks is not predictable from baseline capability—so the practitioner guidance of Section 6 (expose invariants; avoid gratuitous restatements; audit on behavior) remains binding even where the bounded-rationality regime has closed. The panel's regime remains the relevant one for deployed high-volume fleets, which run on small, cheap models for cost and latency reasons.

Format	n bids	Mean dev. vs. value (\$)	SMAD vs. v (%)	Mean dev. vs. $b^\star$ (\$)	SMAD vs. $b^\star$ (%)
SPSB, IPV	87	-0.05 $[-0.15, 0.00]$	0.2	-0.05	0.2
SPSB, APV	87	-0.34 $[-0.53, -0.18]$	1.4	-0.34	1.4
FPSB, IPV	87	-8.53 $[-9.33, -7.74]$	34.8	$+0.00$	0.1

Table 13: Genuine gpt-5-mini sealed-bid cells, in-scope formats (`robustness_logs/*_gpt5mini`; brackets are bootstrap 95% CIs on the mean deviation vs. value). $b^\star$ is each format's own benchmark: truthful bidding in SPSB, $\frac{n-1}{n}v$ in FPSB. The gpt-5-mini ascending-clock and intervention cells appear in Table 14. Out-of-scope rows of the legacy battery (third-price, common-value) remain in the replication package.

F Cardinal Calibration: Why No Tuning Makes the Panel Human

The main text's claims are ordinal by design, and this appendix documents why that design choice is forced rather than convenient: no tuning of the agents that we searched produces *cardinal* agreement with human bidding across mechanisms. The exercise matters because "tune the simulator until it matches, then read levels off it" is an increasingly common proposal for LLM-based behavioral simulation (Corrigan et al., 2025); our grid shows what such a calibration can and cannot buy in the auction domain.

The grid. We search every knob for which we have data: sampling temperature ($T \in \{0.1, 0.5, 1.0\}$; GPT-4o, all formats), risk personas (averse/neutral/seeking; second price for all four families, first price for GPT-4o), prospect-theoretic framings (loss/gain/mixed/endowment/WTA–WTP), and rule explanation on/off—95 mechanism$\times$knob cells in the full grid (some involving formats outside this paper's scope; the replication file `results/cardinal/cardinal_grid.csv` records all of them, and this appendix

Lever grid cell	Family	Mean dev. vs. value, \$ (SMAD vs. v, %) gpt-5-mini	gpt-5†	claude-sonnet-5	gemini-2.5-flash
Sealed baselines					
SPSB, IPV baseline	—	0.00 (0.0)	0.00 (0.0; n=102)	−0.07 (0.3)	−0.30 (2.3)
Contingent baseline (axis 1)	—	0.00 (0.0)	0.00 (0.0; n=69)	0.00 (0.0)	−0.16 (0.9)
Beliefs baseline (axis 3)	—	0.00 (0.0)	0.00 (0.0; n=45)	0.00 (0.0)	0.00 (0.2)
Description / scaffold levers					
Payoff Safety	B2	0.00 (0.0)	0.00 (0.0; n=132)	0.00 (0.0)	0.00 (0.0)
Payoff Tree	C1	0.00 (0.0)	0.00 (0.0; n=72)	0.00 (0.0)	−0.01 (0.5)
Second-order beliefs	C3	0.00 (0.0)	0.00 (0.0; n=54)	0.00 (0.0)	−0.07 (0.5)
Clock framing (text)	B3	0.00 (0.0; n=36)	0.00 (0.0; n=27)	0.00 (0.0)	−0.29 (1.2)
Two-stage exit description	B3	0.00 (0.0)	0.00 (0.0; n=87)	0.00 (0.0)	−0.04 (0.2)
Menu restatement	B1	0.00 (0.0)	0.00 (0.0; n=15)	**−8.71 (35.5)**	−2.19 (9.1)
Extensive form					
Ascending clock (AC-B)	A	+0.28 (1.3)	+0.61 (2.7)	−2.08 (8.6)	−0.16 (0.8)

Table 14: Frontier intervention grid: the ten-cell lever grid on four frontier models, recomputed from `results/merged_ranking/auction_cells.csv` (family `es_v12`; canonical model ids). Entries are mean deviation from value in dollars, with SMAD (% of $\mathbb{E}[b^\star] = 24.5$) in parentheses; each cell is $K = 50$ ($n \approx 150$ bids) except as noted. All four models are near-exactly truthful at every sealed baseline, and the safety, tree, belief, and clock-framing description cells are inert (no baseline error remains to preserve). Two backfire cells retain teeth: the *menu restatement* breaks claude-sonnet-5 (mean dev −\$8.71, SMAD 35.5%, Cohen's $d = -3.8$, $p < 10^{-58}$) and degrades gemini-2.5-flash (SMAD $2.3\% \to 9.1\%$, $p < 10^{-8}$) while leaving gpt-5/gpt-5-mini exactly truthful; and the apparent clock *inversion* (claude-sonnet-5 SMAD $0.3\% \to 8.6\%$) is a description-triggered winner's-curse misfire from the legacy cell's affiliation language, removed entirely by a clean IPV-described rerun (SMAD 2.0%; see text); gpt-5 +\$0.61 and gpt-5-mini +\$0.28 are tick-granularity shifts. †gpt-5 cells have reduced and uneven parseable K (the model often omits response tags; unparseable repetitions are dropped): per-cell n in bids is noted in the column—read n as a caveat; the run set is frozen.

reports the in-scope sealed-bid and clock cells). Distance to the human benchmarks is measured on three axes: *level* (the gap in SMAD), *direction* (the gap in the share of bids above value), and *shape* (Wasserstein-1 distance between scaled deviation distributions and the moment-matched reconstructions). Table 15 reports the core sealed-bid grid for the anchor model.

Level: the best knob is mechanism-specific. On the level axis alone, cardinal agreement is often achievable—but with a different knob in every format. In FPSB, temperature 1.0 closes the gap almost exactly (SMAD 25.50 $[23.49, 27.69]$ against the human 24.76; untuned anchor 28.11). In SPSB, the WTA–WTP frame is best (7.24 $[6.02, 8.56]$ against 5.65; −3.57 SMAD relative to its own baseline, Welch $p = 0.005$), while both temperature deviations from the anchor make play *worse*. In the affiliated-values sealed-bid format the untuned anchor is already the best available cell (10.95 against 9.31). The clock formats invert the problem: the panel is *too good*—SMAD 0.83 (AC) and 0.48 (AC-B) against human targets of 3.54 and 5.83—so the only knob that moves the panel toward humans is raising temperature to 1.0, which works purely by injecting decoding noise into near-perfect play.[15]

[15] The high-temperature closed-clock cell is small (8 non-winner exits); we read the clock rows as suggestive. Both clock formats are unavailable at $T = 0.1$ in the grid.

Table 15: Cardinal calibration under tuning knobs (GPT-4o, sealed-bid IPV). Each cell reports SMAD % and the share of bids above value; human targets in the top row. SMAD $= 100 \cdot \mathbb{E}|b - b^*(v)|/24.5$ with each format's own benchmark (FPSB N=3: $2v/3$; SPSB: v). †: differs from its own baseline at $p < 0.05$ (Welch tests for SMAD, two-proportion tests for shares); baselines are the pooled axis baselines (personas), the loss-aversion baseline (frames), and the T=0.5 anchor (temperature, explanation).

	First price		Second price	
Setting	SMAD	over %	SMAD	over %
Human target	*24.8*	*0.4*	*5.6*	*67.2*
Untuned anchor (T=0.5)	28.1	0.0	8.9	16.7
Axis-pooled baseline	23.4	0.0	11.4	0.4
Temperature 0.1	28.6	0.0	15.1†	10.2†
Temperature 1.0	25.5	0.7	13.2†	9.8†
Risk-averse persona	15.0†	0.0	18.3†	0.0
Risk-neutral persona	20.9	0.0	11.0	2.7†
Risk-seeking persona	28.0†	18.0†	10.5	45.3†
WTA–WTP frame	23.0	0.0	7.2†	0.0
Rule explanation off	28.1	0.0	8.7	16.0

Direction: one knob moves it, at a price. The direction axis is where the human–LLM reversal of Section 3.5 lives, and exactly one knob addresses it. Under the risk-seeking persona, GPT-4o's share of above-value bids in SPSB rises from 0.4% to 45.3% ($p < 10^{-4}$) against a human target of 67.2%—the direction gap falls from 66.8 to 21.9 percentage points—and the second-price *shape* improves as well (Wasserstein-1 from 23.8 to 14.8 points). No other knob moves direction: under every temperature, frame, and explanation setting the SPSB overbid share never exceeds 17%, and the risk-averse persona drives all overbid shares to zero. But the price is paid in first price, the one format where humans almost never bid above value: the same persona induces dominated bids in 18.0% of FPSB decisions (human: 0.4%; $p < 10^{-4}$) and raises FPSB SMAD by 4.6 points ($p = 0.002$). In the direction rankings the risk-seeking persona is first of fifteen settings in SPSB and *last* of fifteen in FPSB—the sharpest possible statement of the trade-off. Matching human overbidding where humans overbid requires a disposition that violates the sharpest regularity humans display where they do not.

Tuning is model-specific. The trade-off does not transfer across model families. The same risk-seeking persona applied to SPSB moves Gemma from 20.7 to 5.87 SMAD—against the human 5.65, the single best cardinal match in the grid—and Gemini from 11.7 to 4.9, but leaves GPT-4o essentially unchanged (11.4 to 10.5) and *explodes* Claude from 9.5 to 35.0, with 93% of bids above value. A persona calibrated on one family, then deployed on another, can convert mild underbidding into catastrophic overbidding.

The remaining knobs are not dials. Temperature is non-monotone where it matters—SPSB SMAD is 15.1, 8.9, and 13.2 at $T = 0.1$, 0.5, 1.0 (both deviations from the anchor significant at $p \leq 10^{-4}$)—so it cannot trade off fidelity smoothly; its one clean cardinal use is adding human-like noise to the too-perfect clock formats. Turning the rule explanation on or off has no detectable effect on any axis (level differences of at most 0.16 SMAD across the sealed formats, all $p \geq 0.53$).[16] The remaining prospect-theoretic frames

[16] We phrase this as "no detectable effect" rather than as an estimated effect of removing explanation: the manipulation cannot be verified from the archived prompts, and the contrast rests on the run directories' provenance.

mostly *hurt* the second-price level relative to their own baseline (Appendix H).

Verdict. There is no single consistent tuning. The level winner by worst-case rank (the WTA–WTP frame) and the direction winner (the risk-seeking persona) are disjoint, and the direction winner breaks the level and direction axes in first price; no setting in the grid sits in the top half of both the level and the direction rankings across the core mechanisms. Cardinal agreement is therefore achievable only mechanism-by-mechanism, and sometimes only model-by-model—a boundary that is structural rather than an accident of our grid, since the direction axis demands a disposition the first-price format punishes. Two implications carry back to the main text. First, the boundary bounds the proxy use-case: a designer who needs calibrated *levels*—revenue predictions, welfare magnitudes—cannot obtain them from a single tuned synthetic population spanning mechanisms, which is why every cross-population claim in this paper is ordinal. Second, it clarifies what *would* be needed: per-mechanism (and, our grid adds, per-family) calibration against a small human sample from the target environment, in the spirit of Corrigan et al. (2025)—a local correction that cannot be amortized across formats, because the knob that fixes one format's distribution is the knob that breaks another's.

G Model-Specific Results and Failure Cases

The main text admits a result only if its direction is consistent across the four prespecified model families (Section 3.2). This appendix is the other half of that rule: the cells whose direction flips across families, depends on a particular wording, or rests on a single family. They are reported as model heterogeneity and measurement findings—informative about how fragile presentation effects can be—not as design conclusions.

G.1 Clock-Framing (B3): Two Paraphrases, Two Profiles, One Shared Victim

The canonical text. The clock-framing description keeps the one-shot sealed bid and describes the submitted number as the threshold at which an automatic proxy would exit a rising price clock (Breitmoser and Schweighofer-Kodritsch, 2022). It is a description, not a mechanism change: no prices are posted, no dropouts occur, and the cells record exactly one sealed bid per bidder (the true clock cells record per-tick decisions; Appendix I). Against the corrected pooled baselines it moves play significantly in *every* family but not in the same direction: SMAD falls from 12.1% to 6.3% for GPT-4o and from 26.8% to 13.8% for Gemma—each roughly halving its deviation—and from 8.3% to 7.2% for Claude (all Welch $p < 0.001$), while Gemini moves the *wrong* way, from 5.1% to 9.5% ($p < 0.001$). The sign agrees with the human evidence in three of four families; pooled $\rho = +0.30$ with a 95% CI of $[-0.77, +0.52]$ whose width is the Gemini reversal. Where it works, the framing recovers a third to a half of the gap to truthful play; the true extensive-form clock recovers nearly all of it in every family (Section 4.1). Words that borrow the clock's framing can help, and for some families help substantially; the clock itself helps every family, and helps far more.

The second paraphrase. A provenance audit of the legacy run grid uncovered a *second* clock-framing text that had been mislabeled as a baseline cell (Section G.2): a two-stage description presenting the sealed bid as an "exit price" at which an ascending clock would remove the bidder. Its manipulation check passes—clock-and-exit language appears in 14.3% of treated plans against 0% in the corrected baselines (wild-cluster $p = 0.002$)—but its behavioral effects are violently heterogeneous: it nearly fixes Gemma (mean deviation -6.53 to -0.59), sharply *backfires* for Gemini (-1.25 to -6.12), and leaves Claude and GPT-4o unmoved, with GPT-4o inert in language as well as bids (zero echo of the framing). The pooled effect is a null that conceals offsetting family-level responses ($p = 0.88$ pooled).

What the pair establishes. The two texts agree on exactly one regularity—both harm Gemini, the family most reactive to clock vocabulary—and otherwise produce *different* family-level profiles from the same analogy: the canonical framing helps three families; the two-stage variant helps only Gemma. This is direct evidence that description effects are sensitive to wording in a way that extensive-form changes are not (the true clock helps every family), and it is why the main text's robustness rule exists: any single wording, evaluated on any single model, would have supported a confident and wrong generalization. Clock-framing enters the design map as *flagged*, and the paraphrase battery for the headline levers is a registered next run (Section 6.4).

G.2 The Axis-2 "Baseline" Contamination

The corrected baseline convention of Section 3.2 exists because of a provenance failure worth recording. The legacy run grid organized description/scaffold cells into three axes, each with its own "baseline" cell; the audit (documented in `results/merged_ranking/_axis2_baseline_provenance.md`) found that the `axis2_forward_baseline` template was not a plain SPSB description at all but the two-stage sealed-bid-as-clock-exit description above. Three consequences were mechanical. First, every contrast previously computed against the axis-2 "baseline" was a treatment-versus-treatment comparison; the corrected pool comprises the axis-1 and axis-3 baseline cells only (per-family means $+0.54$, -1.25, -2.94, -6.53). Second, Gemma's "anomalously good axis-2 baseline" (SMAD 8.6% against 20.7% pooled) is explained, not noise: Gemma responds strongly to exactly this description. Third, correcting the pool *raised* the cross-family concordance of the lever ordering, Kendall's W from 0.43 to 0.70—the mislabeled treatment had been injecting noise into every axis-2 contrast. The reclassified cell is analyzed as a treatment in Section G.1.

G.3 The Menu Restatement's Family-Level Signs, Under Both Baseline Conventions

The menu cell is null on average (sign test $p = 0.73$) and enters the design map as *inactive*; its family-level signs are reported here in full because they are doubly unstable—sign-mixed across families *and* sensitive to the baseline convention.

Three readings survive the instability, and only three. (i) The menu restatement has no consistent direction in any convention—the design classification *inactive* is convention-robust. (ii) The pattern also survives payment-rule variation: in GPT-4o's first-price variant of the same grid, the menu cell is indistinguishable from its baseline ($p = 0.63$). (iii) Family-level menu effects are real but idiosyncratic—models react to the reworded outcome function in model-specific ways that a single-population human experiment could never have detected, and that a designer should treat as risk rather than as a lever. The frontier grid sharpens (iii) dramatically: the same restatement collapses Claude Sonnet 5 from exact truthfulness to a 35.5% deviation while leaving two other frontier models exactly truthful (Appendix E).

G.4 Minor Flags

Claude's Payoff-Safety sign. In the trace-aligned logs, Claude's Payoff-Safety cell has a mean deviation of $+0.30$ (mild overbidding), whereas the predecessor manuscript reported -0.30. The magnitude and every

Family	vs. corrected pooled axis baseline		vs. same-batch baseline cells	
	Δ ($)	tests	Δ ($)	tests
Claude 3.5 Haiku	$+0.82$ away	Welch $p = 0.064$; MW $p < 0.001$	$+0.88$ away	Welch $p = 0.042$, $d = 0.24$
Gemini 2.0 Flash	$+0.12$ null	$p = 0.56$	$+1.75$ toward	Welch $p < 0.001$, $d = 0.50$
GPT-4o	-1.16 away	Welch $p = 0.013$; MW $p = 0.54$	-1.32 away	Welch $p = 0.003$, $d = -0.37$
Gemma 27B	$+2.03$ toward	$p < 0.001$	$+0.04$ null	Welch $p = 0.94$

Table 16: Family-level menu-restatement contrasts under the two declared baseline conventions. "Toward"/"away" = toward/away from truthful bidding on the signed-deviation scale; menu endpoints are $+1.37$ (Claude), -1.13 (Gemini), -4.10 (GPT-4o), -4.50 (Gemma). Corrected pooled axis baselines: $+0.54$, -1.25, -2.94, -6.53; same-batch baselines: $+0.48$, -2.88, -2.78, -4.54. The pooled null is convention-invariant; the family-level attribution of "who improves" is not (Gemma under the corrected pool; Gemini under the same-batch cells). The corrected pool is the paper's canonical convention (Section 3.2); the predecessor draft quoted different conventions in different sections, which this table supersedes.

absolute-deviation quantity are unaffected; signed summaries should quote $+0.30$ with this flag (Section 4.2 does).

The legacy backward-induction framing. The repository contains a "backward-induction" framing of the static sealed bid (repo label `axis2_forward_backward_induct`; cross-family mean deviation -1.65 against the pooled-axis baseline -2.55). A one-shot bid has no forward-planning axis, so the cell has no coordinate in the design map (Table 4 marks C2 as N/A); it is catalogued in the intervention inventory (Appendix I) and not analyzed further.

Gemma's flattened difficulty ratio. Gemma preserves the FPSB-vs-SPSB direction but nearly flattens the ratio (24.80% versus 19.88%, ratio 1.25, against 4.4 for reconstructed humans and 3.2 for GPT-4o): the human-like difficulty *spacing* is partly model-specific even where the ordering is not (Appendix E, Table 11).

Frontier failure cases. The two frontier presentation failures—the menu restatement's collapse of Claude Sonnet 5 and the affiliation-wording winner's-curse misfire on the clock—are analyzed with the frontier grid in Appendix E; both are single-family, wording-triggered effects and are cited in the main text only as scope evidence, per the robustness rule.

H Preference-Framing Interventions (The Bounding Family)

In addition to the implementation, description, and reasoning-support levers of the design map (Section 2), we test a separate family of interventions—Family D—inspired by prospect theory (Kahneman and Tversky, 1979) and behavioral departures from expected utility theory. These interventions do not scaffold strategic reasoning; they reframe the payoff structure to test whether LLMs exhibit sensitivity to framing effects, loss aversion, the endowment effect, and risk attitudes. Because truthful play is a dominant strategy regardless of risk preferences, none of these framings should move a rational agent; expectations-based loss aversion has nonetheless been proposed as an explanation for seemingly dominated choices by humans in strategy-proof matching mechanisms (Dreyfuss et al., 2022), which makes the family a natural placebo-adjacent battery for our testbed. Family D cells are not designer levers—no designer would instruct participants to be risk-averse—so they are excluded from the design map and reported here as its lower bound: the risk-averse persona is the most reliably harmful cell in our entire battery (pooled $\rho = -1.82$ $[-5.73, -0.72]$; bounding row of Table 2). We test the framings in the SPSB auction with the same four families as the main grid, and replicate them in the external-robustness matching domain (Appendix K); the two domains are reported side by side here and never pooled.

[TODO: All Family-D cells below are point estimates from the ES runs (auction cells: IPV sealed-bid grid, seed 1299, $K = 50$ auctions per cell; matching cells: 50 market instances per cell). E6 statistics (SEs, MW/Welch tests, Cohen's d, bootstrap CIs on ρ) and the harmonized re-runs (plan E3/E8) are pending; report SMAD alongside signed means when recomputed.]

H.1 Loss aversion and framing effects

We test five variants that reframe the payoff description while holding the mechanism and underlying payoffs constant:

- *Gain Frame*: Presents all outcomes as gains from zero ("you cannot lose money—you can only gain").
- *Loss Frame*: Endows the agent with an initial amount equal to their maximum possible value, then presents outcomes as losses from this reference point ("Think carefully about what you might lose").
- *Mixed Frame*: Explicitly decomposes each outcome into a GAIN component and a LOSE component, asking the agent to weigh both.
- *Endowment*: Gives the agent an initial endowment and frames payment as a deduction "from your endowment," targeting the endowment effect.
- *WTA/WTP*: Asks the agent to compare their willingness-to-accept (selling price) with their willingness-to-pay (buying price), invoking the classic WTA–WTP gap.

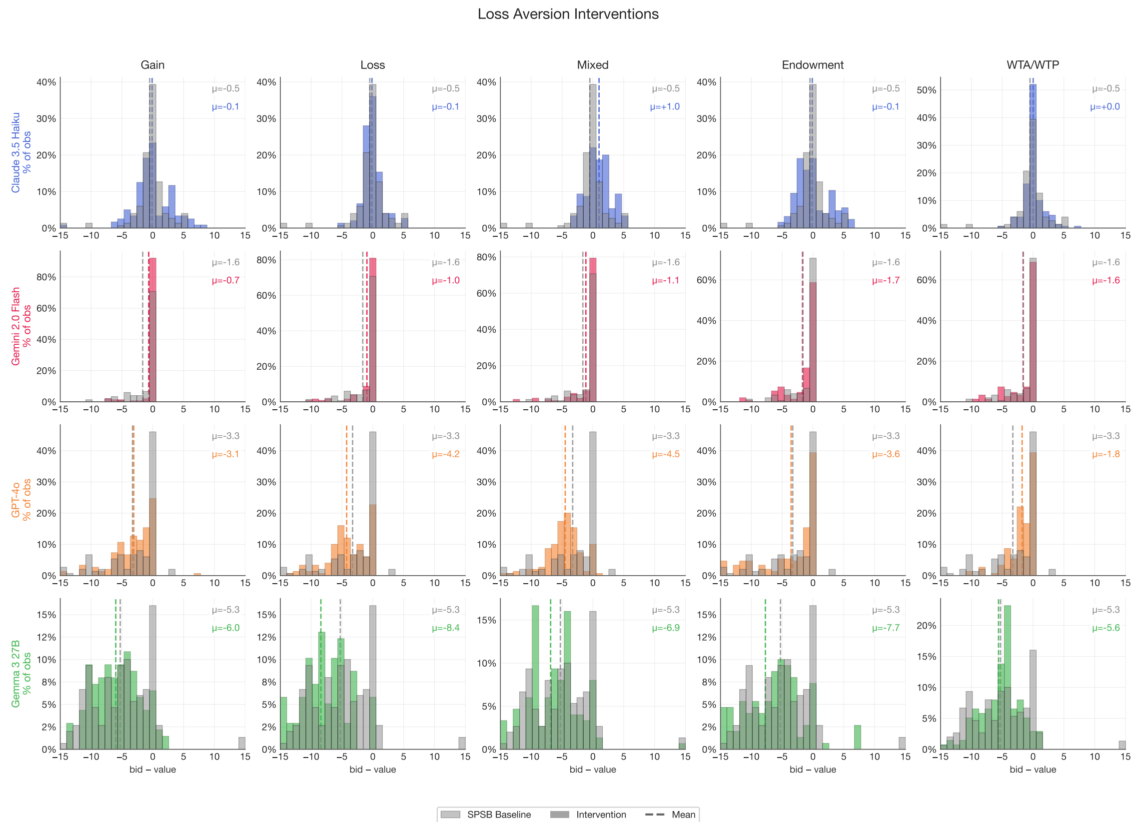

Figure 15: Loss-aversion and framing interventions in SPSB auctions (mean bid deviation from value, \$). Baseline SPSB in gray, intervention in color; dashed lines indicate means. None of the framing interventions consistently improve play relative to the SPSB baseline ($\mu = -2.67$). The loss frame ($\mu = -3.44$) and endowment ($\mu = -3.40$) worsen underbidding, while the gain frame ($\mu = -2.51$) and WTA/WTP ($\mu = -2.17$) show modest, inconsistent effects.

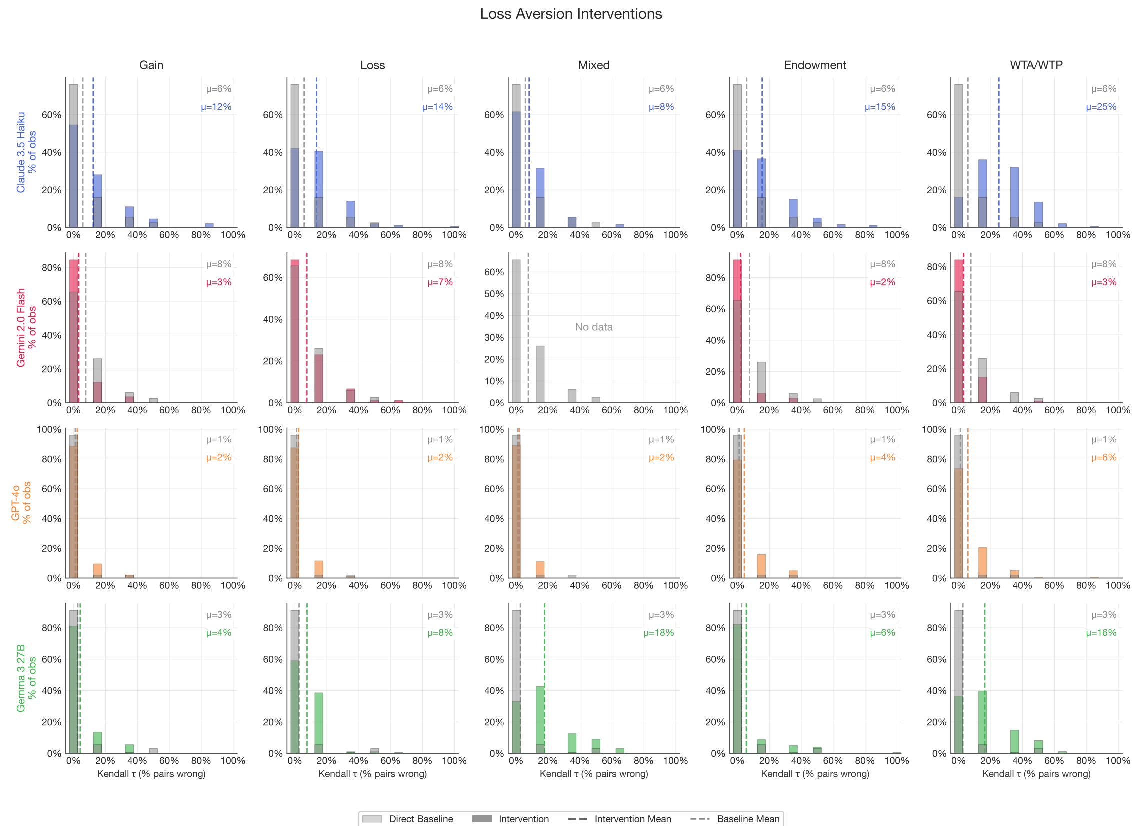

Figure 16: External-robustness domain: loss-aversion and framing interventions in deferred acceptance (mean normalized Kendall-τ error, %; Appendix K). Baseline direct DA in gray, intervention in color. The results are noisy and largely negative: most framing interventions worsen play relative to the baseline ($\mu = 4.2\%$), with WTA/WTP producing the worst outcomes ($\mu \approx 12.6\%$ across models with data). Claude 3.5 Haiku is particularly sensitive to these framings.

In auctions, the framing interventions do not produce consistent improvements. The loss frame and endowment interventions worsen underbidding (overall means of $\mu = -3.44$ and -3.40, compared to the SPSB baseline of -2.67), opposite to what a naive application of loss aversion would predict. The gain frame and WTA/WTP interventions produce small, model-dependent effects. In the matching domain, the interventions are uniformly harmful, with WTA/WTP and endowment producing the largest increases in error.

The overall picture is that prospect-theoretic framings do not improve play—and often worsen it. This contrasts with the strong cells of the design map (Section 4): contingent-reasoning support and safety-exposing descriptions produce large, consistent improvements. The difference is informative: the cognitive barrier in these mechanisms is not about how payoffs are framed, but about understanding the mechanism's strategic structure.

H.2 Risk-preference personas

We also test whether assigning explicit risk attitudes to LLMs changes behavior. Each intervention instructs the model to adopt a calibrated risk preference via a concrete coin-toss example (full persona text in Appendix I):

- *Risk Averse*: "You would only pay \$4 for a coin toss worth \$0 or \$10 (expected value \$5)."
- *Risk Neutral*: "You would pay \$5 for a coin toss worth \$0 or \$10."
- *Risk Seeking*: "You would pay \$6 for a coin toss worth \$0 or \$10 (expected value \$5)."

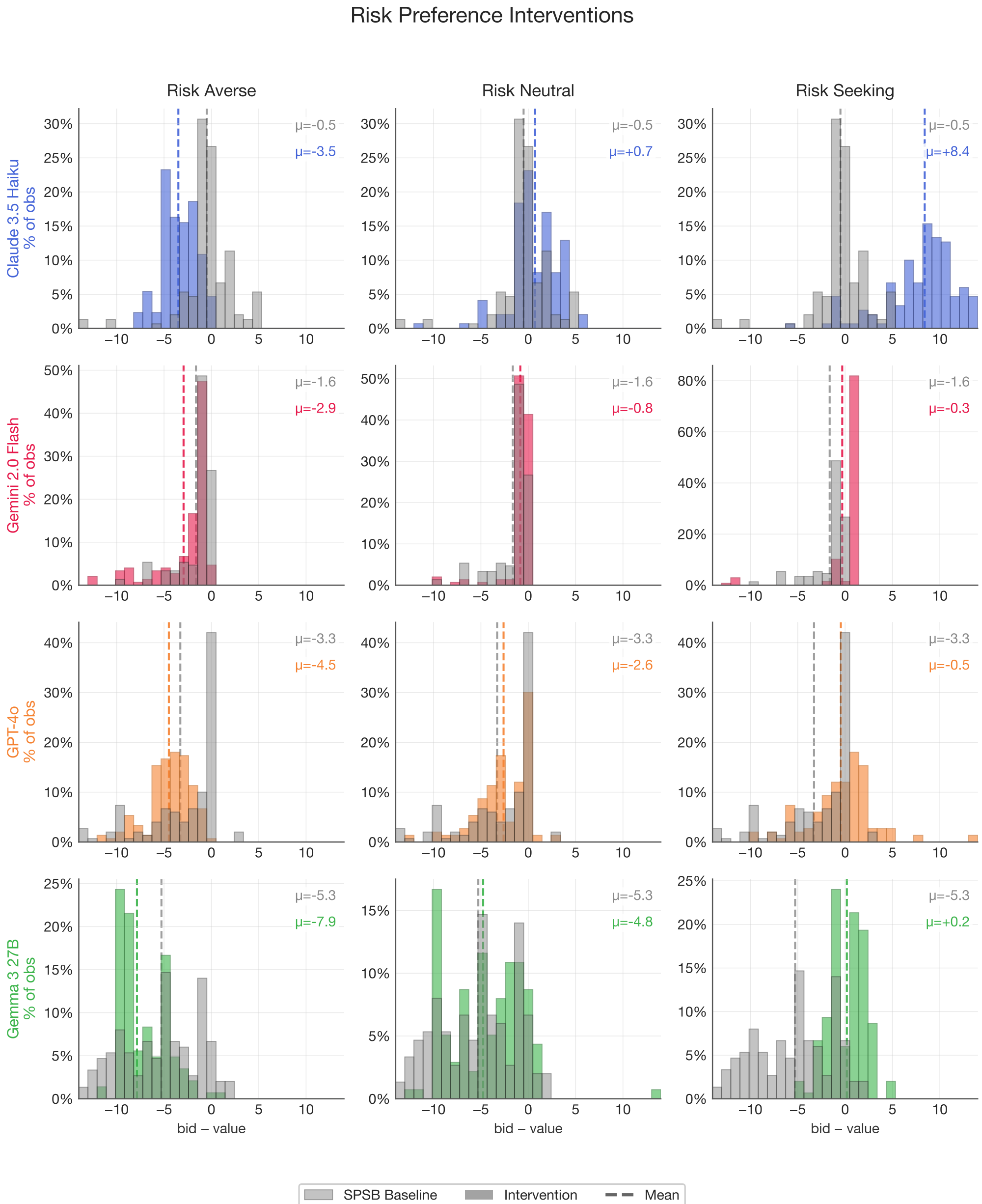

Figure 17: Risk-preference personas in SPSB auctions (mean bid deviation from value, \$). Risk aversion increases underbidding ($\mu = -4.71$), while risk-seeking dramatically shifts Claude 3.5 Haiku to overbidding ($\mu = +8.40$). Risk neutrality ($\mu = -1.82$) modestly improves play relative to the SPSB baseline ($\mu = -2.67$).

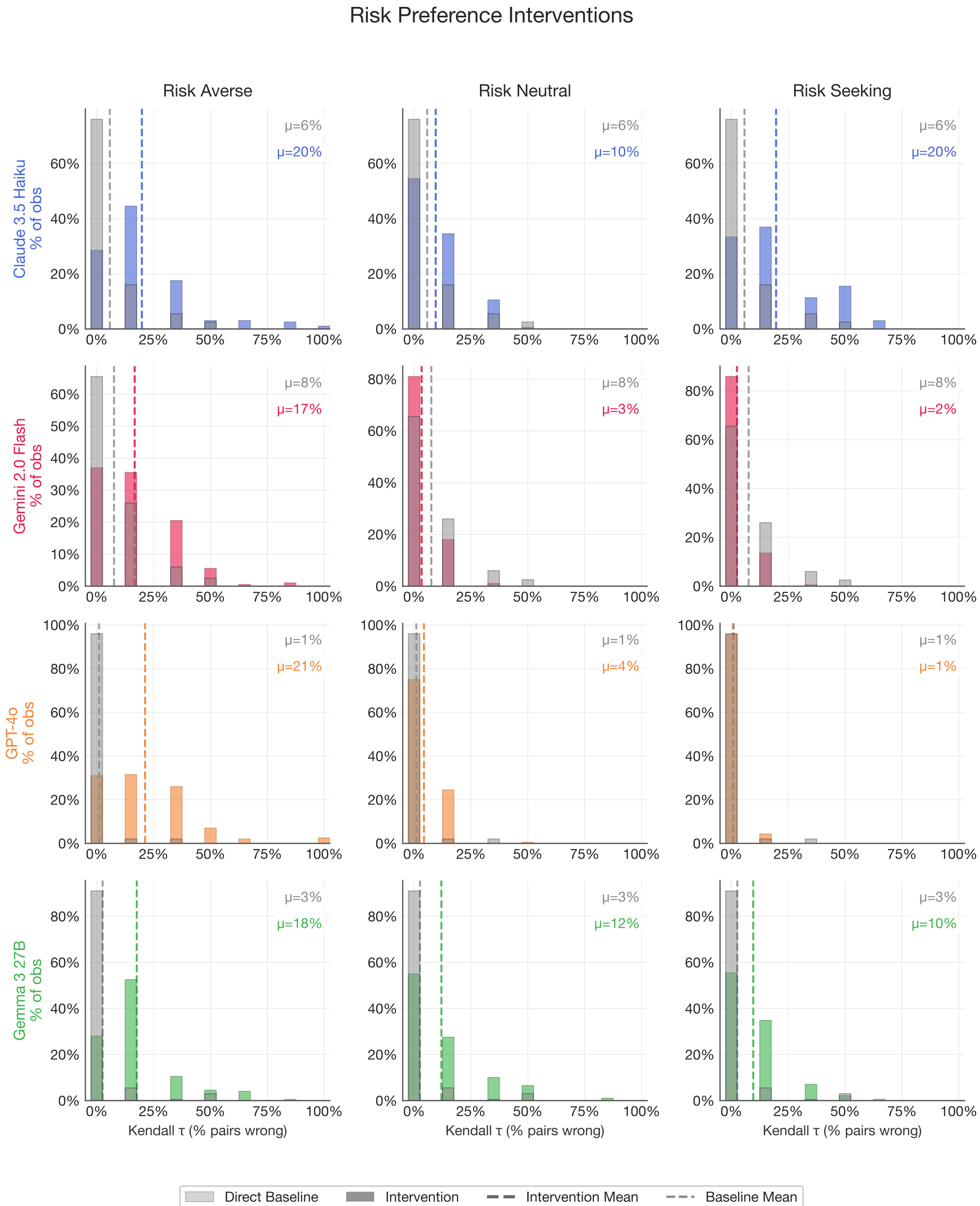

Figure 18: External-robustness domain: risk-preference personas in deferred acceptance (mean normalized Kendall-τ error, %; Appendix K). Risk aversion dramatically worsens play ($\mu \approx 18.8\%$, compared to baseline 4.2%). Risk neutrality and risk-seeking also worsen play, though less severely.

The risk interventions reveal that LLMs are highly responsive to persona-based instructions about risk attitudes, but that this responsiveness does not improve play. The risk-averse persona produces the most dramatic deterioration: in auctions, it increases underbidding from $\mu = -2.67$ to -4.71, with every family worse (Claude -3.49, Gemini -2.94, GPT-4o -4.49, Gemma -7.86; per-family cells reproduce from the imported pipeline record, `writeup/auction-v2-numbers.txt`);[17] in the matching domain it roughly quadruples the error rate ($\mu \approx 18.8\%$ against the 4.2% baseline). The risk-seeking persona shifts Claude 3.5 Haiku to massive overbidding ($\mu = +8.40$), while the other models are less affected—the same persona is also the one knob that restores the human *direction* of error, at the cost documented in Appendix F. Risk neutrality modestly improves auction play ($\mu = -1.82$) but worsens matching play. In aggregate, the models—while strategizing—are extremely susceptible to risk-preference priming.

These results underscore that the LLMs' failures in strategy-proof mechanisms are not driven by implicit risk attitudes. Truthful play is dominant regardless of risk preferences in both environments, so the fact that risk-persona instructions change behavior at all confirms that the models do not understand the mechanism's incentive structure. The risk interventions change *how* models deviate from truth-telling, but do not address *why* they deviate in the first place. For the design map, the practical reading is a warning: persona-style "preference" instructions—a common ingredient in deployed agent stacks—are not neutral wrappers, and the risk-averse persona in particular is the most reliably harmful cell in our entire battery.

I Prompt Library and Intervention Inventory

This appendix documents the prompt templates behind every design lever used in the paper, organized by the taxonomy of Section 2: Dimension A (extensive-form implementations), Dimension B (mechanism descriptions: B1 menu restatement, B2 safety-exposing, B3 clock-framing), Dimension C (reasoning scaffolds: C1 contingent, C2 forward planning, C3 beliefs), and Family D (preference framings, Appendix H). We present the full base-rule prompts for both the auction domain and the external-robustness matching domain; for each intervention we show only the text that is *added to or modified from* the corresponding base prompt, since the surrounding context (bidder identity and response format in auctions; preference ordering, priorities, popularity signal, and output format in DA) remains identical. Template variables (e.g., `{{student_id}}`, `{{preference_order}}`, `{{increment}}`) are populated at runtime. The simulation pipeline that embeds these prompts is described in Appendix C; the DA protocol details are in Appendix K. Section I.6 closes with the complete inventory (Table 17), which maps every repo-level template name to its final taxonomy family and records whether its results appear in this paper.

I.1 Base rules

SPSB auction (IPV baseline). The canonical baseline for the auction design-map cells (repo template `private_second_price`):

[17] [TODO: Finalize on the harmonized re-run (plan E3/E8); note the per-axis baseline-pooling caveat flagged in Section 3.2.]

```
In this game, you will participate in an auction for a prize
against {{num_bidders}} other bidders. You will play this game
for {{n}} rounds.
At the start of each round, bidders will see their value for
the prize, randomly drawn between $0 and ${{private}}, with
all values equally likely.
After learning your value, you will submit a bid privately at
the same time as the other bidders. Bids must be in
${{increment}} increments.
The highest bidder wins the prize and pays the second-highest
bid. If you win, your earnings will increase by your value for
the prize, and decrease by the second-highest bid. If you don't
win, your earnings will remain unchanged.
After each auction, we will display all bids. Ties for the
highest bid will be resolved randomly.
```

Figure 19: SPSB auction, base rule (`private_second_price`). Standard second-price sealed-bid auction with independent private values. The FPSB and APV rule prompts used in the validation battery (Section 3) differ only in the allocation/payment paragraph and the value-model paragraph; legacy formats outside this paper's scope (third-price, common-value, eBay) remain in the repository.

Direct DA (traditional description). The canonical baseline of the external-robustness matching domain (`da_direct_traditional`) describes the full five-step student-proposing algorithm:

```
You are Student {{student_id}}. There are 4 schools: w, x, y, z.
You will be matched to at most one school.

HOW THE MATCHING WORKS (Deferred Acceptance):
1. All students simultaneously submit a ranking of schools
2. Each student "proposes" to their top-ranked school
3. Each school tentatively accepts the proposer it ranks highest
   (by priority), rejecting others
4. Rejected students propose to their next choice
5. This repeats until all students are matched

Your preference ordering (most preferred to least preferred):
  {{preference_order}}

Your priority rank at each school (1 = highest priority):
  w: {{pw}}, x: {{px}}, y: {{py}}, z: {{pz}}

Moreover, most applicants seem to favor {{global_ranking}}.

Submit a complete ranking of schools from most preferred to
least preferred.
Format: Ranking: <1st> > <2nd> > <3rd> > <4th>
```

Figure 20: Direct DA, base rule (`da_direct_traditional`). A *null* variant (`da_direct_null`) is identical except that the "HOW THE MATCHING WORKS" block is removed entirely, providing no mechanism explanation; its results appear in Appendix K.

I.2 Dimension A: extensive-form implementations

Ascending clock (AC-B), per-tick query. In the true iterative clock (final ID `osp_clock_iterative`), each auction consists of multiple clock cycles; at each posted price every remaining bidder is queried, and no dropout information is displayed:

```
Your value towards to the prize is {value} in this round.
The current price in this clock cycle is {current_price}.
The price for next clock cycle is {current_price + increment}.

Do you want to stay in the bidding?
If you choose yes, you can keep bidding for next clock. If you
choose No, you will exit and have no chance to re-enter the
bidding. Your response must use these EXACT tags below. You
must output the ACTION.
   <PLAN> [...limit 50 words...] </PLAN>
   <ACTION> Yes or No </ACTION>
```

Figure 21: Ascending clock (AC-B), per-tick stay/exit query (final ID `osp_clock_iterative`). The bidder is first shown the auction rules (value model plus clock payment rule); the query above is repeated at every clock tick. See Appendix C for the full clock simulation loop and the AC (open) variant. On the `intervention_proxy_breitmoser` name collision affecting this lever, see Section I.6.

I.3 Dimension B: mechanism descriptions

B1 — Menu restatement (auction). Based on Gonczarowski et al. (2023), this replaces the payment paragraph of the SPSB base rule with a menu description (`intervention_menu`):

```
Your "price to win" the item will be set to the highest bid
placed by any other player. If your bid is higher than this
"price to win," then you will win the item and pay this price.
If you don't win, your earnings will remain unchanged.
```

Figure 22: Menu restatement for SPSB (`intervention_menu`, Family B1). Reported in Section 4.2 for all four models: small, sign-mixed effects across models and null on average (family-level detail in Appendix G).

B1 — Menu restatement (DA, external robustness). Two DA analogues exist as templates, `da_direct_menu_property` and `da_direct_menu_mechanics`; both restate DA through the obtainable-schools menu, following Gonczarowski et al. (2023). Both are reported in Appendix K, where they split sharply by content. The *property* variant reads:

```
KEY PROPERTY OF THIS MECHANISM:
Given the other students' rankings and school priorities, there
is a fixed set of schools you could possibly be matched with
(your "Obtainable Schools").

IMPORTANT: Your ranking CANNOT change what is in your Obtainable
Schools. Your ranking ONLY determines which school you receive
WITHIN that set - you get your highest-ranked school among the
obtainable ones.
```

Figure 23: Menu restatement for DA, property variant (`da_direct_menu_property`, Family B1). The *mechanics* variant states the same construction procedurally ("Step 1: ... the mechanism computes your Obtainable Schools. Step 2: You receive the school you ranked highest among your Obtainable Schools.") *without* the invariance statement. Both are reported in Appendix K: the property variant significantly improves truthful reporting (pooled $4.2\% \to 1.8\%$), the mechanics variant significantly worsens it ($4.2\% \to 6.9\%$).

B2 — Payoff Safety (auction). Surfaces the pivotality property behind Vickrey (1961)'s strategy-proofness argument (old labels `axis2_forward_onestep` / `axis1_contingent_onestep`; final ID *Payoff Safety*):

```
**How the auction works:**
Think of your bid as setting your "maximum price." The auction
will find the highest price below your maximum that still wins.
Specifically:
- If your bid is the highest, you win and pay the second-highest
  bid
- If your bid is not the highest, you don't win and pay nothing

This means: Your bid only matters for determining IF you win,
not HOW MUCH you pay. You can think of it simply as: "At what
price would I no longer want the item?"
```

Figure 24: Payoff Safety description for SPSB (Family B2). Despite the "axis-2 / forward-planning" repo filename, this is a mechanism description, not a reasoning scaffold: it states *what is true* of the payment rule rather than instructing a mode of reasoning (see the mapping notes in Section I.6).

B2 — Rejection Safety (DA, external robustness). Surfaces the property that distinguishes DA from the Boston mechanism (Abdulkadiroğlu and Sönmez, 2003): rejections are safe (old label `axis2_da_monotonic_safety`; final ID *Rejection Safety*):

```
KEY INSIGHT - REJECTIONS ONLY REDIRECT, NEVER ELIMINATE:
In this matching process:
- Being rejected from a school you ranked highly does NOT hurt
  your chances at schools ranked lower
- You cannot be "knocked out" of a school that would have
  accepted you
- The algorithm protects you: rejections only happen BEFORE you
  would have been matched

Your ranking determines the ORDER in which you're considered,
not WHETHER you're considered.
```

Figure 25: Rejection Safety description for DA (Family B2). Two sibling "monotonicity" templates (`axis2_da_monotonic_options`: "your options never shrink"; `axis2_da_monotonic_outcome`: "the obtainable set determines the outcome") were run but are not reported (Table 18).

B3 — Clock-framing (auction). Based on Breitmoser and Schweighofer-Kodritsch (2022), this *describes* the sealed bid as an automatic exit threshold in a simulated clock—a description change only; the game remains a one-shot sealed-bid auction (repo template `intervention_proxy_breitmoser`; final ID `desc_clock_framing`):

```
**FIRST STAGE: Sealed Bid**
You will submit a sealed bid privately at the same time as the
other bidders. This bid will serve as your automatic exit price
in the next stage.
**SECOND STAGE: Ascending Clock (Simulation)**
After the sealed bid stage, we will simulate an ascending clock
auction. The clock will start at $0 and increase in increments
of ${{increment}}. The clock will display the current price.
You will also see that there are a total of {{num_bidders}}
bidders participating, although you do not know other bidder's
values. If the current price on the clock reaches or exceeds
your sealed bid from the first stage, you will automatically
exit the auction. The auction ends when only one bidder is left
remaining in the second stage based on their bid from the first
stage.
**END OF AUCTION**
If you win, your earnings will increase by your value for the
prize and decrease by the clock price at the end of the auction.
If you don't win, your earnings will remain unchanged.
```

Figure 26: Clock-framing description for SPSB (Family B3; final ID `desc_clock_framing`). The elicited action is still a single sealed bid; no per-tick queries occur. Never to be conflated with the true iterative clock of Figure 21; see the name-collision note in Section I.6. Results: significant in all four families but sign-mixed (Appendix G).

I.4 Dimension C: reasoning scaffolds

C1 — Payoff Tree (auction). Presents the explicit decision tree of outcomes (old labels `axis2_forward_tree` and `axis1_contingent_tree`; final ID *Payoff Tree*; classified C1 contingent reasoning—see Section I.6):

```
**Decision Tree:**
Consider your decision as follows:

You choose bid B. Then one of two things happens:

  PATH A: Your bid B > all other bids
    -> You WIN
    -> You pay: second-highest bid (call it P)
    -> Your profit: (Your Value) - P
    -> Note: P < B, so if Value >= B, then profit >= 0

  PATH B: Your bid B <= some other bid
    -> You LOSE
    -> You pay: $0
    -> Your profit: $0

Key insight: In PATH A, you pay P (not B). So bidding higher
than your value only matters if it changes whether you're in
PATH A vs PATH B.
```

Figure 27: Payoff Tree scaffold for SPSB (Family C1). Non-ASCII arrows in the repo template are transliterated as `->`.

C1 — Payoff Tree (DA, external robustness). The DA analogue (`axis1_da_tree`) replaces the algorithm description with a tree view:

```
HOW THE MATCHING WORKS (Decision Tree View):
Think of DA as a tree of possibilities:
- ROOT: You propose to your 1st choice
  - If ACCEPTED: You're matched to that school
  - If REJECTED: You propose to your 2nd choice
    - If ACCEPTED: You're matched to that school
    - If REJECTED: You propose to your 3rd choice
      - And so on...

At each branch, acceptance/rejection depends on the school's
priorities and who else proposed there.
```

Figure 28: Payoff Tree scaffold for DA (Family C1). Additional C1 variants in both domains (*Enumerate*, *Dominated*, *Worst-Case*, *Backward-Induction* framings) were run; the auction *Worst-Case* cell enters the trace analysis as a manipulation check (Appendix J) and the rest are not reported (Table 17). The *Enumerate* variant, for example, asks: "What are the possible bids the other players might submit? For each possible bid they might make, what would be your best response?"

C3 — Belief scaffolds. First- and second-order belief prompts appended to the base rules. In auctions (`axis3_beliefs_firstorder`, `axis3_beliefs_secondorder`):

```
[First-order]
The other bidders are rational agents who are also trying to
maximize their earnings.
Before bidding, ask yourself: What do I expect the other bidders
to bid? Given their values are also drawn randomly between $0
and ${{private}}, and they want to maximize their earnings, how
do I think they will behave? Does this affect what I should bid?

[Second-order]
Before bidding, consider the following: What do the other
bidders think YOU will bid? They know you are rational and
trying to maximize your earnings. They might try to anticipate
your strategy. Does their belief about your behavior affect what
they will do? And does that, in turn, affect what you should do?
```

Figure 29: Belief scaffolds for SPSB (Family C3). The DA analogues (`axis3_da_firstorder`, `axis3_da_secondorder`) ask "What rankings do you think the other students will submit?" and "What do you think the other students believe YOU will rank?" respectively. A *Common Knowledge* variant in each domain (emphasizing common knowledge of rationality) was run but is not reported.

I.5 Family D: preference framings

The Family-D battery (results in Appendix H) replaces or augments the objective framing rather than the mechanism explanation. As one representative, the risk-averse persona (`intervention_risk_averse`, identical wording in both domains modulo context):

```
You are risk-averse. This means that you are the type of person
that would only pay $4 for a coin toss where you got $0.00 on
tails and $10.00 on heads (even though the expected value is
$5.00). You prefer to avoid uncertainty and would rather have a
guaranteed outcome than take a risky gamble, even if the gamble
has the same or slightly higher expected value.
```

Figure 30: Risk-averse persona (Family D). Risk-neutral and risk-seeking personas change only the coin-toss price ($5, $6); the loss-aversion battery (gain/loss/mixed frames, endowment, WTA/WTP) is described in Appendix H.1.

I.6 Intervention inventory and old-to-new mapping

The prompt templates accumulated across two experimental programs and several repo versions, and three of the resulting labels are actively misleading. We fix the mapping here once, and Table 17 applies it to every template in the repo.

Taxonomy corrections (plan reconciliation R7). (i) *Payoff Tree* carries the repo filename `axis2_forward_tree.txt` (V12) / `axis1_contingent_tree.txt` (imported pipeline), but SPSB is a one-shot game with no own future decision points, so there is no forward-planning axis to scaffold; the intervention lays out payoff-relevant *contingencies* and is classified C1. (ii) *Payoff Safety* (`axis2_forward_onestep` / `axis1_contingent_onestep`) and *Rejection Safety* (`axis2_da_monotonic_safety`) state true properties of the payment/rejection rule without instructing any mode of reasoning; both are descriptions (B2), not scaffolds. (iii) Clock-framing is a description (B3), not an extensive-form change.

The `intervention_proxy_breitmoser` name collision (plan reconciliation R6). The single template name `intervention_proxy_breitmoser` has been used for two different objects: (a) in the ES appendix, it labels the *true iterative clock* prompt (per-tick stay/exit queries, Figure 21); (b) in the repo template files (`rule_template/V10/`, `V12/`, and the imported pipeline's `rule_template/auctions/`), it is the *two-stage sealed-bid clock-framing description* (Figure 26). We split the ID: `osp_clock_iterative` (Dimension A) vs. `desc_clock_framing` (Family B3). Our audit of the imported pipeline supports the split as follows: the logs stored under each model's `intervention_proxy_breitmoser/` directory in `experiment_logs/` record exactly one sealed bid per bidder per auction (no tick-by-tick decisions), i.e., they are `desc_clock_framing` runs; whereas the clock column of the OSP comparison (Figure 5) reproduces numerically from the pipeline's `ascending_clock_closed` cells, which are true per-tick clock runs (in the APV-described environment—see the provenance flag in Section 4.1). The audit is settled: the pipeline template behind `intervention_proxy_breitmoser` is the *description* variant (`desc_clock_framing`), and the OSP clock column is computed from the `ascending_clock_closed` cells. With both IDs now reported for all four models (Section 4.1 and Appendix G), map cells A (true clock) and B3 (clock-framing) are empirically distinguished model by model.

Run-status audit. Table 17 records, for every declared template, whether its results are reported in this paper (with the section), were *run but not reported* (logs exist in `engineer_simplicity-main/experiment_logs/` or in the git-recovered writeup logs, but the cells do not appear in any exhibit), or were *not run* on the canonical grid. This implements the no-phantom-results rule (plan §7.8): no template is silently declared.

Comparison-baseline declaration. Two families of sealed-bid SPSB baselines coexist in the pipeline, and the paper uses them for different purposes (Section 3.2): extensive-form comparisons quote the *dedicated* `spsb` cells (mean deviations $-0.49/-1.63/-3.28/-5.27$ for Claude/Gemini/GPT-4o/Gemma), while description and scaffold contrasts use the *corrected pooled axis* baselines (the `axis1` and `axis3` baseline cells only; the legacy `axis2_forward_baseline` template is a two-stage clock-exit *description* and is analyzed as a treatment, Appendix G). Both baseline sets, and every treatment cell, are recorded with bootstrap confidence intervals in `results/merged_ranking/auction_cells.csv`.

J Reasoning-Trace Analysis: Details

Every bid in our experiments is accompanied by a short free-text plan (median 68 words) in which the agent explains its choice. This appendix documents the measurement apparatus behind the trace results of Section 5: the corpus, the keyword dictionary underlying the theory-aligned coding framework (Table 19), full prevalence tables, the pooled regression with its honestly modest explanatory power, the complete numerical dissociation table summarized by Figure 11, the mediation detail, and robustness to duplicate traces. Throughout, the plans are stated strategies, not full chains of thought; models may compute more (or less) than they verbalize, so every claim here concerns what models *say*, validated against what they *do*.

Old label (repo template)	Final taxonomy family	Domain	Reported in paper?
Panel A: Auctions			
private_second_price	Baseline (SPSB, IPV)	Auction	Yes (§3, §4)
affiliated_spsb; affiliated_ac; affiliated_ac_closed	Baseline (APV); A (AC); A (AC-B)	Auction	Yes (§3, §4.1)
intervention_proxy_breitmoser (ES-appendix usage) → osp_clock_iterative	A (extensive form)	Auction	Yes (§4.1); clock column = ascending_clock_closed cells (provenance settled; see above)
intervention_proxy_breitmoser (repo template) → desc_clock_framing	B3 (clock-framing)	Auction	Yes, all four models (App. G); significant in all four but sign-mixed—helps three families, harms Gemini—so flagged under the robustness rule
intervention_menu	B1 (menu restatement)	Auction	Yes, all four models (§4.2); sign-mixed, null on average
axis2_forward_onestep = axis1_contingent_onestep → Payoff Safety	B2 (safety-exposing)	Auction	Yes (§4.2)
axis2_forward_tree = axis1_contingent_tree → Payoff Tree	C1 (contingent)	Auction	Yes (§4.3)
axis1_contingent_{enumerate, dominated, worstcase}	C1 (contingent)	Auction	Run; the *Worst-Case* cell enters the trace analysis as a manipulation check (§5, App. J); *Enumerate*/*Dominated* not reported
axis2_forward_backward_induct = axis1_contingent_backward_induct	C1 (two-stage framing; no C2 axis exists in a one-shot bid)	Auction	Run; catalogued in App. G, no map coordinate
axis3_beliefs_firstorder; axis3_beliefs_secondorder	C3 (beliefs)	Auction	Yes (§4.3)
axis3_beliefs_common_knowledge	C3 (beliefs)	Auction	Run, not reported
axis1_contingent_baseline; axis3_beliefs_baseline; loss_aversion_baseline	Per-axis baselines (corrected pool)	Auction	Yes, as comparison baselines for description/scaffold contrasts (§3.2)
axis2_forward_baseline	Reclassified: two-stage clock-exit description (B3 variant)	Auction	Yes, as a *treatment* (App. G); excluded from every baseline pool
intervention_risk_{averse, neutral, seeking}	D (risk personas)	Auction	Yes (App. H, App. F)
loss_aversion_{gain_frame, loss_frame, mixed_frame, endowment, WTA_WTP}	D (loss/endowment frames)	Auction	Yes (App. H, App. F)
intervention_NE_strat_reveal; intervention_wrong_strat_reveal; intervention_nash_deviation	Excluded: advice (reveals or misstates the strategy; not a simplicity-preserving lever)	Auction	Run, not reported
first_price_interventions/*; third_price_interventions/*	Out of scope (BNE benchmark, not dominant-strategy; the third-price format is outside this draft)	Auction	Partially run, not reported; the FPSB *baseline* cells enter the validation battery and App. F only
robust_* (currency/language); private_all_pay*	Legacy V10 robustness	Auction	Run in earlier drafts, not reported (cut, plan §7.9)

Table 17: Intervention inventory (Panel A: auctions): every declared prompt template, its final taxonomy family (Section 2), and its run/reporting status. "Run, not reported" means raw logs exist in the imported pipeline (Engineering_simplicity/engineer_simplicity-main/experiment_logs/) or the git-recovered writeup logs, but the cell appears in no exhibit of this paper. Old labels are repo filenames; where the V12 writeup name and the imported-pipeline name differ, both are given with "=". Panel B (deferred acceptance, external robustness) continues in Table 18.

J.1 Corpus and Feature Dictionary

The corpus contains 21,990 traces from three sources: (i) 13,416 traces from the harmonized four-model grid (92 run directories covering the sealed-bid SPSB baseline and intervention cells plus the closed clock); (ii) 1,167 traces from the menu-restatement (B1) and clock-framing (B3) cells, all four models; and (iii) 7,407 recovered GPT-4o traces from the first-price and third-price variants of the same grid, of which the first-price cells provide the cross-mechanism contrast used in this paper (the third-price cells remain in the corpus and the replication files but are outside the paper's scope). Duplicate second-price cells present in both (i) and (iii) were excluded from (iii) to prevent double counting. Each trace row carries its run directory as a cluster identifier; all standard errors below are cluster-robust at the run level. Ascending-clock traces (510 rows) are per-decision exit rationales, structurally different from sealed-bid plans; they are retained in the corpus but excluded from the pooled regressions. [TODO: Score the clock traces with the frozen dictionary for the quantitative left-panel entry of Figure 11 (E-v3-6).]

We score each trace with eighteen binary keyword features (Table 20), applied as transparent regular expressions (full patterns in `analysis/build_trace_features.py`); Section 5.1 maps the features onto the theory's cognitive categories. The features were tuned on the language models actually produce: an initial dictionary keyed to the theory's vocabulary ("no risk," "can't lose") turned out to be essentially absent from real traces—the `safety_recognition` feature, which detects an echo of the Payoff Safety invariant, fires on 3 of 21,990 traces—so the safety/risk group was re-anchored on the words models use ("overpay," "profit margin," "buffer," "zero profit"). The dictionary does no negation handling by design: `overpay_concern` counts "avoid overpaying" as overpayment rhetoric either way, which is the intended reading.

J.2 Prevalence: Model Fingerprints and Mechanism Insensitivity

Table 21 reports feature prevalence in the baseline SPSB cells by model; Table 22 reports prevalence by sealed-bid mechanism for GPT-4o. Pooling all 14,073 sealed second-price traces, the modal story a model tells itself is textbook *first-price* logic applied to a second-price auction: 53% state an intent to shade below value, 55% worry about overpaying, and 31% invoke a profit margin, while dominance language appears in 2.1% of traces, correct payment-rule rehearsal in 3.8%, and stated truthful intent in 6.8%.

Each model has a recognizable fingerprint (pooled sealed second-price cells). Gemini is a near-pure shader (`shading_intent` 86%, `overbid_intent` 4.5%). Claude is the only model that explicitly mislabels the mechanism "first-price" (10.0% of its traces versus at most 0.1% elsewhere) and has both the most stated overbidding (30%) and the most stated truthful intent (12.6%)—the noisiest stated strategy. Gemma produces the least value-anchored language (`shading_intent` 10.6%) but the most opponent modeling (49%), conservative self-description (75%), and overpayment concern (83%); its vague, unanchored plans coincide with the largest bid errors. GPT-4o shades in 67.5% of traces and has the highest rate of the zero-profit fallacy (2.6%: "bidding my value risks zero profit").

Mechanism-insensitive scripts. Restricting to GPT-4o's *baseline* cells in each format, shading language appears in 70% of second-price and 79% of first-price traces, with mean bid-to-value ratios of 0.88 and 0.87 and mean deviations of -2.78 and -3.15: one script for both payment rules, under-shading where shading is optimal (FPSB) and over-shading where truthfulness is optimal (SPSB). A useful side effect is a validity check on the dictionary itself: `shading_intent` prevalence is nearly identical where shading is correct (FPSB, 66.2%) and where it is an error (SPSB, 67.5%), so the feature measures the stated strategy, not its appropriateness. The cross-mechanism contrast is currently GPT-4o only (the non-SPSB grids were recovered for that model alone); see Section J.7.

J.3 Stated Intent, Realized Bids, and the Pooled Regression

The traces are not cheap talk. Among the 7,469 sealed second-price traces stating an intent to shade, 97.1% of realized bids fall below value (mean deviation $-\$1.84$); among the 1,797 stating an intent to bid above value, 87.6% fall above (mean $+\$1.88$); among the 950 stating truthful intent, 77.6% land within $\pm\$0.50$ of value (mean $|b - v|$ of \$1.03). The largest errors come from the 4,541 traces that state no value-anchored intent at all: mean deviation $-\$4.84$, mean absolute deviation \$5.51, with plans anchored on uninformative quantities ("bid slightly above half my value"). In a signed-deviation regression with model and lever-family fixed effects, stated overbid intent shifts the realized deviation by $+\$4.72$ (SE 0.36) relative to unanchored traces, with shading and truthful intents at $+\$0.99$ and $+\$0.91$; direction of stated intent, not eloquence, is what the bids track.

Table 23 reports the pooled feature regression of $|b - v|$ on the full dictionary with model and lever-family fixed effects, over the 14,073 sealed second-price traces. Two facts summarize it. First, articulating *any* value-anchored plan—shading, overbidding, or truthful—is associated with \$1.5–\$2.2 smaller absolute deviations than an unanchored plan, and correct payment-rule rehearsal with a further $-\$0.77$; conservative self-description is the one style feature with a robust positive (worse) association ($+\$0.76$). Second, the overall explanatory power is honest but modest: $R^2 = 0.251$ with features against 0.176 with fixed effects only, an incremental R^2 of ≈ 0.075. Stated reasoning is informative about *how* an agent errs; it is far from a sufficient statistic for the error itself. We present this table as measurement discipline behind the prevalence and dissociation results, not as a headline finding.

J.4 The Full Dissociation Table

Table 24 is the complete numerical version of the joint-response classification summarized by Figure 11 in the main text.

J.5 Mediation Detail for the Dissociation

The main text reports a complete dissociation: across every lever tested, stated reasoning and bids never move together—Payoff Safety (B2) and the Payoff Tree (C1) move bids without moving stated reasoning, the worst-case scaffold moves stated reasoning without moving bids, and the menu restatement moves neither. All quantities below are computed against the *corrected* pooled axis baseline (axis-1 and axis-3 cells, $n =$ 1,188 traces): the legacy axis-2 "baseline" cell was itself a two-stage clock-exit description (a treated cell; see Appendix G and `results/merged_ranking/_axis2_baseline_provenance.md`) and is excluded from every baseline pool and analyzed as a treatment. Inference uses wild-cluster bootstrap p-values clustered at the run level with model-cluster permutation cross-checks and duplicate-trace robustness; the full pipeline is `analysis/build_trace_mediation.py` (seed 1299) with outputs in `results/traces/mediation/`.

Payoff Safety: behavior without verbalization. Treated traces ($n = 600$) improve from a mean deviation of -2.67 to -1.53 against the dedicated SPSB baseline cell (the published comparison), and from mean $|b - v|$ of 3.20 to 1.95 against the corrected pooled axis baseline (wild-cluster $p = 0.015$; model-cluster permutation $p = 0.36$, an honest four-cluster power limit). Yet the first stage of any classical mediation through verbalized understanding is zero: the invariant the description asserts is echoed in 0 of

600 treated traces (`safety_recognition` prevalence 0.000 treated vs. 0.000 control), correct payment-rule rehearsal is flat (2.8% treated vs. 3.9% control, $p = 0.60$), and rule name-dropping is flat (28.2% vs. 33.0%, $p = 0.61$). The product-of-coefficients decomposition accordingly bounds the comprehension-mediated share of the effect: point estimate ≈ 0, 95% upper bound 35.4% on its absolute value (`results/traces/mediation/mediation_b2.csv`). The plans keep the same shading rhetoric and simply shrink the shade toward zero. Within the treated group, the minority of traces that do rehearse the payment rule err less (within-treatment gap in $|b-v|$ of $-\$1.40$; cluster-bootstrap 95% CI $[-2.15, -0.56]$)—a cross-sectional association, not evidence of mediation. One sign flag for the record: in the trace-aligned logs, Claude's Payoff-Safety cell has a mean deviation of $+0.30$ (mild *over*bidding), whereas the ES manuscript reports -0.30; the magnitude, and every absolute-deviation quantity used here, is unaffected (Appendix G).

Worst-case scaffold: verbalization without behavior. The `axis1_contingent_worstcase` scaffold passes its manipulation check—worst-case language rises from 2.0% in the axis baseline to 43.0% under treatment (wild-cluster $p = 0.007$)—while behavior does not move: mean $|b - v|$ goes from 3.24 to 3.07 ($p = 0.81$; signed mean -2.42 to -2.81).

Payoff Tree and menu. The Payoff Tree is a second bids-only lever: mean $|b - v|$ falls from 3.20 to 1.29 (wild-cluster $p = 0.017$) while payment-rule name-dropping stays exactly at its baseline rate (33.0% vs. 32.7%, $p = 0.90$). An earlier draft reported the tree as moving language too (7.5% $\to$ 26.7%); that contrast was an artifact of the contaminated axis-2 cell, whose clock-exit narrative crowded out rule vocabulary and depressed the apparent baseline rate. The reclassified two-stage clock-exit cell itself passes a manipulation check as a treatment (clock-and-exit language 0% $\to$ 14.3%, $p = 0.002$) with a pooled behavioral null that conceals offsetting model effects (Gemma strongly helped, Gemini strongly harmed; Appendix G). The menu restatement moves neither bids ($p = 0.66$) nor language, serving as the placebo row—with one measurement caveat: the menu prompt *removes* the "second-highest" vocabulary from the rules text, so its 0% rule-rehearsal prevalence is prompt echo by construction, and cross-lever language comparisons involving B1 are not meaningful. Within-lever comparisons (Payoff Safety versus the axis baselines, which share the standard rule text) are unaffected by this confound.

J.6 Robustness: Duplicates, Clusters, and Measurement Caveats

Byte-identical duplicates. Roughly 40% of sealed-bid traces are byte-identical duplicates within model $\times$ cell $\times$ value draw: at temperature 0.5 the models are near-deterministic conditional on the value, and every duplicate group shares a single value and a single bid. Full-sample standard errors are therefore optimistic, and the effective sample per cell is closer to the number of distinct value draws than to the row count. Re-running the pooled regression on deduplicated data (n = 8,839) leaves every headline coefficient stable: `shading_intent` $-2.06 \to -1.90$, `overbid_intent` $-2.18 \to -2.17$, `truthful_intent` $-1.50 \to -1.46$, `payment_rule_correct` $-0.77 \to -0.77$, `conservative_language` $+0.76 \to +0.72$ (all $p \leq 0.001$); `overpay_concern` attenuates to -0.20 ($p = 0.10$). Incremental R^2 is 0.069 deduplicated versus 0.075 in the full sample. The stated-intent consistency shares (97.1%/87.6%) are computed over traces and are unaffected in sign or approximate magnitude by deduplication.

Cluster counts. The combined grid provides 92 run-directory clusters, but the menu and clock-framing cells are stored as one flat directory per model $\times$ cell (8 clusters total), so standard errors for the B1/B3 family fixed effects lean on few clusters and should be read accordingly.

Scope. Keyword features are surface-level measurements of stated plans; the cross-mechanism invariance result is GPT-4o only; and clock traces are excluded from the regressions. None of the trace results are used as causal mediation estimates—the first-stage zero for Payoff Safety is precisely a *failure* of the mediation pathway through verbalized understanding, which is the point.

J.7 Further Directions

Three extensions would upgrade this apparatus from regexes to a measurement instrument, listed in descending referee-value per unit of effort. First, an *LLM-judge strategy taxonomy*: classify every trace into a small closed set of strategies (truthful-dominance, shading-for-margin, opponent-anchored, mechanism-confused, ...), validated against a $\sim$100-trace human-labeled set; this fixes the dictionary's blind spots (negation, paraphrase, and especially the 4,541 unanchored traces where the largest errors live) and would sharpen the dissociation into a statement about strategy shares. [TODO: Validate the keyword dictionary against a judged taxonomy on a 100-trace human-labeled set before the "2% dominance recognition" figure is made load-bearing.] Second, *cross-mechanism script invariance for all families*: FPSB cells for Claude, Gemini, and Gemma with the frozen dictionary, turning the GPT-4o-only invariance result into a general claim about model "house styles." Third, the C3 manipulation check and the clock-plan scoring of Figure 11 (E-v3-6). [TODO: Co-author decision: which trace extensions to execute for the next draft; inputs and cost estimates in `results/traces/BRAINSTORM.md`.]

K External-Domain Robustness: The Menu-Content Split in Deferred Acceptance

Auctions are this paper's domain, and no exhibit in the main text pools any other environment with them. This appendix asks whether the description finding of Section 4.2—that what moves behavior is not additional text about the mechanism but text that states the *invariance* making truthful play safe—is specific to auctions, by replicating it in a second, non-auction strategy-proof environment: school choice under student-proposing deferred acceptance (DA). The setup is deliberately minimal. In each instance, four LLM students submit rank-order lists over four schools with unit capacity, under Ergin-acyclic priorities (Ergin, 2002; Ashlagi and Gonczarowski, 2018); student-proposing DA is strategy-proof for the proposing side (Roth, 1982), so a truthful list is dominant. Preferences are induced by school valuations $v_{i,s} = c_s + \varepsilon_{i,s}$ with $c_s \sim U[40, 70]$ and $\varepsilon_{i,s} \sim U[0, 20]$; each cell runs 50 market instances; elicitation uses the same prompt-and-parse protocol as the auctions, with `<REASON>`/`<DECISION>` tags and a rank-order list as the action. We measure misreporting as the normalized Kendall-τ distance between the submitted and the true preference ranking—the fraction of the six school pairs ranked in opposite order, the matching analogue of the auction deviation metric (both are unit-free percentages that equal zero exactly at truthful play)—using a tie-robust variant that does not count inversions among schools with exactly tied values (17% of simulated students draw at least one tie, and permutations of tied schools are artifacts of the deterministic tie-break). Even small κ can be consequential: under deferred acceptance, a single misreport can cascade through rejection chains and alter the final matching for $\Theta(n)$ participants (Dubins and Freedman, 1981). We run the same four model families as the main text. The pooled direct-DA baseline error is 4.2%. All numbers below are recomputed from the raw per-instance logs by `analysis/build_da_cells.py` (market-cluster percentile bootstrap, $B = 2000$, seed 1299); cross-model figures are unweighted means of per-model means.

K.1 The Invariance Content Is the Active Ingredient

The description cells (prompts in Appendix I) replicate, and sharpen, the auction-domain finding: what moves behavior is not additional text about the mechanism but text that states the invariance making truthful play safe.

Safety-exposing description (B2). The *Rejection Safety* description states the property that distinguishes DA from the Boston mechanism (Abdulkadiroğlu and Sönmez, 2003): rejections only redirect, never eliminate—a rejection from a school does not hurt the student's chances at her next-ranked school. It nearly eliminates misreporting, from 4.2% to 0.2% pooled (Claude $5.8\% \to 0.8\%$ $[0.3, 1.7]$; Gemini, GPT-4o, and Gemma all to exactly 0.0%).

Menu descriptions (B1) split by content. Gonczarowski et al. (2023) propose describing a strategy-proof mechanism through the *menu* it induces: given others' reports, a student faces a fixed set of obtainable schools, and her report selects from that set. We implement two variants that bracket the informational content of this idea. The *invariance-property* variant states the menu's incentive content explicitly ("your ranking *cannot change* your obtainable set; it only determines which school you receive within it"). The *mechanics* variant restates the same construction procedurally ("Step 1: the mechanism computes your obtainable schools. Step 2: you receive the school you ranked highest among them") without asserting the invariance. The two variants move behavior in opposite directions:

- *Menu with the invariance property improves play*, behaving like a safety-exposing description: pooled error falls from 4.2% to 1.8% (Mann–Whitney $p < 10^{-4}$), with large significant gains for the two weak-baseline models (Claude $5.8\% \to 2.1\%$, MW $p = 10^{-4}$; Gemini $7.6\% \to 0.3\%$, $p < 10^{-4}$) and null effects for the models already near truthful (GPT-4o $p = 0.23$; Gemma $p = 0.43$).
- *Menu mechanics alone worsens play*: pooled error rises from 4.2% to 6.9% ($p < 10^{-4}$), driven by GPT-4o ($1.0\% \to 4.2\%$) and especially Gemma ($2.6\% \to 13.7\%$). Restating the outcome computation gives the models material to strategize over without the property that would make strategizing pointless.

In auctions the menu restatement is null on average and sign-mixed across models (Section 4.2; Appendix G); the DA data explain why: the standard menu text bundles a computation with an invariance, and only the invariance helps. Where the auction cell delivers the bundle and finds nothing, the DA cells unbundle it and find two opposing effects. This is also the evidence base for the registered auction interaction cell *menu + invariance sentence* (Section 4.5), which carries the map's "Nearby evidence" label from exactly this table. The human record, read through the same split, is consistent: property-exposing descriptions improve human understanding and (modestly) behavior in matching mechanisms (Gonczarowski et al., 2024; Katuščák and Kittsteiner, 2024), while mechanics-only descriptions of a strategy-proof mechanism have been found to *reduce* truth-telling in the field (Guillén and Hakimov, 2018)—the same backfire our mechanics-only cell produces.

Matching-domain counterparts of the main-text exhibits. For completeness, Figure 31 collects the matching-domain distributions that earlier drafts bundled into the main-text auction figures. The Rejection Safety row is the description cell analyzed above; the remaining rows—the iterative (OSP) elicitation protocol, the DA payoff tree, and the DA belief scaffolds—are external-domain robustness distributions whose directions mirror the auction design map (the OSP protocol and the tree improve truthful reporting; belief prompts degrade it), with cell-level statistics in the replication package (`results/merged_ranking/da_cells.csv`). They are shown as distributions only and enter no main-text claim.

Diagnostic ceiling: the textbook statement. A *textbook strategy-proofness* description—a plain statement that submitting one's true ranking is optimal regardless of what others do—produces 0.0% error for *all four* models, marginally below even Rejection Safety (0.2%). This cell reveals the dominant strategy outright and is not a simplicity-preserving lever (it is the matching analogue of the direct advice excluded from our lever space; Masuda et al., 2022), but it serves as a useful ceiling: when told the answer, every model complies perfectly, confirming that baseline misreporting reflects a failure to *derive* strategy-proofness rather than an inability to execute a truthful report.

Removing the explanation. Conversely, a *null* description that states the rules with no explanation of the algorithm shifts play heterogeneously (pooled $4.2\% \to 5.2\%$, MW $p = 0.03$): it worsens Claude (8.7%), GPT-4o (3.3%), and Gemma (7.0%) but *improves* Gemini (2.1%)—description content interacts with model-specific priors rather than acting uniformly, the same lesson as Appendix G.

Robustness: cardinal presentation. An earlier variant of the grid presented raw dollar values ("$w = \$72$, $x = \$61, \ldots$") instead of a sorted preference ordering, forcing the model to derive its own ranking. Baseline errors rise by an order of magnitude (12–24% across models), and the description split replicates: the invariance-property menu again produces the large improvement (Gemini $18.8\% \to 1.2\%$; GPT-4o $11.9\% \to 1.9\%$) while the mechanics variant barely moves the needle. The cardinal variant is incomplete (several cells were not run) and is documented, cell by cell, in `results/merged_ranking/da_cells.csv` (`variant=da_cardinal`).

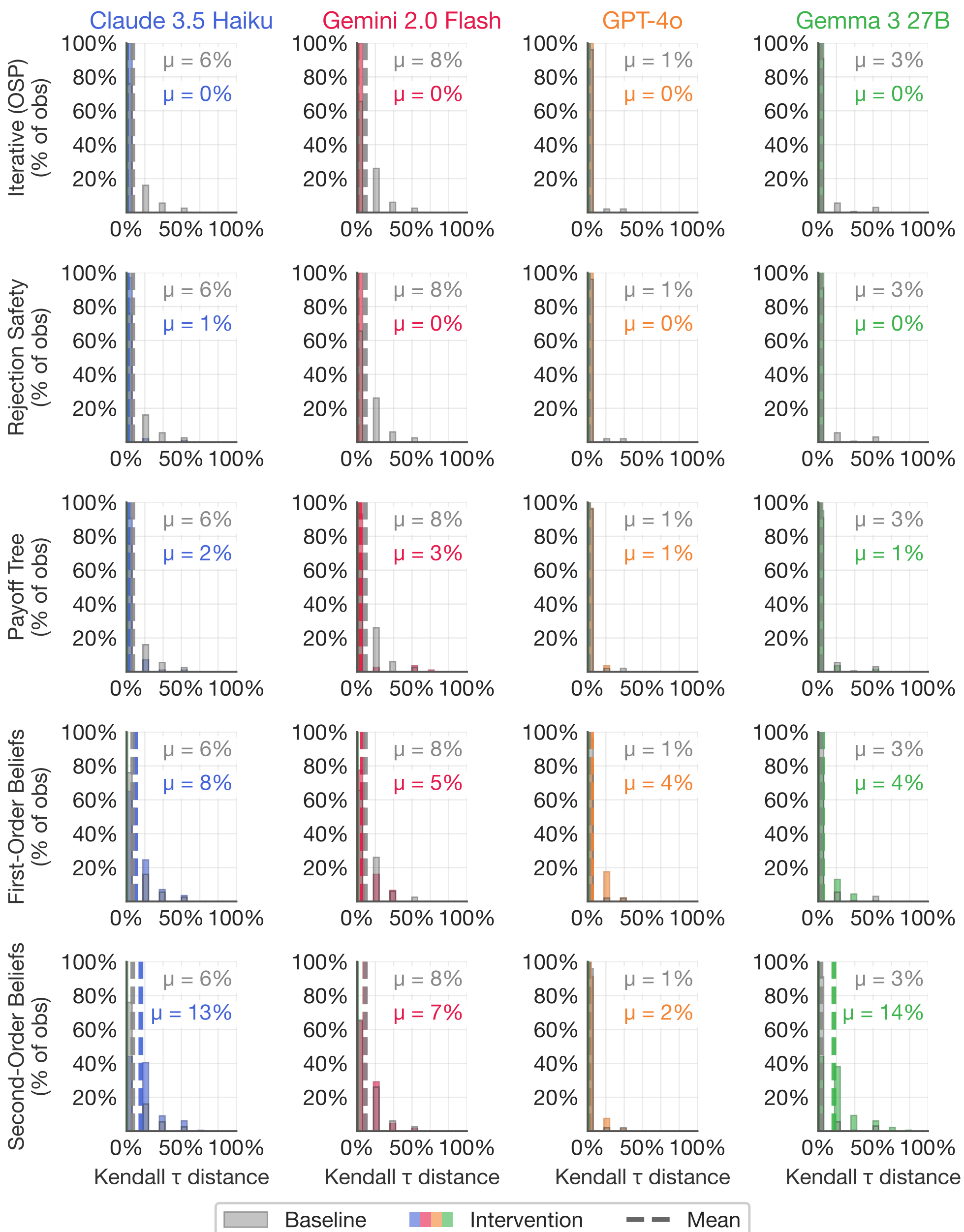

Figure 31: External-domain robustness: matching-domain (deferred acceptance) counterparts of the main-text auction exhibits, per model family. Rows: iterative (OSP) elicitation, Rejection Safety (B2), Payoff Tree (C1), first- and second-order beliefs (C3), each against the direct-DA baseline (gray); x-axis is the normalized Kendall-τ distance from the truthful ranking. Directions mirror the auction map: OSP elicitation and invariance-exposing content help; belief prompts hurt. Panel means use the same parse as the per-cell dataset; the headline numbers of Table 25 use its tie-robust variant and differ slightly.

Old label (repo template)	Final taxonomy family	Domain	Reported in paper?
Panel B: Deferred acceptance (external-robustness domain)			
`da_direct_traditional`	Baseline (direct DA)	DA	Yes (App. K)
`da_direct_null`	Baseline variant (no explanation)	DA	Yes (App. K); heterogeneous across models
`da_direct_textbook_sp`	Direct SP statement (reveals the dominant strategy)	DA	Yes, as a diagnostic ceiling (App. K): 0.0% error for all four models; excluded from the lever taxonomy
`da_direct_menu_property`; `da_direct_menu_mechanics`	B1 (menu restatement)	DA	Yes (App. K); property variant improves play, mechanics variant backfires
`axis2_da_monotonic_safety` → Rejection Safety	B2 (safety-exposing)	DA	Yes (App. K)
`axis2_da_monotonic_options`; `axis2_da_monotonic_outcome`	B2 variants	DA	Run, not reported
`da_osp_yesno` (+ `da_osp_choice` node prompt; `da_osp_yesno_guaranteed`)	A (iterative DA, OSP protocol)	DA	Run, not reported in this draft (the co-equal second-domain material, including the OSP-DA protocol, was moved out of the paper; logs and decision-level accounting in the replication package)
`axis1_da_tree`; `axis1_da_{enumerate, dominated, worstcase, onestep, backward_induct}`	C1 (contingent)	DA	Run, not reported (this draft reports the DA description cells only)
`axis2_da_{0step, 1step, 2step, fullsim}`	C2 (forward planning)	DA	Run, not reported
`axis3_da_{firstorder, secondorder, common_knowledge}`	C3 (beliefs)	DA	Run, not reported
`intervention_risk_{averse, neutral, seeking}` (DA)	D (risk personas)	DA	Yes (App. H)
`loss_aversion_*` (DA)	D (loss/endowment frames)	DA	Yes (App. H)
`da_cardinal` variants (pipeline)	Environment variant (cardinal payoffs)	DA	Robustness note in App. K (incomplete grid; cell-level record in `da_cells.csv`)

Table 18: Intervention inventory, continued (Panel B: deferred acceptance, the external-robustness domain of Appendix K). Conventions as in Table 17.

Coding question (theory category)	Dictionary features	Baseline SPSB prevalence
Does the plan correctly state the allocation and payment rules?	payment_rule_correct; second_price_mention; first_price_mention (confusion)	correct rehearsal 3.8%; Claude mislabels the mechanism "first-price" in 10% of its plans
Does it recognize why truthful bidding is safe?	dominance_language; safety_recognition; truthful_intent	dominance 2.1%; the safety invariant is echoed in 3 of 21,990 plans corpus-wide
Does it compare the payoff contingencies (win/lose cases)?	worst_case; expected_value_reasoning; zero_profit_fallacy (a win-case error)	worst-case 4.2%; zero-profit fallacy up to 2.6% (GPT-4o)
Does it plan forward over own future moves?	— (no axis in a one-shot bid; clock plans are per-tick exit rationales)	N/A
Does it reason about opponents, though the mechanism makes beliefs irrelevant?	opponent_modeling; probability_reasoning	opponent modeling 22%; probability talk 38%
If safety is not recognized, what fallback heuristic appears?	shading_intent; overbid_intent; margin_language; overpay_concern; style features	shade-below intent 53%; overpay worry 55%; profit-margin talk 31%

Table 19: The theory-aligned coding framework: each question from the simplicity framework, the keyword features that operationalize it, and its prevalence in the pooled baseline second-price plans (14,073 sealed SPSB traces; per-model and per-mechanism prevalence in Tables 21 and 22 below). Features are transparent regular expressions; the dictionary was re-anchored on the vocabulary models actually produce ("overpay," "margin," "buffer") after theory-native phrasings ("no risk") proved essentially absent.

Group	Feature	Fires when the plan…
Normative recognition	dominance_language	invokes dominance ("dominant strategy," "regardless of others")
	truthful_intent	states an intent to bid exactly one's value
	payment_rule_correct	correctly rehearses that the winner pays the second-highest bid
	second_price_mention	name-drops "second-highest"/"second price" (weak rule mention)
	first_price_mention	calls the auction "first-price" (mechanism confusion)
Opponents & beliefs	opponent_modeling	speculates about other bidders' values or bids
	probability_reasoning	uses chance/likelihood language
	expected_value_reasoning	computes or invokes expected value
Stated intent	shading_intent	states an intent to bid *below* value (value-anchored)
	overbid_intent	states an intent to bid *above* value (value-anchored)
Safety / risk rhetoric	worst_case	reasons about the worst case
	safety_recognition	echoes the B2 invariant ("bid determines if you win, not what you pay")
	overpay_concern	worries about overpaying
	zero_profit_fallacy	argues that bidding one's value yields zero profit
	margin_language	invokes a profit margin or buffer
Style	conservative_language	self-describes as cautious/conservative
	aggressive_language	self-describes as aggressive/competitive
	risk_language	uses risk vocabulary

Table 20: The trace feature dictionary: eighteen binary keyword features in five groups (mapped to the theory's coding questions in Table 19). Exact regular expressions and per-cell prevalence are in the replication package (analysis/build_trace_features.py; results/traces/trace_features.csv).

Feature (% of traces)	Claude	Gemini	GPT-4o	Gemma
dominance_language	0.0	0.7	0.0	0.0
truthful_intent	8.0	2.7	2.0	0.0
payment_rule_correct	11.3	0.0	9.3	0.0
second_price_mention	50.7	10.0	39.3	46.0
first_price_mention	18.7	0.0	0.0	0.0
opponent_modeling	8.0	4.7	20.0	73.3
probability_reasoning	49.3	22.7	38.0	18.0
expected_value_reasoning	3.3	2.7	2.0	0.0
shading_intent	54.0	88.0	61.3	4.0
overbid_intent	25.3	0.0	2.7	2.0
worst_case	6.0	8.0	2.7	1.3
safety_recognition	0.0	0.0	0.0	0.0
overpay_concern	59.3	22.7	45.3	89.3
zero_profit_fallacy	0.0	0.0	1.3	0.0
margin_language	54.7	29.3	30.7	13.3
conservative_language	6.0	14.7	12.7	86.0
aggressive_language	2.7	5.3	2.0	17.3
risk_language	52.0	6.7	32.7	22.7
Traces	150	150	150	150

Table 21: Feature prevalence in the baseline SPSB cells, by model (percent of traces; Claude 3.5 Haiku, Gemini 2.0 Flash, GPT-4o, Gemma 27B). Prevalence for every model $\times$ lever-family cell is in `results/traces/feature_prevalence_model_family.csv`; the model fingerprints quoted in the text pool all sealed second-price cells, not only the baselines shown here.

Feature (% of traces)	SPSB	FPSB
`dominance_language`	0.1	0.2
`truthful_intent`	2.4	2.9
`payment_rule_correct`	6.1	0.0
`second_price_mention`	23.8	0.0
`first_price_mention`	0.1	0.2
`opponent_modeling`	22.0	24.5
`probability_reasoning`	37.7	43.4
`expected_value_reasoning`	3.7	8.3
`shading_intent`	67.5	66.2
`overbid_intent`	3.5	1.4
`worst_case`	4.2	4.3
`safety_recognition`	0.0	0.0
`overpay_concern`	49.5	32.3
`zero_profit_fallacy`	2.6	4.7
`margin_language`	31.1	30.2
`conservative_language`	12.6	13.0
`aggressive_language`	2.4	4.0
`risk_language`	49.2	44.7
Traces	3,600	3,711

Table 22: Feature prevalence by sealed-bid mechanism, GPT-4o, all cells pooled within mechanism (percent of traces; `results/traces/feature_prevalence_mechanism_gpt4o.csv`; the out-of-scope third-price column of the source file is omitted here). The reasoning script is nearly invariant to the payment rule, although the optimum differs sharply across formats (truthful in SPSB; $\approx \frac{2}{3}v$ in the three-bidder FPSB). The rule-rehearsal rows (`payment_rule_correct`, `second_price_mention`) are mechanically zero outside SPSB, since the phrase they detect describes the second-price rule.

Feature	Coefficient ($)	Cluster SE	p
`shading_intent`	-2.06	0.24	< 0.001
`overbid_intent`	-2.18	0.34	< 0.001
`truthful_intent`	-1.50	0.31	< 0.001
`payment_rule_correct`	-0.77	0.23	0.001
`dominance_language`	-0.84	0.53	0.118
`zero_profit_fallacy`	-0.82	0.37	0.026
`overpay_concern`	-0.29	0.13	0.026
`worst_case`	-0.28	0.21	0.179
`second_price_mention`	-0.21	0.21	0.307
`first_price_mention`	-0.14	0.34	0.688
`opponent_modeling`	$+0.10$	0.20	0.619
`probability_reasoning`	$+0.10$	0.10	0.347
`expected_value_reasoning`	$+0.11$	0.19	0.569
`margin_language`	$+0.17$	0.16	0.277
`conservative_language`	$+0.76$	0.14	< 0.001
`aggressive_language`	$+0.01$	0.59	0.980
`risk_language`	$+0.06$	0.14	0.640
`safety_recognition`	$+8.26$	8.24	0.316

Table 23: Pooled OLS of absolute bid deviation $|b - v|$ (in dollars) on the trace features, sealed second-price cells (n = 14,073), with model and lever-family fixed effects; standard errors cluster-robust by run directory. Intercept $4.05 (SE 0.50). $R^2 = 0.251$; fixed effects only, 0.176; incremental R^2 of the text features ≈ 0.075. The `safety_recognition` coefficient is meaningless noise (the feature fires on 3 of 21,990 traces) and is retained only for completeness. Full statsmodels output: `results/traces/ols_results.txt`.

	Stated reasoning		Behavior	
Lever (family)	target language	Δ vs. baseline	mean $\lvert b - v\rvert$	baseline
Payoff Safety (B2)	payment-rule rehearsal	3.9% $\rightarrow$ 2.8% ($p = 0.60$)	1.95 ($p = 0.015$)	3.20
Worst-case scaffold (C1)	worst-case rhetoric	2% $\rightarrow$ 43% ($p = 0.007$)	3.07 ($p = 0.81$)	3.24
Payoff Tree (C1)	rule rehearsal	33.0% $\rightarrow$ 32.7% ($p = 0.90$)	1.29 ($p = 0.017$)	3.20
Menu restatement (B1)	—	—	3.55 ($p = 0.66$)	3.20

Table 24: Dissociation between stated reasoning and bidding behavior (pooled over four models, sealed-bid second-price cells; wild-cluster bootstrap p-values clustered at the run level, with model-cluster permutation cross-checks and duplicate-trace robustness in `results/traces/mediation/`). No lever moves both columns: Payoff Safety and the Payoff Tree improve bids without changing what agents say; the worst-case scaffold changes what agents say without improving bids; the menu restatement moves neither. The baseline is the corrected pooled axis baseline (axis-1 and axis-3 cells only; the legacy axis-2 "baseline" cell was itself a clock-framed description and is analyzed as a treatment in Appendix G); under the dedicated-SPSB convention the Payoff Safety contrast is the $-2.67 \rightarrow -1.53$ of Section 4.2. No language entry is reported for the menu row because the menu prompt replaces the "second-highest" vocabulary wholesale, so rule-rehearsal prevalence is zero by construction (prompt echo, not comprehension).

		Mean Kendall-τ error, % [95% CI]				
Description	Family	Claude	Gemini	GPT-4o	Gemma	Mean
Direct DA (baseline)	—	5.8 [4.3, 7.5]	7.6 [6.1, 9.2]	1.0 [0.4, 1.7]	2.6 [1.4, 4.0]	4.2
Menu, mechanics only	B1	6.7 [5.3, 8.2]	3.1 [2.0, 4.2]	4.2 [2.9, 5.7]	13.7 [11.0, 16.7]	6.9
Menu, invariance property	B1	2.1 [1.3, 3.0]	0.3 [0.1, 0.7]	0.3 [0.1, 0.7]	4.2 [2.6, 6.1]	1.8
Rejection Safety	B2	0.8 [0.3, 1.7]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.2
Textbook strategy-proofness	(advice)	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0

Table 25: The menu-content split in DA (external-domain robustness): mean normalized Kendall-τ error in percent, with market-cluster bootstrap 95% confidence intervals ($B = 2000$, seed 1299); "Mean" is the unweighted cross-model mean. Models: Claude 3.5 Haiku, Gemini 2.0 Flash, GPT-4o, Gemma 27B. Cells are 50 markets $\times$ 4 students (a few cells have 42–49 markets after parse-failure drops), recorded in `results/merged_ranking/da_cells.csv`. A menu description that carries the invariance property improves play like a safety-exposing description; a menu description that restates only the mechanics worsens it. The textbook strategy-proofness cell states the dominant strategy outright and is included only as a diagnostic ceiling (see text). These cells never enter the auction design map.